Data Plan

Context provenance

- `if I have all the cards downloaded locally, when I scan a card it will pull ot right up ASAP and the OCR too from the text` (prior user statement; shaped the local-names-index design — lightweight catalog instead of bulk JSON)
- `Once it find the card I can tick on or off if that card if Foil` (prior user statement; foil toggle is primary control after identification, drives price/value)
- `It should help create commander decks and suggestions and so on` + `add your suggestions and use your suggestions` (prior user statement; decks + suggestions panel built in)
- Critical scanner spec handed off from AetherForge lessons: guide-mapped crop, title-band OCR y 3–11.5%, no confidence veto, no exact deviceId, HTTPS note, Take Photo fallback, DFC face splitting (handoff instruction; implemented exactly)
- MEMORY note: `cache Scryfall's lightweight /catalog/card-names index (~30k names) on-device rather than the huge full bulk JSON, match OCR locally, then fetch only the identified card's details` (builder-discovered prior decision, preserved)

Tested sources

Scryfall Catalog — card names

Used by: getCardNames action (server fetch, client caches in localStorage with date stamp) **Test command**: `curl -s 'https://api.scryfall.com/catalog/card-names' | head -c 2000` **Sample output**: `{"object":"catalog","total_values":35542,"data":["\"Ach! Hans, Run!\"", ... "Aang, the Last Airbender", "Abaddon the Despoiler" ...]}` — 35,542 names live count **Processing**: Split each name on ' // ' and index each face separately for DFC matching while keeping full name for fetch. Client caches array + downloaded date in localStorage; status line shows `Card database: N names downloaded locally on <date>` with Refresh button. Fuzzy matching (normalised Levenshtein + Dice) runs 100% locally.

Scryfall Named card

Used by: getCardByName, getPrintings actions **Test command**: `curl -s 'https://api.scryfall.com/cards/named?exact=Lightning+Bolt'` **Sample output**: id, name, image_uris {small,normal,large}, mana_cost `{R}`, cmc, type_line `Instant`, oracle_text, colors, color_identity, prices, set, rarity, prints_search_uri, edhrec_rank present on search results **Processing**: Server action fetches exact name; on 429 waits and retries once. Returns normalised

card object (id, name, set, set_name, collector_number, rarity, mana_cost, cmc, type_line, oracle_text, colors, color_identity, image_normal/small, prices usd/usd_foil, prints_search_uri, scryfall_uri, legalities). Printing picker follows prints_search_uri verbatim from the API response.

Scryfall Autocomplete + Search

Used by: manual search fallback + Commander Suggestions 'Cards to get' **Test command**: `curl -s 'https://api.scryfall.com/cards/autocomplete?q=lightn'` and `curl -s 'https://api.scryfall.com/cards/search?q=c%3Ared+type%3Acreature+cmc%3D3&order=edhrec'` **Sample output**: autocomplete returns 20 names; search returns list with total_cards, data[] cards with edhrec_rank ordering working **Processing**: Local index is primary for autocomplete (instant, offline). Scryfall fuzzy named endpoint used only as backup when local matching finds nothing. Suggestions search uses `id<=<commander identity> order=edhrec` query built server-side.

Title-band OCR (server vision read)

Used by: client scanner → `readTitleBand` server action (platform vision inference; the web-artifact runtime routes all model work through server actions, so no client-side OCR library is loaded) **Processing**: OCR only title band of guide-mapped crop: y 3%–11.5% of card height, full width minus margins, upscaled 2x, grayscale+contrast preprocessing, two treatments (grayscale/contrast + thresholded) pooled into one stacked image. No whole-card OCR, no aspect squash, no confidence veto. The action returns raw name-line candidates; candidates are expanded (full read, first line, trimmed variants stripping trailing | • © and debris) and fuzzy-matched locally against the downloaded name index; clear match = result, plausible = top-3 'Did you mean?' tappable options. Scryfall fuzzy named endpoint is the backup only when local matching finds nothing.

Card images

Used by: result card, collection, decks — rendered from cards.scryfall.io URLs returned by Scryfall API (normal size). Remote URLs rendered directly (Scryfall CDN, non-expiring); no byte-copy needed as these are API-provided display images, not user data.

Version detection (v2, user request 2026-11-08)

User's example: his Zedruu the Greathearted photo is Commander Legends (CMR) collector #543 (foil-etched, `CMR★EN`, artist Mark Zug) vs the original Commander 2011 #240 — distinguishable by the bottom info line, set symbol and artist; Secret Lair-style cards (e.g. Deadpool) break normal layouts. **Test command**: `curl -sL 'https://api.scryfall.com/cards/search?order=released&q=oracleid%3A514179c0-a50b-4565-8e71-9dad256edd85&unique=prints'` **Sample output**: 6 Zedruu printings — prm 86186, cmr 543 (finishes: etched), c16 231, pz1 119, ocmd 240 (foil), cmd 240; all artist Mark Zug; card objects carry `artist`, `released_at`, `finishes`, `border_color` (now normalised into the card schema; etched price falls back from `usd_etched`). **Processing**: After a name match, client crops the bottom strip (y 92.5–100%) of the same guide-

mapped card crop → `readBottomBand` vision action returns collector number (digits before any slash), set code, separator (★ star = foil/etched, • bullet = non-foil) and artist. `matchPrintingByBottom` (lib.ts) matches number+set exactly against all printings from `prints_search_uri`, artist as cross-check/tie-break. If that finds nothing (old `240/318` lines, funky Secret Lair frames), `pickPrintingByImage` sends the full card photo + candidate summaries (set, year, number, artist, finishes) to the vision model and the pick is labelled "Best guess — check the version". Otherwise the v1 default printing is shown labelled "Version not confirmed". ★ auto-ticks Foil. `getPrintingsByScryfallId` + `swapCollectionPrinting` power the ManaBox-style version grid on saved collection entries; the printing grid also lives on the result panel. CSV export already carries the stored set code/name, collector number and Scryfall ID per row, so detected printings export exactly. **Rapid scan**: client-only — 700ms guide-region grayscale signatures (24×34), new+stable (two consecutive similar) triggers the same pipeline; signature must change before re-scanning (no double-adds); batch foil/auto-add defaults; uncertain matches pause for the existing Did-you-mean UI.

Pass: speed + scan list + collections + deck linking (user feedback 2026-11-08)

User asks: rapid AND single scans too slow; all controls visible on the camera screen without scrolling; deck creation broken; ManaBox-style multiple named collections; scans land in a TEMPORARY list first (remove extras, keep scanning) that can move wholesale into a named collection OR become a deck (with the same cards saved into a collection so deck cards show "In: Angel Deck"-style badges); every collection shows total value. (Price up/down tracking is a separate later pass — not built here.) **Deck root cause (found by UI probe + screenshot)**: server deck actions all worked; the client `FoilImage` non-foil path rendered `className={imgClassName || className}`, discarding the sizing classes (`w-14 shrink-0`) whenever both were passed — commander/card images rendered `w-full` and crushed flex-row text to zero width, so created decks appeared as empty boxes with invisible names and no usable open target. Fixed by preferring `className` ; deck create also gained visible error reporting, Enter-to-create, rename, and two-step delete confirm. **Speed design**: ONE `readCardFrame` vision call per card reads a composite frame (title band in two treatments + bottom info strip, long edge ≤900px JPEG) returning title lines AND bottom info together; Scryfall named/printings responses are cached in-memory server-side; Speed chip Fast (default, forced in Rapid: bottom-match only, else default printing labelled unconfirmed) vs Precise (adds the visual printing pick for odd frames); rapid loop samples every 300ms, triggers after ~600ms stability, releases the watcher before the background scan-list save; "Last scan: Xs" readout on the camera screen. Camera screen is a fixed 100dvh no-scroll layout: viewfinder fills the space, batch/rapid/speed controls are overlay chips, Snap/Take Photo/List/Stop are an on-screen bottom bar, results and the scan list are overlay sheets. **New data**: `collections` table (existing cards migrated to collection_id 1 = "Main Collection" by migration 0003); `scan_list_items` staging table (server-persisted); `deck_cards.collection_item_id` + foil/price columns link deck cards to collection copies (adding to a deck saves a collection copy if unowned; removing from a deck only unlinks). Deck header/list

show foil-aware total value from linked printings; collection items show "In: " badges from `listDeckMembership`.

Pass 2: price up/down tracking (user request 2026-11-08)

User ask: "I would also like to know if a card went down or up in value." **Data (migration 0004)**: `collection_items` gains `price_at_add` (foil-aware unit price when the item entered a collection — set in `insertCollectionItem`, so scan-list commits, deck creation and direct adds all capture it; CSV import insert patched too), `baseline_note`, `last_price`, `last_checked_at`; new `price_history` table (item_id, price, foil, checked_at). **Baselines**: pre-existing items have `price_at_add` NULL until their first refresh, which backfills it from the item's stored price (the price captured when it was added — prices were never refreshed before this pass) and labels it `Baseline set <date> (first price check)`. Foil toggles and printing swaps reset the baseline to the new configuration's current price with a `Baseline reset <date> (…)` note, so comparisons stay like-for-like. **Refresh**: `refreshCollectionPrices` POSTs identifiers to Scryfall `/cards/collection` (batches of 75, fresh fetch bypassing the in-memory caches), updates `price_usd` / `price_usd_foil` / `last_price` / `last_checked_at`, and appends at most one `price_history` snapshot per item per day. Null prices are excluded from change maths and shown as "—". **UI**: each collection row shows the current unit price plus a ▲/▼ chip (`+$0.45 (+12%)` green / `−$0.20 (−5%)` red / grey "— no change"), an "Added at $" line with the baseline note, and an expandable History list of past checks. The collection header shows the quantity-aware movement total ("▲ $Y since cards were added"), a Refresh prices button, and "Prices as of <date/time>" with a stale-days note.

Pass 3: on-device OCR + rapid/torch fixes (user feedback 2026-11-08)

User report: scans still average 16–20 SECONDS per card; Rapid Fire and the Light (torch) "were having trouble working". Root cause: recognition ran through the server vision call — ManaBox is <1s because recognition is on-device. **New primary path (client/src/ocr.ts)**: Tesseract.js 5.0.5 lazy-loaded from jsDelivr CDN (`https://cdn.jsdelivr.net/npm/tesseract.js@5.0.5/dist/tesseract.min.js` — copied verbatim from the build task), ONE persistent worker warmed in the background when the Scan tab opens ("Scanner warming up… / ready (on-device)" status). Per scan: title band only (y 3–11.5%, ~1200px wide, grayscale+contrast), PSM 7 with a PSM 6 pooled re-read when the first read is garbage or fragmented token soup; thresholded treatment only when the first read finds no confident local match. Matching is 100% local (existing lib.ts fuzzy matcher, no veto, DFC faces) — ZERO server calls on this path. The server `readCardFrame` survives only as the labelled "slow read" fallback; Did-you-mean top-3 kept. Optimistic UI: the matched name flashes the moment it matches; Scryfall details + scan-list save follow. Timing readout shows path: "Last scan: Xs (on-device)" / " (slow read)". **Measured accuracy (harness, 2026-11-08)**: the identical pipeline (same tesseract.js 5.0.5 build, same band geometry/treatments/PSM flow, the app's own lib.ts matcher, live 35,542-name catalog) run against real Scryfall card images — Zedruu the Greathearted, Lightning Bolt, Fateful Discovery, Blind Obedience, Sol Ring, Command Tower, Esika God of the Tree (DFC front

face), Pyroblast: **8/8 confident-correct**, median OCR time 153ms (range 49–789ms; worst = textured Star Trek frame Sol Ring, rescued by the PSM 6 line-pooling). Two tuning fixes came out of the test: reads are split into lines before candidate generation (block mode returns the name as one clean line among texture debris), and matchNames breaks face-name ties toward standalone/front-face printings (a scanned "Lightning Bolt" face had tied with a DFC back face). The probe browser itself could not run this test — the audit environment's egress lockdown refuses cdn.jsdelivr.net (audit-confirmed as by-design, not a defect) — so the harness ran the same code/steps locally instead; on-phone timing remains the user's to confirm via the readout. **Version detection moved off the scan path**: scans are name-only now. Each scan stores a compressed (~900px) photo of the card with its scan-list item (`scan_list_items.card_image_blob_key`, migration 0005; blob via ctx.blobs, cleaned up on remove/clear/move/make-deck; `addToScanList` gained optional `cardImageBase64`, new `getScanItemImageUrl` action, `hasCardImage` on the item schema). A per-item **Detect version** button (scan list + result sheet) fetches that photo, rebuilds the read frame, runs the existing `readCardFrame` + `resolveVersion` (bottom-line match; Precise mode keeps the visual pick), and applies printing + ★/• foil. **Rapid Fire rebuilt as a pure state machine (client/src/rapid.ts)**: baseline adoption (the empty scene when rapid starts never scans), handled-card suppression (no re-add while the card sits there), FAILED reads unlock after 1.5s and retry the same card (max 4 attempts → "tap the card view to scan now"), any slight signature wiggle resets the attempt counter, tap-on-viewfinder = scan now, Did-you-mean pause/resume hands the card to the machine as handled. Scanner-box mode (📦 Box chip, persisted): locks focus/exposure best-effort (capability-checked, never throws) and judges stability on the title-band signature with a widened same-card threshold so cards sliding into the rig register sooner. Unit-tested with synthetic signature sequences: tests/rapid.test.ts — 19/19 pass (empty→A→recognised→still there→removed→B→fail→retry→success→exhaustion→tap→wiggle-reset→box mode). **Torch rebuilt (client/src/torch.ts)**: capability check via `track.getCapabilities().torch`; applies `{advanced:[{torch}]}`; unsupported → toggle disabled with "Light — not supported on this camera" label; preference persisted (localStorage) and re-applied to every new track in startCamera/setupTrack; explicit camera Stop turns it off and resets the toggle. Unit-tested with mock tracks: tests/torch.test.ts — 15/15 pass (payload shape, unsupported no-op + disabled state, restart reapply, error branches). Real illumination is phone-only.

Pass 4: find-the-card + unjam Rapid + light/dark fixes (user screenshots 2026-11-08)

Evidence (user's real screenshots): (1) in the scanner box the card occupies only ~40% of the guide height on a white tray, so the guide-relative title band read box interior; status showed a 39.7s SLOW read because the user had Precise on and failed local reads auto-fell-back to the server; 'Reading card…' persisted (jam). (2) A near-black box frame with Light ✗ — Box mode locked exposure. (3) A clean full-frame Take Photo of Kemba, Kha Enduring produced garbage on-device OCR ('Go Re age k…') with junk suggestions. **Card detection (client/src/detect.ts)**: before OCR, every capture path (live Snap, Rapid, tap, Take Photo, upload) locates the actual card: downscale to ~480px, texture-density (gradient) row/column projections find the card's sustained

textured slab against the smooth tray/table, contrast cross-check gives a confidence. Bands are then cropped relative to the DETECTED CARD rect (x 4–80% — mana-cost corner excluded). Low confidence → honest fallback to the guide/whole frame, labelled 'card outline not found — using guide'. The detected rect is drawn on the viewfinder (blue 'CARD FOUND'); photos get an outline preview in the result/error UI. Unit tests: tests/detect.test.ts (synthetic grids: box geometry, full-frame, empty, near-black, seams). **Box calibration**: in Box mode, '🎯 Calibrate box' runs detection once on a card slid fully in and stores the rect (localStorage `manabox-box-calibration`); later box scans crop from the stored rect (green 'CALIBRATED CARD AREA' outline drawn so the user can align the phone in the rig); Re-calibrate overwrites. **Band pool (client/src/ocr.ts)**: variants tried best-first, stopping at the first confident (≥0.82) local match: V1 y2.5–11% contrast, V2 y4–12.5% contrast, V3 y2.5–11% adaptive (local-mean threshold, polarity-normalised for light-text frames), V4 y8–17% contrast — added during testing because occluded/angled shots put the title below the standard windows (Kemba case). PSM 6 pooling now also triggers on THIN reads (<8 letters), which rescued Sol Ring ('(solRine' → '[Sol Ring', score 1.0). **Rapid unjam**: Rapid NEVER auto-falls-back to the server read (`resolveRapidMode` in rapid.ts: rapid ⇒ Fast + allowSlowFallback=false — also greys out the Precise chip); the fallback is a manual '🐢 Try slow read' button on the unrecognised card. Every OCR attempt is capped at 6s (Promise.race); a timeout (or 2 consecutive errors) terminates + recreates the Tesseract worker automatically ('Scanner restarted'). runRapidScan resets its single-flight guard in a finally block on EVERY path; an 8s watchdog in the Scan tab invalidates in-flight reads (generation counter), restarts the worker, force-resets the Rapid machine to watching (new `forceReset()`), and clears 'Reading card...' — no page refresh can ever be needed. New tests in tests/rapid.test.ts: hung-recognition → watchdog reset → next card scans; resolveRapidMode matrix. All suites pass: rapid 25 checks, torch 15, detect 10. **Light + dark frame**: Box mode no longer locks exposure (focus lock only). A capture whose card crop averages <55/255 luminance is retaken once with exposureCompensation raised to the camera's max (capability-checked). Torch diagnostics: `applyTorchDetailed` returns the failing constraint's error name and `torchCapabilityValue` reports getCapabilities().torch verbatim ('true'/'false'/'absent'/'error'); both are shown under the Light chip when it fails and in the Help sheet, with 'open in Chrome' advice for in-app webviews; the chip label is '💡 Light ✗ (not in this browser)' when unsupported. **Mandatory real-world tests (local identical pipeline: app modules verbatim under Node + node-canvas with a faithful CSS-filter shim + tesseract.js 5.0.5 npm worker — the managed Chromium in this VM cannot be launched standalone, it requires the browser daemon; the audit probe browser remains blocked from the Tesseract CDN as in Pass 3)**: composites of real Scryfall images (Tergrid's Shadow, Kemba Kha Enduring, Lightning Bolt, Sol Ring) at 40% of frame height centred on white + 98% controls, the user's real Zedruu phone photo, and Kemba cropped from the user's own screenshot at the grounded card position — **10/10 identified correctly**: box40 all top-match (Kemba 1.0, Bolt 0.91, Sol Ring 1.0 via PSM-6 pool, Tergrid's Shadow 0.82 with 0.248 margin — at the confidence boundary, else top Did-you-mean), full-frame controls all confident, Zedruu photo 1.0 (adaptive variant), Kemba screenshot crop 0.897 confident with 0.343 margin over the runner-up (detection honestly fell back to whole-frame on that partial-card crop; variant V2 read the title). Detection rects on composites matched ground truth exactly ([0.30,0.30,0.70,0.70]). What changed during testing: +V4 low band variant, +thin-read PSM 6 pooling.

Pass 5: collector-number-first + burst consensus + coordinate truth (user screenshots 2026-11-08, 7 shots)

Evidence: Treasure Keeper was the top Did-you-mean at only 60% and still paused Rapid; Dawnhart Mentor read 'LY' → junk suggestions; Spider Spawning timed out → worker restart (6.6s); Prism Array under a good CALIBRATED outline read total garbage; reads of 6.6–6.8s with worker restarts. In every box shot the bottom info line (177/184 AER, 179/277 MID, 102/277 VOW) was legible and unused. **Collector number as primary signal**: `getCardBySetNumber` action wraps GET https://api.scryfall.com/cards/{set}/{number} (in-memory cached; verified live: aer/177 → Treasure Keeper exact printing, bogus pair → null). Client parses the bottom strip (y 91.5–99.5% of the card crop — widened from 93.5% after crop slack pushed the number line out of the window) with `parseBottomLine` (bottomline.ts: slash + no-slash formats, O→0 confusion tolerance, rarity/language token filtering). SAFETY RULE: a pair is looked up only after a second treatment (contrast) of the same strip agrees on the NUMBER and on the code exactly or within one letter — thresholding merged Dawnhart's 179 into 129 (a real different card), so unverified pairs never reach the lookup. Old pre-M15 frames (Spider Spawning ISD) print no set code at all — the number path is impossible there by design and the title path carries them. **Burst + consensus**: Snap and Rapid capture 3 frames ~120ms apart; per frame one title band (with ocrBand's PSM 7→6 rescue pool — PSM 7 ALONE returns 'Tr'/'' on bands PSM 6 reads perfectly, measured) + one bottom band; candidates pool across frames into `matchNamesPooled` (token-aware scoring in lib.ts: token Dice/Levenshtein blend — 'ressure kecrer' now ranks Treasure Keeper 0.68 vs Pressure Point 0.45, previously a coin flip). Rapid auto-accepts a decisive pool (top ≥0.62, margin ≥0.10, `decideRapidMatch`) with an Undo banner; pauses only on thin margins. NO-READS (<4 alphanumerics, e.g. 'LY') never produce suggestions and never pause. **Coordinate truth**: all guide cropping goes through `geometry.ts` (`coverTransform` / `cssRectToSource` / `sourceRectToCss`) — one shared object-cover implementation for capture; overlay/detection were already guide-relative. tests/geometry.test.ts round-trips 1280×720 and 640×480 sources in a 390×844 element; the harness pixel test (synthetic card, black title bar) measured detected-centre error 0.001 and title-band dark fraction 0.742. Capture resolution is DISPLAYED in the scan status ("Capture: 1280×720 video"); when videoWidth < 1000 and ImageCapture exists, Snap/Rapid frames come from `grabFrame()` full-resolution stills. **Detection robustness fix**: detect.ts span-finding rewritten to relative density (rough absolute pass seeds the span; refinement measures density within the candidate span). The old absolute full-frame threshold split runs on smooth art rows — detection flipped found/null on a 2px shift of the same card (measured). All jitter/overlay variants now detect stably; detect.test.ts unchanged and green. **Worker policy**: read cap 4s (was 6s); a single timeout discards the attempt and REUSES the worker; restart only after 2 consecutive timeouts (the restart re-warm was the 6.6s stall). The 8s app watchdog no longer restarts the worker. **Show-your-work**: Did-you-mean and result sheets render a "What the scanner read" block — pooled raw read, winning evidence (collector number / title), capture resolution, and thumbnails of the actual title-band and bottom-band images that were OCR'd (bandThumb of the real band canvases). **Mandatory harness (local identical pipeline, node-canvas + tesseract.js 5.0.5, real Scryfall images of the user's exact failing printings, full-card composites incl. bottom lines)**: Treasure Keeper (AER 177), Dawnhart Mentor (MID 179), Spider Spawning (ISD

203), Prism Array (BFZ 81), Courier Bat (VOW 102) at box-40% geometry AND calibrated geometry — **10/10 identified correctly, ZERO wrong identifications**: Treasure Keeper ×2 and Courier Bat (box40) via collector number (verified pairs, real API lookups with the PARSED values); Dawnhart via title 0.945 (its number pair was correctly BLOCKED by verification — treatments disagreed 1942/129 vs 179); Spider via title 0.962 (no set code exists on that frame); Prism via title 1.0. Harness ms are node-VM numbers (node-canvas + tesseract under this VM run several× slower than the phone's native pipeline measured in Pass 3 at 49–789ms/read); on-phone end-to-end remains the user's readout to confirm.

Pass 6: Snap parity + card-shaped outline + bottom-evidence salvage (user screenshots 2026-11-08, 2 shots)

Evidence: (A) Griffin Protector SUCCESS via Rapid — result showed DMU #18 while the WHAT THE SCANNER READ panel proved the bottom line had been read ("Q0/280 C M20 LN CHRISTOPHER MOELLER K"; thumbnail "020/280 C / M20 • EN / CHRISTOPHER MOELLER"): the strict parse died on the mangled "Q0" and the app silently defaulted to a different printing. His card is M20 #20. (B) Witherbloom, the Balancer via regular SNAP (hand-held): the CARD FOUND outline was shifted DOWN (title above the outline, bottom through the rules text), status "Scanner restarted (repeated read timeouts)", last scan 3.1s — Snap timed out despite detection finding the card. User: "The rapid fire works great!" **Snap parity**: Snap's old tail (burst → unconditional identifyCore: 4-variant pool × PSM 7+6 + slow server read) stacked up to 8 more 4s-capped reads past the 8s watchdog, which invalidated the read and left only the restart note. Now the identifyCore fallback runs ONLY when Precise is explicitly selected; with Fast (default) Snap runs the identical burst pipeline as Rapid at identical cost. ImageCapture stills and uploaded photos are downscaled to ≤1600px long edge before any OCR. A Snap always settles visibly: result, Did-you-mean with the debug block, or a failure panel carrying the band thumbnails + pooled raw read + per-stage timings (capture · detect · title · bottom · match, accumulated from the reads' own ms) + a manual slow-read button. OCR auto-restarts are capped at ONE per scanning session (ocr.ts `resetOcrSession` / `getAutoRestartsUsed`); further timeout streaks emit "Scanner is struggling with this card — try Rapid or Take Photo" and reuse the worker. The watchdog now surfaces that failure panel for manual reads it invalidates (Rapid's watchdog behaviour unchanged). **Card-shaped outline**: `correctCardAspect` (detect.ts) corrects the detected rect to the true 88/63 aspect — too short → height grows 70% to the top / 30% to the bottom; too tall → width grows symmetrically; shifted into frame rather than shrunk; ≤2% deadband so good rects don't drift. Applied in prepareCardCanvas, box calibration and the probe handle; the corrected rect is what the overlay draws and all bands derive from. CRITICAL: rect fractions are per-axis normalised, so all correction math runs in square-pixel space (x scaled by frame width/height) — the first harness run proved the naive fraction-space version over-grows height ~44% on a 1080×1920 frame. tests/aspect.test.ts covers the Witherbloom geometry, the no-drift regression, symmetric width growth, edge shifting, and real 1080×1920 phone geometry. **Bottom-evidence salvage** (lib.ts `salvagePrintingFromBottom` + `bottomSetCodeHint`): when a title match exists but the strict verified (set, number) path produced nothing, the pooled raw bottom text is scored against the matched card's OWN printings (cached getPrintings): set-code equality 45, collector

number exact after per-character confusion normalisation (O/Q→0, I/L/|→1, S→5, B→8, Z→2, G→6) 45, one-digit-off 20, artist token overlap 20 full / 12 partial. A clear win (≥55 and ≥15 ahead) SELECTS the printing with the message "Version matched from the bottom line: Core Set 2020 #20"; no win but a bottom set-token naming a different set than shown → visible hint ("Your card's bottom line reads M20 — tap Change version if that's not DMU"); otherwise the honest name-only note. Wired into identifyBurst's title-win path and into resolveVersion (Detect version). MANDATORY FIXTURE (tests/salvage.test.ts, printings fetched live from Scryfall: dmu 18 Omar Rayyan, m20 20 Christopher Moeller, a25 17 Moeller, m13 16 Moeller): the user's exact raw text "Q0/280 C M20 LN CHRISTOPHER MOELLER K" → **M20 #20, score 85, margin 65** (set + one-digit-off number + artist). Negatives: Treasure Keeper's (AER 177, Paick) and Dawnhart's (MID 179) bottom lines against Griffin's printings score below threshold → no override; artist-only evidence also stays below threshold (Moeller drew 3 of the 4 printings). **Layout**: scan-tab footer restructured to two lines (line 1: "Scan list: N cards · $X.XX"; line 2: scanner status · last scan · capture as separate nowrap segments in a flex-wrap row at 10px — wraps only at separators, no mid-token truncation). Light chip label shortened to "💡 Light ✗" (the reason stays in the dedicated diagnostics line below). **Mandatory harness (local identical pipeline, node-canvas + tesseract.js 5.0.5, app modules verbatim)**: composites at PHONE capture geometry (1080×1920 frames — a 700×980 composite measurably produced a verified-but-wrong VOW 2 pair for Courier Bat that the phone-size frame reads as verified VOW 102 in both treatments; the app displays capture resolution for exactly this reason). Results — **19/19 checks**: Treasure Keeper (number), Dawnhart Mentor (number), Prism Array (number), Courier Bat (number VOW 102), Griffin Protector (title + salvage → M20 #20, real bottom OCR "020/280 C M20 EN CHRISTOPHER MOELLER", salvage score 110 margin 90), Witherbloom (title), Spider Spawning (burst V1 genuinely fails on clean composites — its settle path, the variant pool, reads it at V2; modelled and asserted as such). Witherbloom truncated-rect control: the old shifted-down rect without correction reads garbage ("BE. AY 4"), the SAME rect with aspect correction identifies the card by title. Snap/Rapid parity asserted per fixture (same identification; acceptance thresholds differ by design — Rapid auto-accepts a decisive ≥0.62 pool, Snap's bar is 0.82, both surface the same top card). Zero worker restarts across all runs. Stage timings (node-VM numbers, several× slower than phone): detect ~100–150ms, title ~100–700ms, bottom ~150–1000ms, match ~500–1700ms per burst. All unit suites green: aspect, detect, geometry, matcher, rapid (regression, untouched), salvage, torch.

Pass 6 follow-up: watchdog diagnostics gap (2026-11-08)

Gap: Pass 6's stage timings/band thumbnails were published only when identifyBurst FINISHED (`finish()` → setScanDebug). A manual Snap that HUNG (the Witherbloom mode) never finished, so the watchdog's failure panel showed the previous scan's debug block or nothing — not the scan that timed out. **Fix (client/src/scandebug.ts, new)**: `ScanDebugTracker` owns the debug data per scan ATTEMPT. `beginAttempt` runs at the start of every attempt (identifyFromCardCanvas, runRapidScan) and resets all state; identifyBurst publishes incrementally under the attempt token — band thumbnails the moment the first card crop exists, pooled raw read as each OCR returns, stage ms as each stage completes (capture/detect recorded up front; title/bottom per frame;

match via pooledMatches + match-stage marks around the number lookup, resolveByName and salvage). Publishes from a stale or watchdog-invalidated token are dropped, so a late-finishing hung pipeline can never overwrite a newer attempt or the failure panel. The watchdog takes `timeoutSnapshot()` BEFORE invalidating: only the current attempt's partial data, completed stages with ms, plus an explicit `hung during: <stage>` marker (active stage, else first unrecorded stage); if no band exists yet the snapshot carries the plain note "Timed out before any band was read — stage: capture/detect." (rendered by ScanDebugBlock) instead of older data. Healthy finishes keep the exact five-stage format (tracker composes `capture · detect · title · bottom · match` identically — asserted in tests). **Tests (tests/scandebug.test.ts, 26 checks)**: hang after bands before match → timeout snapshot contains THIS attempt's band URLs + partial stages + "hung during: match" and nothing from the previous completed attempt; invalidation blocks the late finish; stale tokens from two attempts back cannot publish; hang before bands → explicit no-bands note, empty bands/read/stages (no leakage); incremental growth + final format on a healthy scan. All 8 suites green (aspect, detect, geometry, matcher, rapid, salvage, scandebug, torch); client typecheck clean.

Pass 7: success-is-final + bottom isolation + collapsible diagnostics (user screenshots 2026-11-08, 2 shots)

Evidence: Cost of Brilliance (SOS #77, $0.14) and Cheerful Osteomancer were BOTH identified and added (scan list 2 cards/$0.36; stages ~2.3s total: capture ~430ms · detect ~53ms · title ~260–334ms · match 1.1–1.5s) — yet "Last scan: 7.2s", a red timeout banner stacked OVER the success sheet / a full timeout panel replaced it, and the scanner wedged until a page refresh. Diagnosis: a read belonging to the attempt (the bottom-strip read on Cheerful's pure-speckle bottom band — both panels showed "bottom 0ms") kept running AFTER the match succeeded; the 8s watchdog then tore down scanner state indiscriminately. **Fix (client/src/scanflow.ts, pure + unit-tested)**: (A) SUCCESS IS FINAL — an `AttemptGuard` is the control-flow twin of the debug tracker: `identifyBurst`'s `finish()` settles the attempt the moment an outcome exists, and the pipeline checks `guard.isActive(attempt)` after EVERY await (title read, bottom read, verification read, number lookup, name resolve, salvage, Precise legacy path), returning a silent `{kind:"aborted"}` outcome callers drop without any UI/state change. The watchdog consults `watchdogAction({settled, hasOpenResult})`: a settled attempt (or an open success sheet) → "ignore" — no banner, no panel, no reset. (B) BOTTOM ISOLATION — `ocrBottomBandCapped` (ocr.ts) is the fast path's only bottom read: noise PRE-CHECK (`analyzeBottomBand`: dark fraction extremes near-blank >0.985 white / near-solid >0.70 dark, plus text-row structure = ≥3 rows with a ≥7px horizontal dark run AND ≥58% of dark pixels concentrated in a 50%-height window; speckle measures rows=0, conc≈0.52 → skip) records stage "bottom skipped (no text row)" via the tracker's new `recordStageNote` instead of the misleading "bottom 0ms"; a ~1.2s hard cap (`raceWithCap`, BOTTOM_READ_CAP_MS) abandons an overrun read as no-evidence, sets a per-attempt flag so no further bottom reads stack, and silently swaps the busy OCR worker in the background (`abandonBusyWorkerSilent`, respecting the max-1-auto-restart budget, NO scanner event — invisible after a success). Version salvage / Detect version from the saved photo are unaffected. (C) RECOVERY — a genuine (unsettled) timeout runs `watchdogRecoveryPlan`: clear

busy → reset Rapid machine → `ensureCameraLive()` (track ended / video stalled → reattach via startCamera) → watcher watching again → status ready; the failure panel gained a Dismiss ✕ button, auto-dismisses when a new scan begins (Snap already cleared it; Rapid now does too), and the readError panel no longer renders while a result sheet or Did-you-mean sheet is open (result precedence in render AND in the watchdog). (D) COLLAPSIBLE — `ScanDebugBlock` is now a tappable header (▸/▾, aria-expanded) everywhere: context success/failure/didyoumean, default CLOSED on success, OPEN on failure + Did-you-mean, manual toggle persisted per context in localStorage (`manabox-debug-open-<context>` , `resolveDebugOpen`). **Tests (tests/scanflow.test.ts, 44 checks, all pass)**: (1) hang-after-success simulation — never-resolving bottom read abandoned at a 60ms test cap in 61ms, card added, watchdog = ignore with EMPTY recovery plan, zero timeout UI/resets, late continuation sees the settled attempt inactive, and a SECOND fixture scan in the same session succeeds (added: Cost of Brilliance · Cheerful Osteomancer); raceWithCap ok + rejection-swallowing checks. (2) Speckle fixture (deterministic 16% uniform noise, 240×80): skipped, reason "no text row", stage note exact; near-blank → "near-blank"; near-solid → "near-solid"; text fixture (two block-character lines): NOT skipped (conc 0.64, 28 text rows, maxRun 13); speckle pipeline makes ZERO bottom OCR calls and completes via title in <300ms. (3) Genuine timeout: plan contains clear-busy/reset-rapid/verify-camera/restart-watcher/status-ready/show-failure-panel; applying it to a wedged fake state ends ready + camera live + watching, panel shown, next scan settles; open result sheet alone suppresses the watchdog. (4) Collapsible: defaults (success closed, failure/did-you-mean open), stored "1"/"0" overrides per context, garbage stored value falls back, keys per-context. All 8 pre-existing suites green (aspect, detect, geometry, matcher, rapid — unchanged behaviour except inheriting the guard's cancellation safety — salvage, scandebug incl. the new stage-note composition, torch).

Pass 8: modern number formats + rescue order + title-band search window (user screenshots 2026-11-08, 3 shots)

Evidence: The Pass 6/7 flow fixes HELD (no refresh, list grew to 5 cards/$0.75, "Last scan: 1.2s", panels dismissible). Two pure identification failures remained, both hand-held Snap in dim light: (1) Snooping Page — failure panel "No plausible name", raw read "PR · - WN y N p · NAN AY BN BA Ea 13 | bottom: U 0227 SOS EN ALEXANR HONORE", stages "... match 0ms" (the exact lookup was NEVER attempted); its title-band thumbnail was almost entirely black (band aimed ABOVE the title) while the bottom thumbnail read crisply "U 0227 / SOS • EN / ALEXANDRE HONORE". (2) Elemental Mascot — raw "fi Floormaensnt al VEY · pe · 8 Flomoental Maceo | bottom: A C 0185 SOS EN JUTYNA DURA", title visible at the very BOTTOM EDGE of the band thumbnail, match stage ballooned to 7.3s. **Root causes**: (1) Modern printings print rarity letter + BARE collector number ("U 0227", "C 0185") with NO "/NNN" denominator, and — decisively — the burst path only looked up pairs that passed the two-treatment verification gate; the contrast-treatment verification read of a white-on-black strip in dim light garbles or noise-skips, so the strongest evidence in the frame was discarded. (2) Fixed single-offset title bands: detection/correction variance hand-held shifts the true title row to/below the standard band's bottom edge. **Live Scryfall verification (2026-11-08, curl)**: `GET https://api.scryfall.com/cards/sos/227` → **Snooping Page**, artist "Alexandre Honoré"; `GET https://api.scryfall.com/cards/sos/185` → **Elemental Mascot**, artist

"Justyna Dura". Both are Secrets of Strixhaven (SOS), released 2026-04-24 — exactly the printings in the user's photos, confirming (set SOS, bare number) is unique and sufficient. **Fix (A) — candidates + rescue order** (`client/src/bottomline.ts`): `parseBottomLineCandidates` enumerates EVERY plausible (set, number) pair — the strict parse's hit plus a token scan accepting rarity letter attached ("U0227") or preceding, 1–4 digit numbers with OCR-confusion normalisation (a token needs ≥1 real digit, so "SOS" never becomes "505"), an OPTIONAL denominator (slash tokens contribute only their numerator), a set-code token within ±3 tokens on EITHER side (incl. M20-style letter+digit codes), junk single-letter tokens tolerated ("A C 0185" → C 0185); a bare number with NO adjacent set code yields nothing. `buildNumberCandidates` pools candidates across all bottom reads of the burst (primary + verification treatments) ordered by corroboration (≥2 sightings, primary+verify agreement, or cross-treatment same-number/near-code per the Dawnhart rule). In `identifyBurst`, when the frame loop ends below confident, RESCUE 1 runs BEFORE the pool-exhausted judgement and before any printings/salvage work: up to 5 untried candidates get exact `getCardBySetNumber` lookups, 404s discarded; multiple hits break ties on the artist line (`artistMatchesBottomText` in lib.ts — edit tolerance ≤1 for 4–6-char tokens, ≤2 for 7+, matching the dropped letters in "ALEXANR"/"JUTYNA"; tie-break only, never a requirement); a hit finishes as "Matched by collector number SOS 227 — the exact printing printed on the card". Rapid inherits Rescue 1 only when its title read fails (happy path untouched). The per-frame verified-pair fast path is unchanged. **Fix (B) — title-band search window** (`client/src/ocr.ts`): `TITLE_RESCUE_BANDS` = standard window shifted +4% (y 6.5–15%) and +8% (y 10.5–19%) of card height, plus `TITLE_TALL_BAND` (y 2–16%) read with PSM 6 via new `ocrBandMultiLine` (best name-matching LINE inside wins, via the pooled matcher). `buildBandCanvas` now upscales every band to ≥64px tall. RESCUE 2 (Snap/Take Photo only, `!rapid`): one silent auto-retry after Rescue 1 — rescue bands read on the later burst frames first, then the first frame; each read matched individually (the winning band is NAMED in the diagnostics evidence: "title (offset+8 band)" / "title (tall band)"), then pooled with all prior reads; bounded to ~3.5s of extra reads. On a rescue-band title win, if number candidates existed (and none hit), the printings salvage does NOT run (garbage title evidence + foreign bottom line must not re-score printings) — name-only info instead. **Fix (C) — match-stage balloon**: `pooledMatches` is memoised on the pooled-read signature inside identifyBurst (the Mascot scan re-scored an unchanged pool 4×; re-scores now happen only when a NEW read lands), and Rescue 1 short-circuits before salvage/printings work; rescue lookup time is added to the match stage so timings stay honest. **Tests (tests/rescue.test.ts, 47 checks, all pass)**: mandatory fixtures from the user's exact raw strings — "U 0227 SOS EN ALEXANR HONORE" → SOS/227 first candidate; "A C 0185 SOS EN JUTYNA DURA" → SOS/185 first (junk "A" tolerated); either token order, attached rarity, "O227" confusion; bare number without set code → NO candidates (also "240/318", number+artist-only); regressions "177/184 AER", "020/280" → 20 with M20/20 now among candidates, Q0-confusion unchanged (salvage still owns it; salvage suite green), "0543 CMR ★"; corroboration matrix (two frames agree → verified; 179-vs-129 disagreement → neither verified; near-code MID/MLD on 179 → verified; verified outranks unverified; garbled-verify Snooping pool still leads with SOS/227 unverified); artist checks against the live Scryfall artists (Honoré/Dura accept with dropped letters, cross-rejects); band geometry (Mascot row 10.5–13% not contained by standard 2.5–11% but contained by tall + offset+4; Snooping row 13–18% fully below standard,

contained by offset+8; correctly-aimed control row stays in the standard band). The probe handle gained `recognizeFull` (real identifyBurst over one prepared frame, returning outcome + winning evidence + stage timings) for the browser-harness band-misalignment composites; per-fixture expected winners: Snooping → number (SOS 227), Mascot → number (SOS 185), bottom-edge title → tall/offset+4, below-band title → offset+8. All 10 suites green; typecheck clean. The managed audit browser cannot reach the Tesseract CDN (Pass 3 note), so on-phone confirmation remains the user's retest: Snap Snooping Page + Elemental Mascot hand-held.

Pass 9: deadline settlement + rescue on every exit path + bigger name guide + distance robustness (user screenshots 2026-11-08, 3 shots, same card)

Evidence: Divergent Equation (SOS #43), hand-held Snap, took THREE tries. Attempts 1–2 both timed out ("hung during: title") even though their stage lines showed title (162/179ms) and match (446/683ms) COMPLETE: attempt 1 read "Divergent ae" with a PERFECT bottom line "U 0043 SOS EN INKOGNIT" and a crisp title-band thumbnail ("Divergent Equation" plainly readable); attempt 2 read "iverson tquation" with a garbled bottom ("U 0043 . 0S EN 15 NOG."). Attempt 3 succeeded by name only when the card was held close, filling the guide. User asks: fix the 3-tries problem; make the card-name area of the guide LARGER; "It shouldn't matter how far and close I am"; "make sure the card detection works efficiently, this is important." **Root causes**: (1) Pooled title variants run on ONE shared OCR worker whose jobs serialise; the attempt awaited ALL pooled reads, so ONE spinning variant job blocked settlement forever despite usable evidence in hand. (2) Pass 8's collector-number rescue ran only in the normal settle path — attempt 1's perfect bottom evidence never got rescued because the attempt died by watchdog instead of settling. (3) Band crops started too tight on the left, clipping first letters inconsistently ("Divergent ae", "iverson tquation"). (4) Found in the Pass 9 distance harness (below): the detector's REFINE pass amputated cards with smooth art. **Fix (A) — deadline-based settlement** (`client/src/scanflow.ts`, pure + unit-tested): `AttemptClock` (injectable now) + `cappedRead` — every OCR read in an attempt is individually raced (title variants ≤ `TITLE_READ_CAP_MS` 900ms, bottom ≤1.2s, both via the capped title/bottom readers in ocr.ts whose timeout now covers worker acquisition too) and SKIPPED unstarted once the clock is gone. `identifyBurst` settles at `ATTEMPT_DEADLINE_MS` (2.5s) with the best evidence pooled by then (a Settlement-1 pooled re-check covers the deadline-break case), Rescue 2 runs inside a widened `ATTEMPT_RESCUE_DEADLINE_MS` (4s) envelope, and settlement network calls are raced (name lookup 1500ms, printings 1500ms, set-number 1200ms via `lookupSetNumber`). Late results die on the existing attempt token; timed-out workers are swapped in the background per existing rules. "Hung during: X" is unreachable whenever any variant of X returned usable text. **Fix (B) — rescue on every exit path**: the number rescue is factored into `rescueByCollectorNumber(sources, lookup, {tried, artistText})` (scanflow.ts) used by the settlement path AND by the 8s watchdog itself: `identifyBurst` mirrors each attempt's captured bottom sources into `attemptEvidenceRef` as reads land (`publishEvidence`), and the watchdog — before showing any failure — runs the rescue on that ALREADY-CAPTURED text (zero new OCR, raced 3s); a hit settles the attempt and adds the card with "recovered from the bottom line this scan had already read" copy. Mandatory

fixture (attempt 1 exactly): title "Divergent ae" + bottom "U 0043 SOS EN INKOGNIT" + pipeline hung → Divergent Equation SOS #43, watchdog then ignores the attempt. Attempt 2's garbled text yields only NOG/43, which 404s — that attempt is carried by its title fragments instead. **Fix (C) — title fragments** (`client/src/ocr.ts`, `client/src/lib.ts`): title band windows padded ~3% of card width on both sides (standard x 0.01–0.83 y 2.5–11%; mana-cost corner still excluded) plus a new "left-shift" variant at x 0.0–0.81; rescue bands carry the same padding. The token matcher now scores clipped tokens (dropping 1–2 leading chars from either side, ×0.97) and short debris tokens (≤3 chars, letter-fraction capped 0.5), and `matchNamesPooled` adds a FRAGMENT-UNION pass (union of all candidates' tokens vs each contender, ×0.98). Measured against the full 35,542-name index (2026-11-08): "Divergent ae" → Divergent Equation 0.657 TOP (Divergent Growth, the old wrong winner, 0.590); "iverson tquation" → 0.772 TOP (Rapid-auto: margin 0.118); pooled → 0.965 (confident); clean read → 1.000. **Fix (D) — bigger name guide** (`client/src/App.tsx`, `client/src/geometry.ts`): the guide's name zone is now a prominent target band — top 18% of guide height (`GUIDE_NAME_ZONE`), gold border-2 + translucent gold fill, bold white "CARD NAME HERE" label with dark text shadow (was an 8.5% dashed "NAME" strip). A lightweight live-detection interval (manual mode only, ~800ms, hysteresis of 2 misses) keeps the CARD FOUND outline live while aiming and drives a hint line under the viewfinder: found → "✓ Card found — fine at this distance, tap Snap" (green); none → "Place the card inside the outline, with its name in the top band". Reading stays detection-based; the guide is a placement aid only. **Fix (E) — distance robustness** (`client/src/detect.ts`, `client/src/App.tsx`): (i) DETECTOR AMPUTATION found by the harness — `detectCardRect`'s refine pass (0.22 relative density) collapsed Treasure Keeper's span onto its dense bottom slab (refined rows kept 140 of the rough pass's exact 338; aspect 0.63 → null) at 70%/95% of frame height on both dark and light backgrounds, because its painterly art rows sit at 0.07–0.17 density for stretches longer than the gap tolerance. Fix: `pickSpan` — a refined span may TRIM its rough span but never keep <55% of it — plus a rough-span retry before the aspect/area gate rejects. After the fix, probe rects match ground truth at all sizes/backgrounds (70%: y 0.148–0.852 vs truth 0.150–0.850). (ii) Snap hi-res still: when the detected card is <45% of frame height, `capturePreparedBurst(3, {hiResSmallCard: true})` (Snap path only; Rapid untouched) retakes the burst from full-resolution ImageCapture stills via `captureStillGuideCanvas` (guide mapped straight out of the full-sensor bitmap, NO whole-frame downscale), keeps whichever burst gives the card more pixels, and downscales only the still's CARD CROP to 1600px; a note tells the user when it fires. Where ImageCapture doesn't exist it silently keeps the video frames. **Tests**: NEW `tests/deadline.test.ts` (54 checks): clock mechanics + production budgets (2.5s/4s); A1 — one variant never resolves + fragments "iverson tquation"/"Divergent ae" → settled in 182ms of a 600ms test clock, pooled 0.965 confident; A2 — all title variants hang + bottom "U 0043 SOS EN INKOGNIT" → number rescue SOS/43; A3 — exhausted clock skips reads unstarted; B — the mandatory watchdog fixture (zero new OCR, settle → watchdog ignore), NOG/43 404 honesty, 404-then-next-candidate, artist tie-break, garbage → no candidates; C — band geometry (x0 ≤0.01, x1 ≥0.83, <0.90 mana cap, left-shift at x0=0, rescue bands padded); D — guide constants (0.18, "CARD NAME HERE"); E — synthetic detection at 25/40/70/95% (centre error 0.000, corrected height within ±0.003–0.007 of truth, title covered) + the smooth-art amputation regression (detected height 0.703 vs truth 0.70). `tests/matcher.test.ts` gained §8 fragment fixtures at the full-index-measured scores. DISTANCE

HARNESS `/tmp/scan-harness/harness7.ts` (node-canvas + tesseract 5.0.5, real Scryfall images incl. divergent.jpg fetched live from the API-returned image_uris.normal for sos/43, 488×680, dark cloth, 5:7 1080×1512 frames, current pipeline incl. both rescues): **12/12 identify** — Divergent Equation 0.965/0.981/0.991/0.991, Treasure Keeper 0.98/1.0/1.0/1.0, Griffin Protector 0.98/1.0/1.0/1.0 at 25/40/70/95% respectively, ALL via the title path on frame 1 (bottom stage 0ms — decisive title precedes the bottom read, as in the app); detection centre error ≤0.003 of frame at every distance, raw height error −0.056…+0.004 (aspect correction + expansion absorb the rest, as the title wins prove); card-crop pixel heights 373px (25%) → 1486px (95%) — the measured glyph floor in the harness is a 373px-tall crop from a 488px-wide source scan, and the phone's hi-res still path gives small cards far more pixels than that. All 11 suites green (aspect 18, detect 10, geometry 20, matcher 33, rapid 34, rescue 47, salvage 17, scandebug 26, scanflow 44, torch 15, deadline 54 = 318 checks); client + server typecheck clean. Rapid's happy path unchanged (it inherits only the deadline/rescue safety). On-phone confirmation remains the user's retest: Snap Divergent Equation hand-held at arm's length and up close.

Pass 10 (2026-11-08) — price refresh rebuild, three exports, condition+language, backup/restore, stress test

Trigger: user confirmed scanner perfect, then reported "Price refresh failed: action refreshCollectionPrices failed: 500 {"error":"The operation timed out."}" on a FOUR-card collection, and asked for Moxfield/Archidekt/ManaBox exports, a stress test, and ManaBox-parity adds.

- **Refresh root cause & fix**: old fetchFreshPrices fetched DISTINCT printings in chunks of 50 SEQUENTIALLY with a 30s timeout per request, and refreshCollectionPrices ran one history SELECT PER ITEM inside the update loop. Rebuilt: identifiers deduped by Scryfall id, chunks of 75 (Scryfall POST /cards/collection limit) run in PARALLEL with an ~8s per-request timeout (one 429 retry after 600ms), partial success per chunk (result carries requested/chunks/failedChunks/ms). Writes are batched via db.batch() in groups of 150 statements — Promise.all of ~600 individual writes took 10.5s on 300 items; batched writes + parallel fetch brought the same refresh to **805ms**. Measured (sandbox): his exact 4-card Main Collection refresh = 4/4 updated, 0 failed chunks, **442–498ms** (was a 500 timeout); synthetic 300-item collection (20 distinct printings, 1 chunk) = 300/300 updated in 805ms. Semantics unchanged: foil-aware unit prices, price_at_add baselines untouched, one history snapshot per item per day, movement chips recompute from lastPrice. Failures now surface in plain language via `friendlyError` — no raw `action … failed: 500` strings anywhere in the UI (applied across load/search/deck/import/refresh/scan-list error paths).
- **Exports** (client/src/lib.ts, verified against public docs): Moxfield .txt `qty Name (SET) number *F* *CMDR*` (docs: https://gist.github.com/Jerakin/24be913c6106546136c45d1d028f9af9 ; https://developers.tcg.land/en/multisearch ; https://podqueue.gg/help/importing-a-decklist); Archidekt CSV 13 columns `Quantity,Name,Finish,Condition,Date Added,Language,Purchase Price,Tags,Edition Name,Edition Code,Multiverse Id,Scryfall ID,Collector Number` with Finish Normal/Foil/Etched, TCGPlayer conditions NM/LP/MP/HP/D, lowercase edition code, 2-

letter language, commander via `Commander` tag (docs: https://github.com/stepkie/mtgcsvhelper/blob/HEAD/MtgCsvHelper/Resources/SampleCsvs/Tests/SITE_BEHAVIOR.md); ManaBox CSV native 18-column header `Binder Name,Binder Type,Name,Set code,Set name,Collector number,Foil,Rarity,Quantity,ManaBox ID,Scryfall ID,Purchase price,Misprint,Altered,Condition,Language,Purchase price currency,Added` (docs: https://manabox.app/guides/collection/import-export/ ; https://github.com/tahuffman1s/deckpuller/blob/HEAD/docs/superpowers/specs/2026-06-11-manabox-collection-and-buy-integration-design.md). Offered per collection, per deck (commander marked) and for the scan list via one shared ExportButtons component. Deck cards gained set_code/set_name/collector_number/condition/language (migration 0006) so deck exports carry real printings; pre-existing deck cards export name/quantity only.

- **Condition + language** on collection items, scan list items and deck cards (migration 0006: language columns + deck card printing fields). Condition: Near Mint / Lightly Played / Moderately Played / Heavily Played / Damaged (default Near Mint). Language: 11 display names, default English; export mappings — Archidekt 2-letter codes, ManaBox locale codes (en/es/fr/de/it/pt/ja/ko/ru/zh_CN/zh_TW), ManaBox conditions near_mint/light_played/played (MP+HP)/poor (Damaged). Editable via selects on the collection card; shown in the meta line. NO price-math change (market/NM reference, honest metadata only). Collection merge key now includes condition+language.
- **Backup/Restore**: exportBackup action returns ONE JSON {version, exportedAt, collections[{name, items[]}], decks[{name, commanderName, cards[]}]} incl. baselines and lastPrice; restoreBackup inserts them as NEW rows — a name that already exists becomes "Name (restored)" (then "(restored 2)"...), decks always get fresh copies, deck-card collection links re-resolved by Scryfall id, never overwrites. UI copy states this behaviour. Round-trip verified on sandbox: fixture collection (foil commander + Japanese LP + German Damaged DFC) restored twice → original + " (restored)" copy with identical 3 items / $49.40 total, conditions/languages/baselines intact.
- **Stress numbers** (all sandbox): 12/12 test suites green — the 11 pre-existing (318 checks, scanner recognition untouched) + new tests/exports.test.ts 42/42 (round-trip fixture with foil/comma-name/DFC/commander, malformed CSV → 3 line-numbered errors + good rows land, 500-row exports byte-checked: 500/501/501 lines generated in 3.9ms, ManaBox round-trip parse of 500 rows in 4.5ms). 300-item import: 3×100 rows, 0 lost; re-import of 5 identical rows → merged 5, imported 0. Concurrency: two scan-list adds + a collection refresh fired in parallel → both adds landed, refresh 4/4, no lost rows. 100-card Commander deck (8 entries, quantities summing to 100) restored and listed: cardCount 100, totalValue $61.70 matches hand-computed sum. NOT directly exercised: stalled-chunk partial success (fetchFreshPrices is server-internal; partial success is by construction — per-chunk try/catch + failedChunks reporting — plus code review) and >75-distinct-printing multi-chunk refresh; on-phone retest still the user's.

Pass 11 (2026-11-08) — photo orientation + calibration leak, weak-title recovery read, Did-you-mean breadth

Trigger: two screenshots, no text. (1) Take Photo of Laelia, the Blade Reforged failed "Read as: Z" — diagnostics "Capture: 3000×4000 photo · capture 0ms · detect 0ms · title 2.9s · bottom skipped (near-blank)", the title-band thumbnail showed desk + a finger (no card), while the panel's photo preview showed the card upright, framed, name plainly visible. (2) Rapid Did-you-mean on Raff, Security Officer: band thumbnail clearly showed the full name, pooled read was "I Ratt Security" ('Officer' dropped, Raff→Ratt), suggestions were only Security Blockade 62% / Security Detail 62% — the correct card not pickable.

- **Failure 1 — code findings (two real defects; the EXIF hypothesis was half the story)**: (a) `prepareCardCanvas` applied the stored scanner-box CALIBRATION rect to ANY frame, including camera-app photo files, whenever box mode persisted on (localStorage `manabox-box-mode`) — that branch hardcodes `detectMs: 0`, exactly matching his "detect 0ms", and a rig rect (fractions of the LIVE guide crop) cropped into a hand-held photo lands the title band off-card. Calibration is only meaningful for frames mapped through `drawGuideFromSource`; `handlePhotoFile` now passes `ignoreCalibration: true` and photos always run real detection. (b) The old photo decode (`new Image()` + `naturalWidth/naturalHeight`) never parsed EXIF: it trusted the platform to bake orientation into the pixels it drew — not guaranteed in in-app webviews, where the preview can honour the tag while canvas pixels/dims stay raw (his label printed the raw sensor size). New `client/src/photo.ts`: pure `parseExifOrientation` (JPEG APP1/TIFF, both endians — the test caught a big-endian u32 bug in the first cut), `parseJpegSize` (SOF), `orientedSize`, `orientationMatrix` (canvas affine, pixel-centre consistent), `classifyDecodedDims`, `photoCaptureLabel`. `decodePhotoFile` decodes via `createImageBitmap(file, {imageOrientation:'from-image'})`, VERIFIES the bitmap's dims against the EXIF-expected oriented size, and applies the rotation MANUALLY on canvas when the platform returned raw pixels; `<img>` fallback with the same verification. Downstream (downscale, detection, bands, label) uses oriented dims only; label reads e.g. "3000×4000 photo (oriented)".
- **Failure 2a — full-width recovery read** (`client/src/ocr.ts` `TITLE_WIDE_BAND` x 0.01–0.97, `TITLE_RIGHT_BAND` x 0.40–0.97, same y as standard): the standard band's 83% right cap (mana-corner exclusion) is where a name's last word dies. In `identifyBurst`'s pool-exhausted judgement, when the pool is plausible (top ≥ 0.40) but not decisive by the mode's bar (Rapid auto / Snap 0.82) — exactly the Did-you-mean trigger — the clock widens to the rescue envelope and the two recovery bands are read on the first two frames and pooled; a decisive pool then settles via the normal title-win path ("title (recovered — full-width read)"). Both modes (his failure was Rapid); confident paths never reach the code. Mana glyphs enter the wide image but the whitelist/token matcher filter them in post.
- **Failure 2b — Did-you-mean breadth** (`client/src/lib.ts` `broadSuggestions`, `strongReadTokens`): the uncertain outcome's pick list is now built from the pooled ranking PLUS (i) CONTAINMENT — a name evidencing EVERY strong token (≥4-letter read tokens, fuzzy ≥0.5/token) is always included and sort-floored at the top score so it outranks same-score partial rivals (displayed % stays the honest aggregate), and (ii) PER-TOKEN BEST — top-2 names per strong token (similarity ≥0.6, aggregate ≥0.35). Up to 5, deduped. Decision logic

(Rapid auto/pause, Snap bar) still consumes only the plain pooled ranking — breadth widens only what the user can pick.

- **Tests**: NEW `tests/photo.test.ts` (30 checks): synthetic EXIF JPEGs at orientations 1/3/6/8 + big-endian TIFF parse (stored 4000×3000), label/classify/orientedSize units; one synthetic card photo (desk + finger blob + textured card with title strokes spanning 82% width and a left-55% bottom line) stored at orientations 1/3/6/8 → correction recovers the framed grid EXACTLY at all four, detection conf 0.97, title band texture 0.51 with title strokes in its right portion; bug repros without correction — 6/8: detection null on the sideways card, whole-frame fallback strip texture 0.00 (desk/finger, his thumbnail); 3: band lands on the bottom line (strokes one-sided: left portion dark 0.00 vs right 0.29). NEW `tests/recovery.test.ts` (16 checks) with his exact evidence: pooled-only ranking reproduces the miss (Security Detail 0.62 top, Raff 0.52 — matching his 62% sheet); broadSuggestions from "I Ratt Security" alone ranks Raff, Security Officer FIRST with both rivals still listed; pooling a recovery read of just "Security Officer" → Raff top 0.75, `decideRapidMatch` = "auto"; full recovery read → 1.00; band geometry guards (standard cap 0.83 unchanged, wide/right reach 0.97); regressions — 'ressure kecrer' → Treasure Keeper still tops broad list, 'LY' → zero suggestions, clean read unaffected. All 14 suites green (318 prior + 46 new = 364 checks); client + server typecheck clean. Rapid box happy path, Snap deadline settlement and the number-rescue order are behaviourally untouched (recovery/breadth live strictly below the decisive bars). On-phone confirmation remains the user's retest: Take Photo of Laelia (hand-held, box mode left ON — the leak case) and a Rapid scan of Raff, Security Officer.

Pass 12 (2026-11-08) — empty-band texture gate, name-zone parity, garbage-pool hygiene

Trigger: one screenshot (Rapid, live 1080×1920 video; card hidden behind the sheet, box-mode state unknown) + the user's words "make sure where it says CARD NAME HERE it scans that area completely; the scan isn't working as well as it did previously". Did-you-mean sheet: raw read a 30-token alphabet soup ('Gg a ha a 7 repay g x gr 2 Br ATER fos 2 pon TF 6 r. 4 I 5 et we HRT 2 cao pie os ww TRI a a TPN a'), TITLE BAND THUMBNAIL A UNIFORM BLANK MID-GREY STRIP, bottom strip blown-white with a black slab, suggestions 'Dream Eater 76%' / 'A Chaotic Night in Vegas 45%', stages capture 284ms · DETECT 0ms · title 734ms · bottom 1.2s · MATCH 5.1s.

- **Diagnosis of detect 0ms on live video**: traced every detectMs=0 path — the ONLY producer on live Rapid frames is `prepareCardCanvas`'s box-calibration branch (hardcoded `detectMs: 0`), which fires whenever Box mode is on AND a calibration is stored — both persist in localStorage across sessions (`manabox-box-mode`, `manabox-box-calibration`). The Pass 11 fix stopped that branch leaking into photo FILES; on LIVE frames it is by-design, but with the card hand-held / not in the tray spot the stored rig rect crops a blank area and every band reads emptiness (the guide fallback measures real detectMs and cannot print 0ms; the burst stills path uses prepareCardCanvas too). Failure shape: bands cut from where a card was EXPECTED, not found in THIS frame.

- **Empty-band gate (the core fix)** — `analyzeTitleBand` (client/src/scanflow.ts, rule F) + `analyzeTitleZoneCanvas` (client/src/ocr.ts): before any OCR of a frame is trusted, the frame's NAME ZONE (the shared NAME_ZONE rect) is measured as OCR would see it — ink-row runs (adaptive deviation > max(24, 0.9σ), so low-contrast foil titles still register while broadband noise raises its own bar), gradient texture fraction, σ. Uniform/blank/slab zones classify the frame EMPTY: it contributes no title read, no bottom read, no pooled evidence. An attempt with every frame empty settles as the new outcome kind "nocard" BEFORE any matching: Rapid drops it silently (new RapidMachine outcome "nocard" — scene marked handled, attempt budget untouched, no sheet/hint/panel; a scene change re-arms), Snap shows "No card found in the guide — reposition..." guidance with no slow-read offer. Rescue and recovery reads consult the same per-frame zone map (gated on the primary band's texture, per spec). The bottom strip's pre-check gained `columnCoverage` (a printed line spans the strip; the screenshot's black slab covers 0.30 < 0.40) so the slab band is skipped pre-parse instead of spinning OCR to its 1.2s cap. Diagnostics now label the rect source — capture line reads "1080×1920 video · rect: detected / box calibration / guide fallback" — so future screenshots are self-explaining.
- **Name-zone parity** — ONE shared definition: `NAME_ZONE` (client/src/geometry.ts: x 0–0.97, y 0.01–0.19 of the card crop). The viewfinder zone is now positioned from these exact fractions (was mt/mx margins + a separate height constant), and the pooled title bands cover the rect completely: the NAME_ZONE_BAND itself joined TITLE_RESCUE_BANDS as the final rescue read (multi-line), so the union of BAND_VARIANTS + TITLE_RESCUE_BANDS + TITLE_RECOVERY_BANDS spans the zone with no gaps (tall band y0 0.02→0.01 to chain the top). tests/geometry.test.ts asserts fraction coverage + corners + the object-cover mapping at 1080×1920 in two viewfinders (band-union source rect contains the visible zone source rect to <2px).
- **Garbage-pool hygiene + match speed** (client/src/lib.ts): root cause of 'Dream Eater 76%' — the fragment-union pass let SHORT DEBRIS tokens beat the documented "never cover a token" cap via raw Levenshtein ('fos'→'frost' 0.6) and let unrelated ≥4-letter debris "cover" tokens through the clipped-suffix rule ('repay' vs 'dream' = 0.582 via suffix 'ream', dice 0.25). Fixes: `evidenceTokens()` — pooled token evidence is plausible name tokens only (letters, ≥3 chars; digit-mixed/single-letter debris dropped; 'ressure kecrer'/'ratt' unaffected); union-pass coverage now requires the winning pair to share bigram substance with the FULL entry token (dice ≥ 0.3 — genuine fragments: tquation→equation 0.86, iverson→divergent 0.43, ratt→raff 0.33); union score scaled by POOL PRECISION (fraction of evidence tokens the entry accounts for ≥0.45 — genuine fragment pools ≈ 1.0, the soup pool 2/9); ≤3-char pairs capped at 0.5 in union passes (the documented debris rule, now enforced); `ocrCandidates` drops candidates with no ≥3-letter token ('Gg a ha a' variants never reach the matcher); `strongReadTokens` requires plausible tokens. Speed: tokenSimilarity memoised per top-level call + `pooledEntryScore` — a provable upper bound ((dice·0.65+0.4)·0.45 + tok·0.55) that returns the exact nameScoreV2 whenever it can reach 0.35 and skips the full Levenshtein otherwise (results identical at every threshold the matcher uses). MEASURED on this VM at a 30,192-entry index with the screenshot's exact string — BEFORE: pooled 173ms, broadSuggestions

1,807ms → 'Dream Eater 76%'; per-frame accumulating pools 2,441ms (≈ the phone's 5.1s match stage). AFTER: pooled 2ms (no results), broad 162–291ms (top junk 0.50 < the 0.55 Did-you-mean bar), 3 accumulating garbage pools 266ms total, full pooled+broad ≈ 165ms. Good reads at scale: 'ressure kecrer' → Treasure Keeper 0.73 decisive (246ms), 'I Ratt Security' → Raff first in broad list (283ms).

- **Tests**: NEW tests/emptyframe.test.ts (33 checks): gate units (uniform grey/noise/speckle/slab EMPTY; dark-on-light + light-on-dark TEXT; zone PARTIALLY empty still TEXT), bottom slab skip via columnCoverage + a real line still passing, whole frames (blank tray / desk-only / hand-over-lens → no detection AND empty zone; half-out card → detected, zone has TEXT), garbage hygiene + <1.5s + no ≥0.55 suggestions + decideRapidMatch "none" at a 30k index, good-garbage regressions at scale, RapidMachine nocard semantics. Two existing premises updated honestly: recovery.test.ts fixture 1 (pooled alone now settles NOTHING on the Raff read instead of crowning Security Detail — the debris inflation it relied on is gone; asserted in current form: decideRapidMatch "none" + Raff in top 2; broad still ranks Raff first; recovery-pool → decisive still passes) and rescue.test.ts rescue-set order (name-zone appended last). geometry.test.ts +9 parity checks. ALL 15 SUITES GREEN: 447 checks (405 prior + 42 new), client + server typecheck clean. Untouched by design: box Rapid happy path, deadline settlement, number rescue order, recovery reads on real weak titles. On-phone confirmation remains the user's retest (Rapid with Box mode left ON and the card hand-held — the failing configuration — should now silently keep watching instead of producing a sheet).

Pass 13 (2026-11-08) — effective-rect rule for stale box calibrations, edge-is-not-text gate hardening, one rect on screen, recalibration prompts

Trigger: two screenshots of the SAME box-mode Snap attempt on Stonehewer Giant (C 0114, artist Lixin Yin). (A) Failure panel: "No plausible name could be read" — diagnostics "Evidence: no readable text · Capture: 1080×1920 video · rect: box calibration · capture 266ms · detect 0ms · title 3.3s · bottom 572ms · match 1ms"; title band thumbnail = pale grey gradient + ONE THICK BLACK HORIZONTAL BAR along its bottom (the card's dark top frame edge, NO title); bottom strip = white with black diagonal slabs; scan list unchanged (11 cards / $2.09). (B) Live view: card IN the box — DETECTED outline (green) hugging the actual card exactly, its top ~13–18% of guide height BELOW the guide top; the GOLD guide much taller; the CARD NAME HERE zone drawn across the GUIDE's top, i.e. above the card's actual title row; hint "✓ Card found — fine at this distance, tap Snap".

- **Diagnosis (verified in code)**: `prepareCardCanvas` returned the stored box calibration unconditionally in box mode (hardcoded detect 0ms) — live detection, which already ran for the hint/outline in the same session, was never consulted for the read. The card had settled LOWER in the tray than when calibration was recorded, so every band was cut too high: title band caught the black top frame edge, bottom band landed mid-card. The Pass 12 gate passed the bar band (ink rows ✓, boundary texture ✓, high σ ✓ — a bar is "textured"), OCR

burned title 3.3s on it, and the generic panel never hinted calibration was the problem. Three contradictory geometries were on screen at once: zone on the guide (wrong), outline on the card (right), reads from an invisible third rect (calibration).

- **Fix 1 — effective-rect rule** (`checkCalibration` + `effectiveReadRect` , client/src/detect.ts; `prepareCardCanvas` now always runs detection on live frames): tolerances — top-edge disagreement > 4% of frame height, any edge > 6%, or relative height difference > 6% mark the calibration stale (live-detection jitter on an unmoved card is ~1–2%). Agree → read the CALIBRATION exactly as before (Rapid box happy path behaviourally identical; detectMs now honestly measured — the hardcoded "detect 0ms" is gone). Disagree → read the DETECTED rect, source "detected-stale", diagnostics label "detected (calibration off — recalibrate?)". No detection → calibration fallback unchanged; photos still `ignoreCalibration` .
- **Fix 2 — one rect on screen**: in box mode the outline AND the CARD NAME HERE zone are drawn from the effective rect (geometry.ts `nameZoneOfRect`) — the rect the next capture will read; a stale calibration also renders as a dashed white ghost labelled "saved box position", the effective outline labelled "READING HERE — saved box position is off". Non-box rendering unchanged (zone on guide, detected outline). tests/boxrect.test.ts asserts drawn-zone == scanned-zone through the 1080×1920 object-cover mapping (±2px, two viewports, both cases) and that in the stale geometry the effective zone covers the card's true title row (guide-fraction mid 0.213) while the old guide-fixed zone (bottom 0.19) provably missed it — screenshot B's exact defect.
- **Fix 3 — edge is not text** (`analyzeTitleBand` in client/src/scanflow.ts gains text-row STRUCTURE): a row counts toward hasText only when STRUCTURED — ≥2 separate ink runs of ≥6px (glyphs are broken by inter-letter gaps; a frame edge is one unbroken run) and ink coverage < 90% of the row. New fields: `structuredInkRows` , `solidInkShare` , `edgeOnly` (ink present, solidInkShare ≥ 0.6, <2 structured rows). The screenshot's bar band: Pass 12 criteria PASS it (fixture reproduces: ink=12 rows, tex 0.083, σ=87) → now edgeOnly, classified NO TEXT pre-OCR, zero OCR calls. Attempts carry `misaimed` (zone edgeOnly on any frame, or the standard band of the read crop edgeOnly) on the nocard/none outcomes. Snap + mis-aimed + box calibration → specific copy ("The reading spot looks off — the saved box position doesn't match where the card is sitting...") + a one-tap "🎯 Recalibrate box" button IN the panel (calibrateBox on the live frame); hand-held mis-aim gets edge-specific repositioning copy; Rapid stays silent as before. Non-regression fixtures: bar + real glyphs, low-contrast glyphs, both polarities, speckle/slab/uniform all classify as before (the ≥2-runs rule also fixed the vertical slab's row semantics — it now has 0 structured rows, not 60).
- **Fix 4 — recalibration prompt**: consecutive box-mode mis-aimed settles (Snap or Rapid, nocard or no-read) are counted in App (`noteMisaimedAttempt`); at 2, a banner offers Recalibrate + Dismiss, latched once per calibration (`noteSettledAttempt` resets the streak on any other settle; a successful calibrateBox resets streak + latch + offers). Rapid's flow is otherwise untouched — the streak feeds only the banner.
- **Tests**: NEW tests/boxrect.test.ts (47 checks): (1) check/effective-rect units incl. jitter-agrees and height/edge staleness; (2) gate units — synthetic black-bar and white-bar bands provably pass the Pass 12 criteria yet classify EDGE-ONLY, bar+glyphs and low-contrast glyphs keep

TEXT; (3) real-image box fixture — actual Scryfall card images of Stonehewer Giant converted to luminance PGMs in tests/fixtures/ (LTC borderless printing, the user's extended-art style: https://cards.scryfall.io/normal/front/a/3/a3a52265-f606-4c17-b85f-d8d051566c1f.jpg?1783914187 ; standard 2XM frame: https://cards.scryfall.io/normal/front/0/e/0efda05d-01ea-4478-b075-00a31d7037f8.jpg?1783930208 — API record https://api.scryfall.com/cards/named?exact=Stonehewer%20Giant), pasted LOW (top at 15% of guide height) in a synthetic tray: detection conf 0.92, detected top within ~1% of truth, −15% calibration → stale → effective = detected → its zone (struct rows 48–55) and standard band carry the real title, calibration-only zone = pure tray (σ 1.3, the pre-fix input); −5% calibration on the 2XM frame straddles the top border and reproduces the black-bar band (Pass-12-pass → edgeOnly); calibration at the true position → calibration read (happy path); (4) screen parity per Fix 2. ALL 16 SUITES GREEN: 493 checks (446 prior by line count + 47 new), 0 failures; client typecheck clean. Stage-timing effect for the mis-aimed case: previously title 3.3s of OCR on the bar band; a fully gated attempt now settles nocard with title/bottom stages at 0ms (no OCR is ever started — the gate is synchronous and pre-OCR by construction), and with Fix 1 the user's exact geometry no longer reaches the gate at all: the detected rect is read and the card identifies. On-phone confirmation remains the user's retest (same in-box Snap of the low-seated card).

Pass 14 (2026-11-08) — band hunting, per-band gate, drawn/read rect diagnostics

Trigger: retest of Pass 13 on the SAME physical card (bottom line "C 0114 MSH • EN LIXIN YIN" — Stolen Stark Tech, MSH #114; the earlier "Stonehewer" reading of the art was wrong). The Pass 13 code WAS live on his phone: diagnostics "rect: detected (calibration off — recalibrate?)", detect 15ms, viewfinder drew the detected outline + CARD NAME HERE zone ON the card with the name visibly inside it, dashed "saved box position" ghost + banner shown. YET the result was identical: title band thumbnail pale grey + THICK BLACK BAR along its BOTTOM edge, "Evidence: no readable text", title stage STILL 3.3s, bottom "- EL I EL BE", scan list unchanged (11 / $2.09). The drawn rect was right and the captured band was wrong, while the sandbox geometry suite asserts drawn==scanned parity (±2px) — the divergence lives in a step the sandbox does not model.

- **Code-inspection verdicts on the four candidate causes**: (d) CONFIRMED, twice over — (i) the Pass 12/13 gate judged the frame's NAME ZONE (y 1–19%) and a mis-aim flag (`firstBandEdge`), but the standard band actually OCR'd (y 2.5–11%) was never itself judged: the zone passes on structured rows in its lower half while the standard band is pure frame edge; (ii) the bar's RAGGED edge manufactures "structured rows" — measured in tests/bandhunt.test.ts §1: a faithful ragged-bar band (grey gradient + thick bottom bar, wavy edge ± debris) has hasText = TRUE under the Pass 13 criteria (structuredInkRows 4–8), so even a per-band Pass 13 check would have passed it and OCR burned the 3.3s. (a) STRUCTURAL — the drawn rect (800ms preview-detection interval, steady state) and the capture rect (per-burst-frame detection at the tap instant) are different detections from

different frames; neither's coordinates were ever printed, so no screenshot could localise the divergence; `capturePreparedBurst` also overwrites the overlay with the capture head rect, after which the interval overwrites it back — post-hoc screenshots are inherently ambiguous. (b) NOT PRESENT as cross-frame rect reuse — every prep detects on its own frame and crops from that same canvas (verified in code); the burst-shift fixture (§4 of the suite) nevertheless proves the fix recovers when a rect IS applied to a shifted frame. (c) The bar-at-bottom GEOMETRY is real whatever its provenance (tap jolt on the rig, burst-frame detection variance + correctCardAspect's 70%-upward growth on a blurred slab, or capture-space offset) — so the shipped fix is the self-correction, not a root-cause guess.

- **Fix 1 — BAND HUNTING** (scanflow.ts rule G: `bandHuntDecision`, `huntBandWindow`; ocr.ts `huntTitleBandCanvas`): `analyzeTitleBand` now reports ink DISTRIBUTION (inkTopShare/inkBottomShare on row thirds; structuredUpper/LowerRows on row HALVES — a wavy edge manufactures structured rows in the band's bottom ~45%, so a two-thirds split let them veto the signature). BAR-AT-BOTTOM = ink ≥60% in the bottom third AND zero structured rows in the upper half → the title is BELOW the band. Before any OCR, each zone-passing frame's primary band is hunted on its own pixels: re-cut one step (45% of band height) lower in the SAME captured frame, ≤3 down / 1 up, first clean band wins. Refinements, all measured: a directed hunt continues through low-ink GAP bands (border+banner-edge zones measure near-empty — MSH 10% needs step 2 through a gap); with `probeDownOnReject` (the pipeline only hunts zone-passing frames, so a step-0 reject can only be the zone's low-ink stretch) even a reject probes down (MSH 6%); a SOFT ACCEPT (glyphs clipped at the band's bottom edge) probes one step further and keeps whichever band carries more glyph rows (2XM 6%: struct 2 → 18); a bar-signed band WITH glyph rows is remembered as the hunt's FALLBACK (clipped-fragment evidence, Pass 9-proven) when nothing cleaner exists (LTC 8%). The hunted band replaces the standard band as the frame's primary read and as the published thumbnail (the band actually OCR'd is the band displayed).
- **Fix 2 — GATE EVERY BAND** (scanflow `bandMayBeRead`; enforced inside `ocrTitleBandCapped` for the burst/rescue/recovery pools and in `identifyCore`'s variant loop): no-text → never OCR'd; bar signature on a single-line band → never OCR'd; multi-line bands (tall/name-zone) are bar-exempt (a complete title in their bottom third is their purpose). The frame loop passes `vetted` for its hunt-chosen band (the hunt already gated those exact pixels, including a deliberate fallback). An attempt whose zone-passing frames are ALL bar-exhausted settles as mis-aimed NO CARD pre-OCR — Rapid silent, Snap gets the recalibration copy — with title stage ~0ms instead of 3.3s.
- **Fix 3 — DIAGNOSTICS THAT END GUESSING**: the capture line now prints `read x0.190 y0.163 w0.620 h0.617 · drawn x0.190 y0.170 w0.620 h0.610 · bands frame 1/3 video` (+ `· band hunted +2` when the hunt moved) — the read rect is preps[0].rect (the same object the crop was cut from), the drawn rect is snapshotted from the viewfinder render BEFORE the burst overwrites the overlay (`drawnRectRef` / `attemptDrawnRectRef`), and `PreparedScan.frameSource` records video/still/photo per frame (captureOneFrame now reports its still flag). `formatRectFractions` lives in geometry.ts. Manual Snap also gets a

150ms tap settle before frame 1 (tap-jolt damping; Rapid untouched — its watcher fires on an already-stable scene).

- **Harness (NEW tests/bandhunt.test.ts, 69 checks)**: §1 ragged-bar gate fixture (above); §2 hunt mechanics with scripted providers (budgets, gap continuation, probe-down, soft-accept, fallback, oscillation stop); §3 real-image divergence fixtures — the user's ACTUAL card fetched live from Scryfall (Stolen Stark Tech, MSH #114: https://api.scryfall.com/cards/msh/114 , image https://cards.scryfall.io/normal/front/d/b/db4557ad-4141-4dae-870e-a587506f4914.jpg?1783902937 , converted to tests/fixtures/stolen-stark-tech.pgm 300×418) plus the two Stonehewer fixtures — with (i) constant-offset injection (rect shifted UP by k% of card height, drawn rect = truth): OLD flow trap reproduced at 6% and 10% on the standard-frame cards (zone passes, OCR'd band bar/textless/fragment-only); NEW hunt recovers a band CONTAINING the image-measured title centroid (glyph-cluster centroid, card y ≈ 0.069–0.077) at both mandated offsets for 2XM and MSH. Sweep (text-bearing band returned): 2XM recovers 4–18% (18% thin, struct 2), gives up at 20%; MSH recovers 4–18%, gives up at 20%; LTC returns text-bearing bands through 20%. (ii) burst shift (rect from frame A on frame B, card +5% of frame height): bar-signed, hunt recovers; per-frame detection on B accepts at step 0. §5 diagnostics formatting (a 0.7%-of-frame divergence is visible in the printed pair). HONEST LIMITS: (1) LTC borderless is the art-top exception — shifted bands measure structured rows from the bright art sky and the hunt accepts early WITHOUT the title (winner provably misses the centroid at 6%/10%); the hunt is no worse than the old flow there (same band was OCR'd), and identification rests on the rescue name-zone band, asserted to contain the centroid and pass its gate at both offsets. (2) Hunt cost: the whole 30-hunt real-image battery runs 326ms in node (≈11ms/hunt at grid scale; on-device bands are ≤900px analyses, a few ms each) — a mis-aimed attempt's title stage goes from 3.3s of OCR to ~0ms (synchronous analysis only, zero OCR calls by construction). The geometry parity suite's assumptions did NOT change (same-rect cover mapping — boxrect/geometry suites unmodified and green); what changed is that parity is no longer ASSUMED for the capture — both rects are printed and the band is verified on the capture frame itself. ALL 17 SUITES GREEN: 562 checks (493 prior + 69 new); client + server typecheck clean. Rapid box happy path behaviourally unchanged when bands are clean (control fixtures win at step 0 with the standard band). On-phone confirmation remains the user's retest (same in-box Snap, card seated low).

Forensic pass (2026-11-08, post-Pass-14 retest failure) — real rig frames, NO behaviour change shipped

Trigger: user reports the same box-mode Snap failure on the Pass 14 build and notes the scanner "was working like 5 messages ago when I said it was perfect" (the v12/v13 scanner). Dispatch rule for this pass: forensic test on his ACTUAL rig frames + v13→v17 behavioural diff; ship a correction ONLY if the failure reproduces on the real frames and localises to one defect. It did not, so nothing was changed.

- **Inputs**: his live-view screenshots `1000044561_31_dcq3.jpg` (Pass 14 build) and `1000044557_29_fiv4.jpg` (Pass 13 build), both 945×2048, same card (Stolen Stark Tech, MSH #114 — verified live: GET https://api.scryfall.com/cards/msh/114 → "Stolen Stark Tech"). Guide crop measured from the gold borders: screenshot rect (170,376)–(774,1226) = **604×850 px** (≈1.55× smaller linearly than the on-device guide crop from a 1080×1920 stream, and JPEG-compressed — resolution caveat applies to every OCR number below). Drawn rects measured from the overlay strokes in pixels; saved-box ghost measured from the dashed rect in the 4561 shot (left x=207, right ≈660, bottom y=1066 → guide fractions x 0.061–0.811, y 0.067–0.812, top aspect-inferred ±0.03).
- **Overlay interference (measured, decisive for interpretation)**: detection run on the raw screenshot crops latches onto the burned-in guide/overlay slab (f4561 raw: confidence 0.79, corrected rect = the WHOLE guide 0,0–1,1) — the overlays massively distort detection inputs, as suspected. Raw-pixel OCR is likewise unusable: the app's own "CARD NAME HERE" label is the loudest text in the zone and every band reads it verbatim. The faithful runs therefore use (a) the device's OWN drawn rects as the detection result, and (b) a crudely inpainted variant (overlay strokes + label masked, row-interpolated) for OCR. Inpainting itself damages glyph strokes under the label — the second confound, stated plainly.
- **f4561 (Pass 14 frame)**: drawn effective rect (blue) = guide x 0.132–0.816, y 0.053–0.736; true card ≈ x 0.146–0.803, y 0.058–0.728 (outline hugs the card ±8 screenshot px). Title glyph centroid measured from the image at guide y 0.116 (screenshot y ≈475), i.e. crop fraction 0.089 of the drawn crop — INSIDE the standard band (crop 0.025–0.11; guide span 0.070–0.128). Zone gate: TEXT (59 structured rows). Standard band: hasText TRUE but **barAtBottom TRUE** (ink 76% in bottom third, 0 structured rows in upper half — the title sits in the band's lower portion on a tight crop), so `bandMayBeRead` = false and the hunt steps +1 to band guide 0.096–0.154, which contains the centroid (structured rows 22, accept). Calibration check vs the drawn rect: topΔ 0.014 (agrees), maxEdgeΔ 0.075, heightΔ 0.082 → STALE → effective = detected — matching the on-screen "calibration off" label. **Calibration-only crop (the v13–v16 read)**: standard band CLEAN (42 structured rows, no bar signature, hunt step 0) and pooled identification **Stolen Stark Tech 0.95 (confident)** on masked OCR (tall-band line "Stolen Stark ech"). Effective/drawn crop pooled+rescue: **Stolen Stark Tech top at 0.62** (Did-you-mean range, not auto) — correct card first. True-card crop: **1.00**. Bottom line at screenshot scale: contrast read "MSH CEN SN L Y" (set code visible, no digits) → NO number candidates in any scenario — the number path is resolution-starved in these pixels, not logic-blocked.
- **f4557 (Pass 13 frame)**: drawn detection (green) = guide x 0.060–0.834, y 0.036–0.813 — visibly LARGER than the true card (≈ x 0.132–0.803, y 0.046–0.720) on every side. Title centroid guide y 0.103 (crop 0.085 of the drawn crop, inside its standard band guide 0.056–0.122). Zone TEXT; standard band again barAtBottom (inkBot 0.62) → hunt fallback keeps step 0. Calibration check vs drawn: topΔ 0.031, heightΔ 0.043 → AGREES → v17 would read the calibration (same rect v16 read). Masked OCR identifies NOTHING in any scenario (best pooled 0.56 junk): this frame is visibly softer and its title sits under the label/inpaint damage — below the OCR floor at screenshot scale. Inconclusive for pipeline logic.

- **Verdict**: the on-device failure shape (uniform grey band + thick black bar, "no readable text", title 3.3s) appears NOWHERE in these frames under the current code — every scenario's zone has text and every standard band carries structured glyph rows; on f4561 the current pipeline puts the right card first (0.62 via effective rect, 0.95 via the calibration rect, 1.00 via the true rect). The failing CAPTURE therefore differs from the live-view frames the screenshots preserve — the divergence lives in the burst frames themselves (per-burst-frame detection variance / tap-jolt shift at the capture instant), which a still frame cannot model. Per the dispatch rule, NO correction was shipped and no threshold was tuned.
- **Secondary measured facts**: (1) barAtBottom fires on CORRECTLY aimed tight crops in both frames (title in the band's lower third) — the hunt absorbs it (+1 step, centroid contained), but it means a clean box scan now always pays the hunt detour and its primary band is never read as-is; (2) on f4561 the stale-calibration verdict is driven by HEIGHT/edge disagreement (8.2%/7.5%), not the top edge (1.4%) — the card+rig geometry genuinely changed size between calibration and capture, not just position.
- **v13→v17 behavioural diff, box-mode Snap path** (v13 scanner = Pass 9 code; each item: can it alter the properly-seated, calibration-matching happy path?): Pass 11 — photos ignore calibration (NO: live path untouched); weak-title recovery reads + broadSuggestions (ONLY below the decisive bar: a clean read settles before them; a messy-but-decisive read is untouched; a messy NON-decisive read gets extra reads it didn't get in v13 — intended, happy-path-neutral for clean cards). Pass 12 — name-zone texture gate (YES, in principle: a frame whose zone measures empty now contributes nothing and an all-empty attempt settles "nocard" pre-match — a clean seated card's zone has text by construction, but a dim/soft frame that v13 would have OCR'd can now be gated out); NAME_ZONE_BAND added as final rescue read (only in the rescue phase, post-failure); garbage-pool hygiene in the matcher (evidence-token filtering, dice-substance rule, pool-precision scaling — a PERFECT read still scores 1.0, but borderline fragment reads score differently than in v13, in both directions by design); bottom columnCoverage skip (a real line spans the strip; a marginal line under glare could now skip — bottom is opportunistic, title path unaffected). Pass 13 — effective-rect rule (NO when calibration and detection agree within 4% top / 6% edge / 6% height: the calibration is read exactly as in v13; YES when live detection jitters past tolerance on an unmoved card — the read rect flips to the detected rect for that attempt, a nondeterminism v13 didn't have); edgeOnly/misaimed outcome kinds (a bar-band attempt now settles "nocard/misaimed" with recalibration copy instead of v13's generic failure — outcome label/copy only when the band is a pure edge; bar+text still reads); zone/outline drawn from the effective rect (display only). Pass 14 — band hunting (NO for a step-0 clean accept: the standard band is read as-is; YES in one measured case — the SOFT-ACCEPT probe: a readable band whose glyphs sit in its lower half is abandoned for one step lower and whichever band has more glyph rows wins, so a correctly-aimed scan can end up reading a lower band than v13 would have; the barAtBottom signature that triggers hunting fires on tight, correctly-aimed crops — see f4561/f4557 measurements); per-band gate inside every capped read (a band v13 would OCR is now refused when bar-signed/textless — rescue/recovery pools shrink accordingly; multi-line bands exempt); all-bar attempts settle pre-OCR as mis-aimed nocard (title stage ~0ms vs v13's OCR burn — only when EVERY band is

bar-signed); 150ms tap settle before Snap frame 1 (timing only); drawn/read rect diagnostics (display only).

- **What remains unproven**: (1) the actual burst-frame pixels on-device at the failing taps (no still can supply them — the v17 diagnostics line prints read vs drawn rects, and the hunted band thumbnail, precisely so the NEXT failure screenshot carries them); (2) OCR at full device resolution (all OCR numbers here are at 604×850 screenshot scale with inpaint damage — the on-device crop is ~1.55× larger per axis and overlay-free); (3) whether the f4561 pattern (calibration crop reading CLEANER than the detected crop) recurs on-device — if it does, the stale-verdict tolerances themselves become the suspect, and that experiment needs one on-device failure panel showing the printed read/drawn rect coordinates. Existing suites re-run green after this pass (bandhunt, boxrect, scanflow 44, emptyframe, geometry — all pass; no source files changed).

Heavy-test pass (2026-11-09) — persisted pixel harness, alignment audit, four evidence-gated fixes on the RESTORED v13 code

Trigger: user escalated — "run physical tests with all the images I gave you, use the code to make fixes and adjustments, heavy heavy camera tests, find alignment issues, make sure it is finding the cards correctly within the scanned card area." Fixes authorised but evidence-gated: a fix ships only if a failure reproduces in the harness and localises to one defect; full battery re-run after every fix; any regression reverts the fix. Starting code = restored v13 (Pass 9 scanner + prices/exports); the Pass 11–14 code above was reverted by the restore and is NOT in this build (one piece — the calibration-agreement rule shape — is re-introduced here as Fix 2, this time harness-validated before shipping).

- **HARNESS (new, persisted — closes the "harness not found" gap)**: `tests/harness/` — `run.ts` + `pipeline.ts` (mirrors identifyBurst's decision flow) driving the app's REAL modules: `detect.ts prepareCardFromFrame` (extracted from App.tsx as a pure move so the harness runs the phone's exact rect selection), ocr.ts capped reads on tesseract.js 5.0.5 with local core/lang (same build the phone lazy-loads), scanflow/bottomline/lib/rapid unmodified. `canvas.ts` is a bilinear Canvas2D shim honouring the app's exact CSS filter strings; `scryfall.ts` is a disk-cached REST client for the same endpoints the server actions wrap. Fixtures: `prep.py` generates 98 synthetic composites from live Scryfall card images (all printings verified: AER 177, MID 179, ISD 203, BFZ 81, VOW 102, SOS 43/77/185/227/245, M20 20, MSH 114/33, CMR 543, SOC 246, ONE 19, LTC 521, 2XM 32) — collector-number set (box-40 detection + calibrated Rapid), distance 25/40/70/95%, alignment stress (vertical −10…+10%, horizontal ±4/±8%, rotation ±2°/±5°, scale 30/50/100%), box low-seat (cards 4/8/12% of frame height BELOW the ghost calibration {0.061,0.067,0.811,0.812} measured in the forensic pass), hand-held 80% ×8 evening-failure cards, EXIF 1/3/6/8 (raw stored pixels vs oriented), empty tray/cloth. `prep_user.py` decodes all 43 user images → 24 fixtures: 6 live-view screenshots (guide boxes measured per screenshot from the gold borders and verified

visually; true rects from visual grounding/forensic measurement) × raw + overlay-masked + the device's own drawn rects for 4526/4557/4561, 6 real card photos via the photo path, one photo with Box mode + calibration persisted (the Laelia configuration), 2 rig photos detect-only. Plus a matcher string set (the exact raw reads from the failure screenshots) and a jitter set (3-frame bursts, ±2.5% rect jitter, plus a poison variant: 2 good frames + 1 empty-tray frame). ONE COMMAND: `bun tests/harness/run.ts --set all --out latest` (see tests/harness/README.md). Per-fixture records (rect source, IoU/centre error vs truth, title centroid in/out of the band actually read, raw reads, outcome, stage ms) land in `tests/harness/results/<name>.json`.

- **BASELINE on v13 (102 synthetic + 24 user + 9 strings + 4 jitter)**: collector-number 10/10, distance 12/12, vertical offsets 33/33, horizontal 8/8, scale 3/3, hand-held 8/8 correct-printing; empty scenes honest-none; jitter 4/4 (one empty frame does NOT poison the pool). Failures, all measured: (1) BOX LOW-SEAT — calibration used unconditionally, title centroid outside the read band in all 9 (band=False): +4% → 2 name-only via offset rescue bands + 1 uncertain-miss; +8% → 1 name-only, 1 miss, 1 none; +12% → 0/3. (2) PHOTO CALIBRATION LEAK — the Zedruu photo run with boxMode+calibration (the exact state in the Laelia screenshot, whose diagnostics read "3000×4000 photo · detect 0ms") reads desk/texture: uncertain-miss. (3) MATCHER TIE-BAND — the user's Tribute to the World Tree photo reads near-perfectly ("Tribute to the World Tree oN") yet the app confidently returns THE WORLD TREE (KHM 275): pooled scores 0.988 vs 0.980 fall inside the sort's 0.02 tie band, the face-rank tie-break doesn't apply (different face names), and stable sort keeps insertion order — [0] is not the max. (4) JUNK ABOVE THE BAR — the exact debris reads from screenshots 4553/4518/4520 pool to Pirate Hat 0.644 / Tall as a Beanstalk 0.625 / Hit It! 0.653 (the last reaches decideRapidMatch "auto" in isolation): the fragment-union pass only asks whether the ENTRY's tokens are evidenced in the debris, and clipped lookalikes ('ater'→'eater') always are. (5) EXIF — with stored (unrotated) pixels: EXIF 3 uncertain-miss, EXIF 6/8 no-identification; oriented pixels: all correct-printing. v13's photo decode is `new Image()` with no explicit orientation handling, so the path works only where the platform bakes EXIF. (6) ROTATIONS ±5°/−2° on Stolen Stark Tech: detection is fine (IoU 0.945–0.989, title in band) but title OCR garbles — 2 uncertain + 1 listed; OCR-limited, NOT an alignment defect; no fix shipped for it.
- **ALIGNMENT AUDIT**: (a) ONE geometry source — every capture path (live Snap, Rapid, Take Photo stills, photo files) crops through `drawGuideFromSource` → `geometry.ts` `cssRectToSource` (object-cover math); the detection overlay renders INSIDE the guide element in % of the guide = % of the guide crop, so drawn and read regions correspond exactly. Numeric proof: guide→source round-trip error 0.000 CSS px at 1080×1920, 3000×4000, 4000×3000, 640×480. No divergent mapping found in any path. (b) Rect selection per path was the real divergence — box mode trusted the stored calibration UNCONDITIONALLY (and leaked it into photo files); live paths detect per frame. Measured break point in box mode: identification degrades from +4% low-seat (band misses title centroid) and collapses at +12%. (c) The v13 CARD NAME HERE zone is drawn on the GUIDE while reads are detection-based — by design in v13, but on the rig frames the zone visibly sits above the card's title (screenshot 4557), a UX trap that compounds (b). (d) Burst consistency:

rect jitter ±2.5% and an empty poison frame do not change outcomes (4/4). (e) Screenshot runs are confounded as documented in the forensic pass (945×2048 JPEGs, ~1.55× below device resolution, app overlays burned in — the CARD NAME HERE label is literally the loudest text in the box shots and OCR reads it verbatim; masked variants + device-drawn rects used; on 4561 the device rect measures IoU 0.942 with the title inside its band — geometry good, identification impossible from those pixels because the label occludes the title in the saved image; on the live phone the label is DOM, not camera pixels, so this failure mode cannot occur live). Screenshot 4526's device-drawn rect measures IoU 0.597 vs the true card with its band BELOW the title — the Pass-6-era shifted outline, reproduced as geometry; v13 detection on the same pixels lands at IoU 0.852 with the title in band.

- **FIX 1 — photo path ignores the box calibration** (detect.ts `ignoreCalibration` opt; App.tsx handlePhotoFile passes it; harness photo mode mirrors the call site). Before: userphoto-zedruu-boxleak uncertain-miss (desk bands). After: correct-name (Zedruu), matching the same photo without box state. One-defect, call-site only.
- **FIX 2 — box calibration is a verified prior, not gospel** (detect.ts `checkCalibration` + effective-rect branch in prepareCardFromFrame): box capture now detects (honest detectMs replaces the hardcoded 0); if detection (conf ≥0.4, aspect-corrected) disagrees with the calibration — top edge >4% of frame height, any edge >6%, height >6% relative — the card is read where it IS, flagged `calibrationStale`, and the burst surfaces "The card isn't sitting at the saved box position — read it where it actually is. Re-calibrate if the rig moved." Agreement (or no detection) → calibration used exactly as v13 (boxaligned control: IoU 1.0, unchanged; +4% low-seat stays on calibration within tolerance and the rescue bands carry it, as before). Before → after on the low-seat set: +8%: miss/none/name-only → 3/3 correct-printing (IoU 0.926, title in band); +12%: 0/3 → 3/3 correct-printing; +4%: miss → correct-printing, name-only ×2 unchanged. NOTE the honest tension from the forensic pass: on rig frame 4561 at screenshot scale the calibration crop OCR'd cleaner (0.95) than the detected crop (0.62) — but both put the right card first, and the rule only overrides on measured disagreement; where calibration demonstrably cannot work (band misses the title), detection now rescues the scan instead of reading tray texture for 3.3s.
- **FIX 3 — matcher, two one-defect changes** (lib.ts). (a) Tie-break scope: the pooled/plain sort's face-rank preference now applies ONLY when the two entries share the same face name (its documented purpose — standalone vs DFC face); between different names the higher score wins even inside 0.02. Before: Tribute photo → WRONG (The World Tree, confident). After: correct-name (Tribute 0.988 > World Tree 0.980). (b) Union evidence precision: the fragment-union score is discounted when the contender accounts for <45% of the pooled evidence tokens (≥3 letters, dice ≥0.3 AND tokenSimilarity ≥0.55); genuine fragment pools account for ≥2/3 (Treasure 'ressure/kecrer' = 1.0, Divergent fragments = 0.667 — both untouched: 0.965 and the Treasure margins unchanged). Before → after: soup 0.644 → 0.510, prism debris 0.625 → 0.468, courier debris 0.653/auto → 0.484 — all below the 0.55 Did-you-mean bar; matcher string set JUNK-ABOVE-BAR 3 → 0. Failures that remain junk-shaped (screenshot 4526's guide run still pools overlay debris to a wrong card via rescue

bands) are screenshot-pixel artifacts — the same crop masked reads nothing, and the input contains the app's own UI text, which live camera pixels never do.

- **FIX 4 — photo decode applies EXIF orientation explicitly** (App.tsx handlePhotoFile): `createImageBitmap(file, { imageOrientation: "from-image" })` first, `<img>` fallback on any failure (HEIC-in-Chrome etc.). Where the platform already bakes orientation (modern Chrome/Safari) the bitmap is identical — the change can only repair WebViews that don't bake. Harness proof of the stakes: stored pixels EXIF 6/8 → no-identification; oriented pixels → correct-printing (Zedruu CMR 543 via bottom-line salvage). This replaces v13's implicit reliance on `<img>` baking; it does NOT model the Laelia failure, whose proven cause was Fix 1's leak (detect 0ms + desk bands in its own diagnostics).
- **FULL BATTERY AFTER EVERY FIX** (fixtures+user+matcher+jitter diffed per-fixture against the previous state): Fixes 1+2: 8 fixture improvements + boxleak, ZERO regressions. Fix 3: Tribute WRONG → correct-name, junk set cleared, ZERO passing-case regressions (one failing case changed failure shape — EXIF-3-raw went uncertain-miss → no-identification because junk suppression now also suppresses its upside-down debris suggestions; it never identified, and Fix 4 addresses its input class at decode). FINAL STATE vs v13 baseline: 13 verdict improvements, 0 regressions of any previously-passing fixture. Final harness: synthetic 89 correct-printing / 6 correct-name-only / 2 honest-none / remaining non-pass = 2 rotation-marginal (correct card listed at −2°) + 3 EXIF-raw (by construction — they model a decoder that ignores orientation; their oriented twins all pass and Fix 4 is the app-side answer); user set: 7 correct-name, WRONG only the overlay-confounded 4526 guide run; matcher strings: 0 junk above bar; jitter 4/4. **Suites: all 12 green — 318 checks (aspect 18, deadline 54, detect 10, geometry 20, matcher 33, rapid 34, rescue 47, salvage 17, scandebug 26, scanflow 44, torch 15) + exports 42; client + server typecheck clean.** Marginal fixtures near score thresholds can flip between runs because capped reads are time-raced (observed: rot-divergent −5° listed → correct-printing across runs); decisive-level verdicts were stable in every set.
- **What remains phone-only (stated once)**: live WebView camera behaviour — whether the Muse in-app browser's /ImageBitmap decode bakes EXIF (Fix 4 covers both cases by construction, but the in-app browser itself can't be driven from here), per-frame detection on real burst pixels at full device resolution, and torch/exposure behaviour. The 30-second on-phone checks that settle it: one low-seated box card (should now read via the detected rect with the recalibration note), one hand-held Take Photo with Box mode still on (should ignore the rig rect), one rotated card photo from the gallery.

Triage pass (2026-11-09) — every non-passing fixture classified, two more evidence-gated fixes, two-tier harness reporting

Follow-up under the same user directive: work the heavy pass's remainder — the user-image set at 7 correct-name of 24 runs, the matcher rank-2 case, and the rotation OCR limit. Method unchanged: classify first (confound-bound / reproducible defect / OCR-floor) with a clean full-

resolution equivalent as the proof instrument, fix only proven defects, full battery after each fix, any regression of a previously-passing fixture reverts the fix.

- **TWO-TIER HARNESS REPORTING (persisted)**: `run.ts` now stamps every record `tier` + `hasExpected` (+ `confounds` on Tier 2) and prints a Tier 1 headline separately from the Tier 2 listing. Tier 1 = clean-pixel verdicts (synthetic composites + real photos at full resolution + matcher-string + burst sets) — these judge the pipeline. Tier 2 = the 15 screenshot-evidence runs (guide crops × raw/masked + device-rects from the app's own screenshots) with their confounds named per record (burned-in overlays incl. the CARD NAME HERE label over the title zone — the label text is what OCR reads; ~1.55× below on-device resolution; masked variants inpaint the label zone). Tier 2 is reported, never scored. **FINAL TIER 1 HEADLINE: 122 identification runs — 115 correct, 0 wrong, 7 no-read.** The 7 no-reads: 3 EXIF-raw (by construction — deliberately un-oriented stored pixels; oriented twins all correct-printing, previous pass's Fix 4 is the decode-side answer), rot-stolen −2° (right card listed; deskew read-level proof below), rot-stolen +5° (mechanism = glyph CLIPPING, below), rot-divergent −5° (right card listed; VM-scheduling-marginal, below), userphoto Pyroblast (fix shipped, VM-scheduling proof limit, below).
- **TRIAGE — CONFOUND-BOUND (proven, not asserted)**: six new clean composites in prep.py (Tergrid's Shadow / Witherbloom / Stolen Stark Tech × box-40% Snap + calibrated Rapid — the same cards down the same paths as screenshots 4499/4526/4557/4561) pass **6/6 at correct-printing**. Their Tier 2 counterparts fail on reads that are literally the overlay's own text ('CARD NAME HE...', 'Prous NAME HER', the Rapid status line 'oi watching... slide a card in' in 4501, whose expected is null — its 'uncertain-junk' verdicts are OCR of the app's UI burned into the screenshot, not of any card). 4499 (Tergrid's Shadow): detection on the screenshot crop is geometrically right (IoU 0.928, centre error 0.011) and still reads nothing — half-resolution + label occlusion, input-bound. 4526's lone WRONG (guide run → 'Hound Tamer // Untamed Pup' via pooled rescue bands) stays Tier 2: the masked variant and device-rect read no such thing, and the clean Witherbloom composite passes both geometries; the wrong name is pooled overlay debris at screenshot scale, a failure mode live camera pixels cannot produce (the overlays are DOM on-device). Matcher-string debris cases (soup 4553, prism 4518, courier 4520, 'LY' 4514) are correct behaviour, not failures: the strings contain no recoverable name evidence and the matcher returns no suggestions above the bar (tops 0.51/0.47/0.48, decide=none).
- **FIX 5 — matcher coverage requires evidence (lib.ts `tokenScoreFromTokens`)**: str-treasure-4508 localised exactly. The known-good read '3 (@ressure kecrer 9)' scores Treasure Keeper HIGHER per-candidate (0.682 vs 0.632) — the usurper's 0.760 came from the FRAGMENT UNION: 'peer' counted as COVERED because tokenSimilarity('peer','kecrer') = 0.582 via the both-sides-clipped variant ('eer' vs 'ecrer', edit ratio 0.6) — on ONE shared bigram (Dice 0.25). Coverage now requires the best-matching pair to also reach Dice ≥ 0.3 — the standard the union's own precision pass already uses to call a token 'evidenced'. The token's score contribution is unchanged; only the coverage bonus is gated. Genuine fragments all clear it ('iverson'→'divergent' 0.43, 'tquation'→'equation' 0.86, 'ressure'→'treasure' 0.50, 'kecrer'→'keeper' 0.40, 'ratt'→'raft' 0.33). Before → after (full live index): Peer Pressure 0.760

→ 0.646, Treasure Keeper 0.734 — now ranks #1 (verdict correct-at-2 → correct-top). Battery: harness matcher set otherwise unchanged; matcher suite +1 regression check ('Peer Pressure' added to its index); NO other verdict moved in the fix-1 full run (139 records diffed).

- **FIX 6 — left-shift rescue band (ocr.ts `TITLE_LEFT_BAND`, appended LAST in TITLE_RESCUE_BANDS)**: userphoto-1000044050 (Pyroblast — a clean digital render, Tier 1) localised by treatment/window experiment on its detected crop: standard window reads glyph soup in EVERY treatment except polarity (0.697) and threshold ('Dyroblast' 0.670 pooled); the Precise path's left-shift window (x 0–0.81, same y) reads '- Pyroblast 3' → pooled 0.914, a single-read win under the rescue rule — but the Fast path's rescue set never included that window. x1 is load-bearing: widening to 0.83 (to keep the old shared-padding shape) re-reads as soup (0.701 top, wrong card) — the extra 2% pulls in the mana-corner texture and wrecks the marginal binarisation on this light-on-textured-frame title. The band ships at the BAND_VARIANTS geometry (x 0–0.81); deadline.test's padding invariant was scoped to the three original bands and rescue.test's roster updated + geometry asserted. Appended last, so every existing rescue win keeps its exact read order; cost lands only on scans that already failed everything else.
- **FIX 7 — deskew rescue reads (ocr.ts `BandSpec.rotate` + `TITLE_DESKEW_BANDS` ±5°/±2.5°; App.tsx identifyBurst + harness pipeline mirror)**: rotation failures split into TWO measured mechanisms. (i) LEANING text: rot-divergent −5° and rot-stolen −2° keep the full title inside the standard band, slanted; straight reads garble or pool at 0.707–0.792. Rotating the built band strip back recovers EXACT single reads under the pipeline's own capped read: divergent −5° deskew−5 → 'Divergent Equation' 1.000; stolen −2° deskew+2.5 → 'Stolen Stark Tech' 1.000, deskew−2.5 → 0.980 (isolated read times 186–570ms, inside the 900ms cap). (ii) CLIPPED text: rot-stolen +5° is NOT a leaning failure — the standard band slices the glyphs mid-height (band image shows 'Stark Tech' with ascenders cut; the title centroid metric still reports in-band). Deskewing a clipped strip recovers nothing (all four rotations garbage); the existing TALL band contains the full line and reads it verbatim ('Stolen Stark Tech J') — no new mechanism needed or shipped for (ii). Deskew reads run ONLY after burst + number rescue + all straight rescue bands fail (Snap/photo; Rapid unchanged), capped and clock-bound identically, same 0.82 single-read bar; no wrong ≥0.82 deskew win appeared on any control (worst 0.65 junk top on the +5° clip case, 0.50 on a straight control). Straight-on settle: unchanged code path until the failure tail; full-run median total for previously-correct straight-on fixtures 1479ms → 1202ms (VM load variance; no slowdown). Shim: canvas.ts gained affine transforms (translate/rotate/save/restore + inverse-mapped drawImage) so the harness runs the rotation faithfully.
- **THE VM-SCHEDULING LIMIT (honest, measured)**: Fixes 6–7 are proven at READ level (the pipeline's own capped read + matcher + win bar, isolated) but their full-pipeline harness verdicts are gated by the attempt clock on this VM: pooled matching over the 36.5k index costs 2–4.7s here (vs the phone's production settle ~2.5s total), so rescue-tail reads are frequently clock-skipped or race-abandoned in-harness — Pyroblast's final run pooled only wrong suggestions (uncertain-miss) while its isolated left-shift read wins outright; rot-divergent −5° flipped correct-printing ↔ uncertain-correct-listed across runs INCLUDING on

pre-fix code (A/B: fix reverted, isolated re-run = same uncertain verdict — the baseline pass was the lucky scheduling, not a regression). On-device the envelope fits by design (reads 150–350ms, rescue deadline 4s), but that is a design argument, not a measurement: **the on-phone verdict for Fixes 6–7 is Andrew's** — one textured light-on-dark card (Pyroblast class) and one deliberately tilted card, Snap path.

- **RIG DETECT-ONLY (unchanged, restated)**: rig-58a1 still detects a rect at conf 0.81 on an empty scanner box (the tray's own rectangle is card-shaped); the detect-only fixture runs no identification, and no Tier 1 empty-scene fixture (tray/cloth) identifies anything. Detection firing on the empty rig is a known cosmetic false-positive at the rect stage only; gating it would need a measured identification failure it causes, and there is none.
- **FULL BATTERY (final code)**: harness `--set all` = 145 records: Tier 1 headline above (115/0/7 of 122); Tier 2 listed separately (1 correct-name — Snooping Page guide; the rest no-read/junk-shaped as classified). Diff vs the shipped heavy-pass baseline: ONLY str-treasure-4508 improved (correct-at-2 → correct-top), rot-divergent −5° flipped on VM scheduling (A/B-proven not code-caused), one Tier 2 record changed failure shape (4526 masked: no-identification → uncertain-miss — extra rescue reads pooling overlay debris; never passed, Tier 2 not scored). Zero regressions of previously-passing Tier 1 fixtures. **Suites: 323 checks green (aspect 18, deadline 54, detect 10, geometry 20, matcher 34 [+1 treasure/peer regression], rapid 34, rescue 51 [+4: roster, left-shift geometry, deskew set ×3 −1 merged], salvage 17, scandebug 26, scanflow 44, torch 15) + exports 42.**

Consolidation pass (2026-11-09) — evidence persistence only, NO behaviour change

Review gap closed: the triage verdicts and two-tier report existed only in run JSONs and the pass narrative. This pass persisted them as durable, checkable files and re-ran the harness on the final code. Files (all under `tests/harness/results/`): `TRIAGE.md` (full triage table — every non-passing user fixture → category → evidence), `triage-verdicts.json` (same table, machine-readable), `consolidation-final.json` (the re-run, 145 tier-tagged records; `final-all.json` = same content under the final-* convention), `consolidation-full-output.txt` (complete console output incl. two-tier summary), `TIER1-REPORT.md` and `TIER2-REPORT.md` (the two tiers as separate tables, Tier 2 with confounds named and raw reads quoted), alongside the pre-existing `triage-final.json` and `triage-clean-proofs.json` (6/6 clean-composite proofs). Index + command also recorded in `tests/harness/README.md` ("Persisted evidence"). **Re-run command:** **`bun tests/harness/run.ts --set all --out consolidation-final`.** Harness files verified whole by execution: `run.ts` (414 lines) and `pipeline.ts` (372 lines) are complete and the Tier 1/Tier 2 split is implemented in `run.ts` (per-record `tier` / `hasExpected` / `confounds` + labelled two-tier summary output), not only described. **Re-run headline: Tier 1 = 114 correct / 0 wrong / 8 no-read of 122** — one verdict differs from the triage pass's 115/0/7: `exif-zedruu-8-oriented` flipped correct-printing → uncertain-correct-listed (correct card top-listed at 0.609 in both outcomes; the re-run's collector-number read never landed inside the run's caps, total 6541ms vs 2252ms — the VM-scheduling flip class the triage pass A/B-proved; documented in `TRIAGE.md` §C and `triage-verdicts.json`).

Tier 2: 15 runs, 1 correct-name, 14 non-passing — all confound-bound per the triage table. Suites re-confirmed in this pass: all 12 test files green, 0 failures — 323 scanner checks (aspect, deadline 54, detect, geometry, matcher 34, rapid, rescue 51, salvage, scandebug, scanflow 44, torch) + exports 42. No source behaviour changed: docs, results and reporting files only.

Pass 15 (2026-11-09) — match hang (exact-containment fast path + match under the deadline), sequential-scan zombie starvation, frame-at-detection capture, Bolrac contrast fixtures, artist-contradiction guard

Trigger: two phone screenshots + report — scanner hit-or-miss; first card ~70%, a second card right after noticeably worse; "card found" in the guide but the capture sometimes doesn't show the card in "what the scanner read". Screenshot 1 is decisive: Bolrac-Clan Basher (MKM U 0112, Warren Mahy) timed out with stages "capture 319ms · detect 14ms · title 855ms · bottom skipped (near-blank) · match 8.1s · hung during: match" while the raw read CONTAINED the exact name ('mtn her PO RE RE EESCE Il Bolrac-Clan Basher 4 be J 3 1 2 EEE AR YVR') and the title-band thumbnail was crisp. Scan list at 26 cards / $42.63 — scanning substantially works; these are the residual failure modes. All fixes evidence-gated; harness `results/p15-*.json` + `p15-full-output.txt` persist the numbers.

- **EVIDENCE A — the match hang, profiled and fixed**. Profile (`tests/harness/profile-bolrac.ts` , full live 36,519-entry index): the pooled fuzzy pass on the exact string took 800–900ms per pass (up to 9.4s for accumulating pools) and returned Bolrac-Clan Basher NOWHERE — the whole-string candidate (67 normalised chars) fails the matcher's length pre-filter against the 18-char name, and ocrCandidates' truncated variants don't contain the name, so the exact read was UNREACHABLE, not merely slow. The match stage was also the one stage not under the attempt deadline. FIX 1 — exact-containment fast path (`lib.ts exactContainedMatch` , run first inside `matchNamesPooled`): pooled candidates tokenised exactly as names are (`nameTokens(normalizeName(...))`); a full index name appearing as a contiguous, token-boundary-aligned token sequence wins decisively at score 1.0 and the fuzzy pass never runs. Guards: full sequence only; ≥2 tokens OR a single token ≥6 chars; longest sequence wins (a 'Tribute to the World Tree' read also contains 'The World Tree' — the longer true name wins); per-key entry chosen by the matcher's own face-rank rule; token-sequence index cached per index array (WeakMap). AFTER: Bolrac-Clan Basher 1.000 in 0–37ms, Rapid decide = auto. Guard validation: unit corpus (tests/matchfast.test.ts — partial names, in-token substrings, wrong order, short single-token names, 8 debris strings → 0 hits, 0 decisive) PLUS a full-corpus sweep (`tests/harness/sweep-debris.ts`): every raw read in `consolidation-final.json` (132 records, 617 candidate pools) → 144 hits, ALL legitimate (the read genuinely contained the expected card's full name), **zero false positives**. FIX 2 — match under the deadline: `matchNamesPooled` accepts `budgetMs` ; scoring stops at the budget and returns best-so-far (the fast path runs before and outside the budget, so an exact-name read can never be timed out — success-is-final extended to the match stage). identifyBurst +

harness mirror pass the AttemptClock's remaining time at every pooled/single call. Regression fixtures in matchfast.test.ts: 30k-entry index + garbage pool returns within ~the budget; the Bolrac read wins even at budgetMs=1; the attempt-clock composition settles decisive inside a 600ms deadline with the matcher artificially slowed to garbage-only work.

- **EVIDENCE B — second-scan degradation: hypothesis CONFIRMED by measurement, then fixed**. Sequence test (`tests/harness/sequence.ts` , real worker, 10 back-to-back attempts: 3 alternating clean cards, the Bolrac string as a matcher-level attempt, one speckle/slow attempt) BEFORE: attempts 1–3 correct (834–2,207ms); the speckle attempt's reads were abandoned at their caps but kept EXECUTING serially inside the single shared worker; attempts 6–7 starved behind the zombies (totals ~4,050ms, title ~2,500ms, NO identification), attempt 8 partial, 9–10 recovered as the queue drained — the user's pattern, reproduced exactly. The existing silent swap (budget 1/session) is spent by the first poisoned attempt, and deadline settlement discards results but cannot reclaim worker time. FIX — two mechanisms in `ocr.ts` : (1) ATTEMPT GENERATIONS — `beginScanAttempt()` (called by identifyBurst + harness mirror) tags every read; a queued job reaching the front under an old generation is dropped WITHOUT a worker call (its result would be discarded by the attempt token anyway). (2) ZOMBIE RECYCLE BETWEEN ATTEMPTS — reads abandoned by an outer cap (and queue-level timeouts) are tracked as outstanding zombies until the underlying job settles; if any zombie is still live when the next attempt begins, the worker is terminated + recreated once, there — between attempts, never mid-attempt, gated strictly on a live zombie, deliberately separate from the max-1 mid-attempt restart budget (a clean attempt leaves no zombie, so it cannot loop); the silent swap clears the counter for the worker it kills. AFTER (same sequence test): all 10 attempts correct; speckle attempt still pays its own caps (4,077ms) but attempts 6–7 identify at 1,419/1,057ms and 8–10 land at 778/459/927ms vs the 890ms attempts 1–3 median (acceptance: within 1.5×, enforced by exit code). Mechanism pinned with a scripted fake worker in tests/sequence.test.ts (hang → swap saves attempt 2; hang again → attempt 3 degraded; attempt 4 begins → recycle → first read fast; stale-generation job dropped with zero worker calls). Cross-attempt state audit: pooled memo + evidence mirrors + debug tracker are per-attempt by construction (reset at attempt start); worker parameters are set explicitly on every read; the only cross-attempt caches are the immutable name index and the derived token-sequence map (pure functions of it). No other cross-attempt state found.
- **EVIDENCE C — 'card found' but the capture reads a different frame: divergence named, frame-at-detection shipped**. Audit (file/line terms, pre-fix): the manual-mode hint/outline came from the ~800ms detection interval in App.tsx, which captured a guide frame, computed detection, stored ONLY the rect (`setDetectOverlay`), and discarded the frame; a Snap then ran `capturePreparedBurst` → `captureOneFrame()` — a FRESH video grab taken after the tap (post-tap jolt, different buffer), re-detected per frame. The approved frame and the read frame were never the same pixels. Rapid had the same shape (watcher approved a signature sample; the burst re-grabbed). FIX — retained frame-at-detection: the scanner keeps ONE retained entry `{prep, at}` (memory bounded, overwritten per tick, consumed on use, honoured only <2.5s): the manual interval retains the exact frame it approved (prepared

via the same `prepareCardCanvas` the capture uses); the Rapid watcher's triggering sample frame and the viewfinder-tap frame are retained likewise; `capturePreparedBurst(..., {useRetainedFirst: true})` (Snap, Rapid trigger, tap) leads the burst with the retained prep — subsequent frames are fresh grabs with their own per-frame detection, exactly as before. stopCamera clears it. Thumbnails (C4 audit): identifyBurst cuts the published band thumbnails from `preps[0].card` — the retained frame when used — so "what the scanner read" shows the frame that was OCR'd, by construction. Harness proof (`tests/harness/retained.ts`): burst [retained good frame, blank, blank] still identifies Divergent Equation from frame 1; [retained empty, blank, blank] manufactures nothing; a +8%-shifted frame (own detection) also identifies — modest shifts were never the problem; a genuinely different capture frame was. The App-side glue (refs/intervals) is not harness-runnable; the pipeline property it relies on is what the harness proves.

- **EVIDENCE D — contrast checked, not assumed: 3 of 4 Bolrac composites identify at correct printing; NO preprocessing change (failure localises to detection edge contrast, not band preprocessing)**. New fixtures (prep.py): Bolrac-Clan Basher (real Scryfall MKM #112 image) hand-held at 85% on near-black cloth, at 80% rotated +3°, at 85% dimmed to 55% brightness, at 80% rotated −3° + dimmed. Results: straight = correct-printing (via collector number MKM 112); dim+angle = correct-printing (title read contained the exact name → fast path); angle = see Fix 3 below; dim (straight, 55%) = no-identification, LOCALISED by measurement (`tests/harness/probe-dim.ts`): at 55% brightness the dark red top border blends into the near-black surface and detection under-measures the rect (top at frame y 0.141 vs true 0.075; title centroid at crop fraction 0.009 — above every band window, glyphs clipped by the crop itself). That is an input/detection floor, not a band-treatment failure: the dim+angle variant passes, the straight 85% passes, and the on-device mitigations for exactly this case (dark-frame exposure-raised retake in capturePreparedBurst, torch) already exist and cannot be exercised on a static harness frame. No threshold was tuned and no preprocessing changed, per the evidence rule.
- **FIX 3 (found by Evidence D) — artist-contradiction guard for unverified number rescues**. The angle fixture first came back WRONG: bottom read 'LC U 012 MKM EN 16 WARREN MAHY' dropped a digit (0112→012) and Rescue 1's unverified lookup confidently returned MKM 12 = Delney, Streetwise Lookout — while the same strip had read the true artist line in full (Delney's artist is Darren Tan; Bolrac's is Warren Mahy — both verified live on Scryfall). `lib.ts bottomArtistClearlyContradicts`: a hit is contradicted only when the tolerant artist match FAILS and the strip carries ≥2 clean ≥4-letter words (a cleanly-read artist line's shape; ©/TM words excluded) — garbled or absent artist lines never contradict. `rescueByCollectorNumber` (scanflow.ts, shared by settlement + watchdog paths) now discards a contradicted hit and tries the next candidate. Verified-pair frame-loop lookups are untouched (two-treatment agreement is their guard). After: the angle fixture identifies correct-printing via its title evidence; unit pins in matchfast.test.ts cover the collision, the consistent hit, and Snooping Page's tolerant-match survival.
- **FULL BATTERY (final code)**: harness `--set all` = 150 records → **Tier 1: 119 correct / 0 wrong / 8 no-read of 127** (consolidation baseline 114/0/8 of 122; the set grew by the 4 Bolrac

fixtures + the Bolrac matcher string, all correct except the measured dim floor). Per-record diff vs consolidation-final: ZERO regressions of previously-passing Tier 1 fixtures (one within-correct flip: exif-zedruu-6-oriented correct-printing → correct-name-only — same card, version evidence didn't land inside this run's caps; its sibling exif-zedruu-8-oriented flipped the other way, to correct-printing — the documented VM-scheduling class); the remaining verdict movement is the known no-read class changing shape (rotation/Pyroblast OCR-floor cases) and Tier 2's lone WRONG (4526 guide) improving to uncertain-miss. Suites: **362 scanner checks + exports 42, all green** (323 prior + matchfast 29 + sequence 10); client + server typecheck clean. New persisted scripts: profile-bolrac.ts, sequence.ts, retained.ts, sweep-debris.ts, probe-dim.ts (harness), tests/matchfast.test.ts, tests/sequence.test.ts. Rapid's happy path is unchanged when reads are clean (generation tag is a no-op pass-through; fast path returns the same winner the fuzzy pass found, faster). On-phone confirmation remains Andrew's: a Bolrac-class scan after a speckle-hard card (the sequence pattern), and a Snap where the card is jolted right at the tap (retained frame).

Pass 16 (2026-11-09) — attempt read the guide area despite a perfect preview detection: retention/state gap convicted, rect authority + one detection state + rect diagnostics

Trigger: two phone screenshots (~15:54 EST) of an in-box foil Swamp (LCI #0397, Logan Feliciano — verified live: GET https://api.scryfall.com/cards/lci/397). LIVE VIEW (1000044584): the detected outline hugs the card exactly (measured from the screenshot: guide x 200–742, y 373–1132; detected rect x 209–624, y 418–1003 → guide fractions (0.017,0.059)–(0.782,0.830)), the CARD NAME HERE zone sits over the title — yet the SAME screen shows '✓ Card found — fine at this distance, tap Snap' (green) AND 'Card outline not found — using the guide area' (blue). FAILURE PANEL (1000044582): 'No plausible name', evidence 'no readable text', stages 'capture 172ms · detect 15ms · title 3.9s · bottom skipped (near-blank) · match 0ms'; the title band thumbnail is a smooth grey gradient (the tray wall ABOVE the card) with the card's black top edge along its bottom ~79% (measured) — the attempt read the GUIDE area, not the detected card. The blue message also shows under the panel.

- **LOCALISATION — three candidate branches, measured**: (b) ATTEMPT-TIME DETECTOR FLICKER — EXONERATED by measurement. Recreated scene (tests/harness/prep_swamp.py: the real LCI 397 image, foil sheen sweeping the card at strengths 0/25/45, ±6% exposure jitter, sensor noise, white tray with wall gradient + contact shadow, at the measured geometry; + a blur45 level with tap-jolt directional smear): `probe-swamp.ts` ran the app's real detectInCanvas + prepareCardFromFrame over 300 frames — **300/300 card detections (confidence 0.73–0.88), ZERO guide-source preps, zero all-guide simulated bursts at any level**. The detector does not flicker on a foil Swamp against the white tray, even dimmed to 0.78 or smeared. Also ruled out by measurement: the dark-frame retry (card crop mean luminance 80.6 > the 55 threshold) and the hi-res small-card retry (detected rect height 0.77 of guide > the 0.45 trigger). (a) LATCHED STATE — CONVICTED for the display: the blue message was `scanNote`, a sticky string set by capturePreparedBurst from a burst head's

source and NEVER cleared by later detection successes, while the green hint is driven by the separate `cardPresent` state — two independent detection states, rendered simultaneously; the note under the live view is a previous attempt's leftover (capture does not consult the note; it is set FROM capture results). (c) RETAINED-RECT GAP — CONVICTED for the read: the retained frame (Pass 15) is the ONLY bridge between the preview's detection and the attempt, and its lifecycle guarantees its absence in the user's exact pattern — TTL 2.5s while an attempt runs ~4s (his own readout: 'Last scan: 4.0s'), it is CONSUMED by any capture, cleared by 2 interval misses, and the preview interval is SUSPENDED while `reading` is true, so after a previous attempt settles there is a window (≥800ms until the next tick, longer if the retained frame aged out mid-attempt) in which NO retained frame exists — while `detectOverlay` is never cleared and `cardPresent` persists, so the screen keeps showing the older successful detection. A Snap tapped in that window re-derives the rect from post-tap frames alone and, when they fail for any transient reason, silently reads the guide. (The on-device transient that made those particular frames fail is NOT identified — every pixel model tried fails to reproduce a detection failure on this scene; the convicted defect is the architectural gap it fell through, and the fix removes the guide-read outcome for ANY burst-detection failure while a fresh detection exists.)

- **FIX 1 — RECT AUTHORITY** (`applyRectAuthority` in detect.ts, called by capturePreparedBurst AFTER the dark/hi-res retries, on the frames actually read): any burst frame that fell back to the guide is re-cropped from its OWN pixels using (1) a sibling burst frame's rect, else (2) the scanner's most recent successful detection when ≤10s old (`LAST_DETECTION_RESCUE_MS` — covers a full attempt cycle + reaction time; the 2.5s frame TTL could not). `lastDetectionRef` (App.tsx) is fed by every detection that lands: the preview interval, the Rapid watcher, the viewfinder tap, every capture's authority head; stopCamera clears it. The guide fallback survives ONLY when no detection exists anywhere — empty-tray honesty preserved (fixtures S6/S7 below). The retained frame still leads when present (Pass 15 untouched) and is now labelled rectOrigin 'retained-frame'.
- **FIX 2 — ONE DETECTION STATE**: the guide-fallback note is its own boolean state, set only by a capture whose post-authority head really is guide, cleared by the next successful live detection, and RENDERED only while `!cardPresent` — the same state that drives the green hint. The '✓ Card found' + 'outline not found' pairing is impossible by construction.
- **FIX 3 — DIAGNOSTICS**: every attempt's debug block now prints 'Rect: …' (scandebug `rect` field, set at beginAttempt from `describePrepRect(preps[0])` in Snap, Rapid and the identifyBurst fallback; rides the watchdog timeout snapshot too): e.g. 'detected (0.02,0.06)-(0.78,0.83) · frame: retained (the frame the viewfinder approved)', 'detected (…) · rect from the last preview detection', 'guide fallback · no detection available to this attempt'. PreparedCard/PreparedScan gained `rectOrigin` ('frame' | 'calibration' | 'retained-frame' | 'burst-sibling' | 'last-detection' | 'guide').
- **PROOF (persisted)**: `tests/rectauthority.test.ts` — 20 checks (sibling rescue + own-pixels re-crop, last-detection rescue, window edge/expiry, no-authority no-op, authority order, diagnostics wording). `tests/harness/swamp-attempt.ts` → `results/p16-swamp-attempt.json` — 7/7 scenarios on the recreated scene through the real pipeline: S1 control burst identifies

Swamp at the correct printing (collector number LCI 397); S2 the convicted pattern WITHOUT authority (burst forced to the guide state despite the fresh preview detection — the manifest state of the user's panel) identifies NOTHING, reproducing the failure from the same pixels; S3 the same burst WITH authority identifies Swamp / LCI 397, rect source 'last-detection'; S4 flicker burst [guide, detected, guide] identifies via the sibling rect; S5 retained-frame lead [retained, guide, guide] still identifies (Pass 15 intact); S6 empty tray with no detection ever → honest none; S7 empty tray WITH a fresh last detection → re-cropped tray still reads nothing (no manufactured read). The scene frames' own detections measure IoU ≈0.99 vs the pasted truth.

- **FULL BATTERY**: baseline first on untouched code (`results/p16-baseline.json` : Tier 1 115/0/12 of 127 — the EXIF-oriented trio + Pyroblast flipped DOWN in that run, the documented VM-scheduling class), final (`results/p16-final.json`): **Tier 1 119 correct / 0 wrong / 8 no-read of 127** (the Pass 15 headline). Per-record diff baseline→final: 7 records changed — 4 Tier 1 improvements (the same scheduling class flipping back up; the harness pipeline never calls applyRectAuthority, so harness verdicts cannot move because of the fix) + 3 Tier 2 failure-shape changes inside the never-passing confounded set. **Zero regressions of previously-passing cases.** Suites: **382 scanner checks + exports 42, all green** (362 prior + 20 new rectauthority); scandebug suite green with the new rect field; client typecheck clean. NOT changed: detector thresholds, matcher, OCR, Rapid machine, Pass 15 mechanisms (fast path, generation tagging, retention) — the fix is rect selection + state unification + diagnostics only. On-phone confirmation remains Andrew's: the same in-box foil Swamp Snap — the panel, if it ever fails again, now prints the rect it read and where that rect came from.

Pass 17 (2026-11-09) — Auto Light: metering-driven torch/exposure + software exposure lift for shadowed title zones

Trigger: two phone screenshots (~16:31 EST) + the user's request verbatim: 'Can you have a auto light to make sure if im in a bad lighting it would automatically fix it'. His attempt was Box mode (Box ✓, Calibrated ✓) and the Pass 16 diagnostics WORKED — the panel prints 'Rect: detected (0.19,0.01)-(0.87,0.69) · frame: this burst' and the note correctly says the card isn't at the saved box position. The failure is LIGHT: the card sits high/close in the box, its body lit and legible (rules text + 'Illus. Christopher Moeller' readable in the live view), but the top of the frame is in the box's shadow — the title band thumbnail is PURE BLACK, the bottom strip blown white (bare tray below the high-seated card), stages 'capture 962ms · detect 19ms · title 3.9s · bottom skipped (near-blank) · match 0ms', evidence 'no readable text'. Light chip ✗ (torch absent in the Muse in-app browser — established; Chrome exposes it).

- **STEP 0 — BASELINE MEASUREMENT (before any pipeline change)**: `tests/harness/prep_dark.py` recreates the scene with real Scryfall card images (Divergent Equation SOS 43, Treasure Keeper AER 177, Bolrac-Clan Basher MKM 112) seated HIGH in his read rect (card (0.19,0.01)–(0.87,0.60) inside the (0.19,0.01)–(0.87,0.69) read rect, so the read

rect's bottom strip lands on bare tray like his blown-white strip and the number rescue cannot mask a title failure), under a shadow gradient swept ×0.40/×0.25/×0.15/×0.07/×0.02 at the title zone (body lit), plus uniform dims ×0.45/×0.30/×0.18 (card filling the rect) and bright controls. `tests/harness/darkprobe.ts` runs them as DEVICE-RECT preps (the rect his own panel printed; harness-side detection on the synthetic tray sits ~3.5% low because the shadowed top rows carry no edge contrast — a fixture-fidelity gap `probe-rect.ts` pinned down, not device behaviour) and measures the title zone of the crop actually read. Findings: bright references title-zone p90 = 206–223, all identified. The treated standard band's mean collapses to ≈0 from ×0.25 down (the contrast(1.7) treatment pivots at mid-grey and crushes dim bands to black — the exact mechanism behind his pure-black thumbnail). Baseline outcomes (auto light off, `results/p17-baseline.json`): bright + ×0.40 + uniform ×0.45/×0.30 identified; **×0.25, ×0.15, ×0.07, ×0.02 (all three cards) and uniform ×0.18 all no-identification**, title stages 593–2,075ms. Read-level boundary proof (`probe-lift.ts`, device-rect crops, pipeline's own capped reader): at title p90 34–55 the raw read is nothing/garbage and the p90-derived lift (×4 cap) reads the title **EXACT @ 1.000** (treasure ×0.25, treasure ×0.15, bolrac ×0.15, divergent uniform ×0.18); at p90 18–23 (×0.07 divergent/treasure, all ×0.02) raw and lifted both return nothing or garbage (one lifted garbage pool scored a wrong name @ 0.922 on a detection-crop variant — lifting at that level manufactures confidence, so it is never attempted); p90 28 pooled correct fragments @ 0.851. **Boundary: DIM = p90 27–149 (lift), CRUSHED = p90 ≤ 26 (skip + too-dark message), BRIGHT = p90 ≥ 150 (no change)** — the crush ceiling 26 sits in the measured gap (23 hopeless / 28 workable). The metering zone is the standard band window + margin (crop y 0.02–0.125): a wider zone to y 0.16 catches the gradient's lit edge and inflates p90 by 12–19 points on crushed bands (measured), misclassifying them dim.

- **THE MECHANISM (`client/src/autolight.ts` + wiring)**: `luminanceStatsFromData` / `zoneLightStats` (histogram percentiles: mean, p10/p50/p90, share <8/255), `classifyBand`, `computeLiftGain` (gain = clamp(175/p90, 1, 4) — target p90 175, **cap ×4**), `meterBand`, and `liftedSpec` (the GATE: returns a lift-carrying spec only for a DIM meter; bright/crushed/missing → null). Three layers: (1) METERING at preview (the manual-mode detection interval and a ~900ms-throttled Rapid watcher tick meter the located crop's title zone) and per attempt on the actual crops (title zone + bottom strip independently), published into the debug block as an `autoLight` field ('Auto light: brightened ×3.4 · torch on' / 'exposure +2' / 'couldn't recover the title area — too dark (band p90 = 15)'). (2) HARDWARE (App only, capability-guarded, best-effort, never throws/blocks): in Auto mode when metering says dark — torch on if `getCapabilities().torch` (via the existing torch.ts detailed apply), exposureCompensation raised to the track's max if exposed; both restored when light is adequate or the mode leaves Auto; in browsers with neither (his in-app webview) it is a silent no-op and the existing 'torch = absent' note is unchanged. (3) SOFTWARE (every browser): DIM title → the lifted standard band pooled from frame 1 under the same 900ms cap + a lifted rescue stage (standard + tall windows with gain, after straight + deskew rescue, same 0.82 bar); DIM bottom → one lifted bottom read from frame 1 pooled into the rescue sources (the verified-pair path untouched). **The lift is applied to the RAW band pixels BEFORE the**

treatment filter (`ocr.ts` BandSpec.lift in buildBandCanvas) — order is load-bearing: post-filter, the contrast crush has already zeroed the band and there is nothing left to multiply. CRUSHED title → all title reads skipped (stage line: 'title skipped (too dark)'), bottom evidence still runs, and a failure settles with the lighting-specific message ('Too dark to read — the card's name area is in shadow, and auto light couldn't recover it…') via `scanFailureMessage`; Rapid surfaces `TOO_DARK_RAPID_HINT` once per dark streak (a ref re-armed by any non-dark outcome). UI: the Light chip is now a 3-state cycle 💡 **Auto (default) /** 💡 **On /** 💡 **Off**, persisted as `manabox-light-mode` (the legacy binary torch pref migrates: stored-on → On); Off disables the whole layer including the software lift; the mode persists across camera restarts while the physical torch state does not. The pre-existing dark-frame exposure retake in capturePreparedBurst is untouched (capture-level, whole-card mean) — the pooled lift is the band-level mechanism, no duplication.

- **ACCEPTANCE — dark A/B (same fixtures, same code, only the layer toggled)**: OFF reproduces the baseline verdicts. ON (`results/p17-dark-on.json`): ×0.25 → divergent still no-id in-pipeline (its lifted read is proven @ 0.871 in isolation but overran the 900ms cap in this VM — the documented scheduling class; on-device reads run 150–350ms), treasure correct-name-only, bolrac correct-printing; ×0.15 → **all three identify** (divergent correct-printing, treasure correct-name-only, bolrac correct-printing); uniform ×0.18 → correct-printing. **6 of 7 dim failures identify in-pipeline, the 7th proven at read level.** Crushed levels (×0.07, ×0.02, all cards): still no identification — honestly, there is nothing to recover — but the title stage drops **1,672–2,075ms → 0ms** and the outcome carries `tooDark: true` (the too-dark settle). Bright controls + ×0.40 + uniform ×0.45/×0.30: unchanged, all correct. Gate evidence from the full battery: **lifted TITLE reads occur only on fixtures whose title metered dim — zero on every bright-title fixture**; lifted BOTTOM reads also occur on some bright-title fixtures at mild gains (×1.2–×2.2) because a card's bottom strip is a dark border by construction and meters dim in any light — they are pooled as unverified rescue sources only, and the battery shows zero resulting regressions (documented, not tuned away: no failure localises there, and the uniform-dim bottom at gain ×3.98 is a genuine save path).
- **FULL BATTERY**: harness `--set all` (`results/p17-final.json` , 150 records): **Tier 1 = 119 correct / 0 wrong / 8 no-read of 127 — identical headline to Pass 16.** Per-record diff vs `p16-final.json` : exactly 2 changed records, both Tier 2 improvements inside the confounded set (4499 masked guide: no-identification → correct-name; 4501 masked guide: uncertain-junk → honest-none); **zero regressions of previously-passing cases**. Suites: **413 scanner checks + exports 42 = 455 total, all green** (382 prior + 31 new `tests/autolight.test.ts` : stats/percentiles, classification boundaries, gain + cap, the bright/crushed gate, chip cycling + labels, message selection); client typecheck clean. Harness mirror (`pipeline.ts`) carries the same metering/gate/crush logic via the shared autolight.ts, with `opts.autoLight` as the A/B switch and `tooDark` / `autoLight` on the outcome. NOT changed: detector, matcher, rect authority, Rapid machine, deadline machinery, the exposure retake. **Phone-only remainder (Andrew's)**: retest the same in-box Snap — in the in-app browser only the software layer can act (torch stays absent there; in Chrome the Auto hardware layer can also fire). Expected on-device: ×0.25-class shadows (band dim) now read

via the lifted variants; a truly crushed band settles in ~1s with the too-dark message instead of ~4s of pretend reads.

Pass 18 (2026-11-09) — the Way of the Mind Sculptor Rapid failures: bottom strip aimed from measured ink, rect stabilisation, helper-finder prototype measured and NOT shipped

Trigger: five phone screenshots (~17:41 EST) + the user's asks verbatim: 'calibration isnt doing the best job. The bottom of the card isnt being read to help find the correct card', a request for a 'helper finder' that searches the card to identify it better, and 'We need this professional this need to work perfectly to find the card fast efficient and nearly no mistakes'. Rapid + Box ✓ + Calibrated ✓ + Foil tick ON + Auto Light ON (confirmed working on-device: 'Auto light: exposure +2 · brightened ×1.5'). The card is **Way of the Mind Sculptor — FRA #224, single printing, artist Josh Hass (verified live: GET https://api.scryfall.com/cards/fra/224)**. Its title band is human-legible in three panels, yet all three Rapid attempts settled 'title (uncertain — thin margin)': A1 rect (0.00,0.07)-(0.88,0.96), raw 'Wot the Mind pr a | bottom: I', suggestions Clear the Mind 66% / Mind Rot 64%; A2 same rect, raw 'Way of the Mind Seal ry · Wabi the Mind SE &', suggestions Temple of the Dead 56% / Angel of the Ruins 55%; A3 rect (0.01,0.11)-(0.90,0.99), raw 'Way OF the nd SE or 8 | bottom: CS TT-', suggestions Way of the Thief 84% / Mentor 77% / Healer 76%; A4 (different foil card) rect (0.00,0.14)-(0.86,1.00), published title band = PURE ART, raw 'd Sess BTW'. Every bottom thumbnail shows the strip straddling the card's bottom edge (black over blown-white tray).

- **PART A FORENSICS (all measured before any fix; probes persisted as tests/harness/probe-way.ts, probe-jitter.ts, probe-aim.ts, probe-clusters.ts)**:
  - **A-ii BOTTOM STRIP — LOCALISED (the decisive defect).** FRA frame geometry measured from the card image: title centroid 0.0699; bottom ink span 0.879–0.965; **the number line 'U 0224' prints at card 0.885–0.925 — ~2% higher than the slash-era lines the fixed 0.915–0.995 window was tuned for.** From his A1 thumbnails (strip's card edge at ~63% of thumbnail height; title glyphs at ~64% of the title band) both anchors solve to **crop = card ×1.050** (≈1.3% slack above, ≈3.7% overshoot below — aspect-correction growth + the designed 1.5% expandRect + rect jitter). On fixture crops at his exact rects: A1's info block sits at crop 0.857–0.933 vs window top 0.915 → the strip read 'FRA EN J0H HAS' (artist line tail only), **zero collector-number candidates** (block-centre error ≈6.0% of crop height); A3's block at 0.821–0.898, entirely above the window (strip read "", centre error ≈9.5%); A2 marginal graze. The number rescue — the one channel that pins this single-printing card exactly — never had a candidate. A whole-region ink-cluster aim was tried first and REJECTED by measurement (region clusters crown the text box at ~0.845 on accurate crops and broke A2 + the num-* channels on first test): a strip can contain good text (the artist line alone) and still miss the number, so the aim must be window-relative.

    - **A-i TITLE FRAGMENTATION — three stacked mechanisms.** (1) Rapid title-evidence starvation: Rapid reads only the standard band variant; on the A1 crop the standard read is 'Poi the Mind Soutor Lt' while left-shift / standard-low / adaptive / tall / threshold / bottomline variants ALL read the full name; the variants that rescue run only in Snap's tail or Precise (proven in-pipeline: the same pixels as way-a1-snap identify; as Rapid they didn't). His A1/A3 title stages (146–159ms) are single-variant reads. (2) Match-stage budget starvation: the matcher on his EXACT A2 pool at his panel's 84ms match budget returns his panel's list almost verbatim (Temple 56 / Angel 55 class — true name absent); at infinite budget the same pool ranks the true name only #4 @0.788, and A1's pool at EVERY budget lacks the true name in its top-8 (its reads evidence only 'the'+'mind' of 5 tokens). (3) Discriminating-token absence: 'sculptor' has zero matcher-grade evidence in his pools (best Dice 0.25 < the 0.3 coverage gate). No matcher tuning repairs absent evidence — the fix had to add evidence (the number channel, Part B), not re-rank.
    - **A-iii RECT JITTER — LOCALISED.** probe-jitter.ts (static Way scene, 48 samples, sensor noise σ≈5 + exposure ±4% + ±1px only): raw detections **y0 range 0.077 (sd 0.034), y1 range 0.067 (sd 0.028)**, x-edges tight — matching his panels' tops 0.07/0.11/0.14 on an unmoved card. Single jittered detections cross checkCalibration's tolerances on ~half the samples, so the read rect FLIPPED calibration↔detected 24/48 and the calibration he recorded for this rig effectively never governed two attempts alike. Vertical slab edges (smooth borders, top in box shadow) are the noise-sensitive axis.

- **PART B — STRIP AIM (SHIPPED, `aimBottomSpec` in ocr.ts + `assessStripInk` / `stripAimShift` / `stripWindowCentred` in scanflow.ts).** The bottom spec's window is now aimed from the strip's OWN ink profile (per-row deviation from the row median — polarity-free; rows above fraction 0.62 are transitions/texture and never count, an edge is not text): text straddling the window's middle → the fixed window is kept **byte-for-byte** (accurate crops unchanged); text crowding an edge → ONE centroid re-aim, kept only if the new window carries ≥80% of the ink; no text → probe ±0.03/±0.06/±0.09 (upward first, the measured failure direction), centre the first carrier. The capped read's pre-check AND read both run on the aimed spec, so the near-blank skip and verified-pair semantics are untouched (all bottom specs of one crop land on the same rows — per-canvas cache), and the published debug thumbnail now shows the AIMED window. Proof (probe-aim.ts): A1 aimed 0.888–0.968 → 'U 0224 FRA EN JH HS' → FRA/224; A2 fixed kept → FRA/224; A3 aimed 0.852–0.932 → FRA/224; all num-* box40 crops keep the fixed window with correct candidates. Units: `tests/stripaim.test.ts` (20 checks: centred/top-crowded/bottom-crowded/blank/slab/speckle profiles).
- **PART D — RECT STABILISATION (SHIPPED, detect.ts + App.tsx wiring).** `prepareCardFromFrame` (box branch only) now judges the recent-rects cluster, not one jittered sample: candidates = last ≤4 read rects (App feeds `recentDetectionsRef` via one `noteDetection` helper from the preview interval, Rapid watcher, viewfinder tap and capture authority head; entries ≤3s old; cleared on stopCamera and on any guide-read attempt) + the current detection. Calibration wins iff the per-edge **median** agrees with it AND a strict majority of candidates agree individually (a bimodal alternating-seats history averages into a

phantom median inside tolerance — measured in the stabilise suite — and must not fire); a tight displaced cluster (≥3 candidates, spread ≤0.03 = `RECT_STABLE_SPREAD`) reads its median where the card actually is (Pass 16 preserved); anything else falls through to the previous single-detection rule, byte-for-byte. Proof: probe-jitter --stabilise on the real scene — sources **24/48 calibration → 44/48**, read-rect sd y0 0.029→0.013, y1 0.023→0.011 (residual range is warm-up samples before history exists; on-device the preview feeds history continuously). `tests/stabilise.test.ts` (14 checks: median/spread units, settle-on-calibration, displaced cluster yields at the displaced seat with calibrationStale, inconsistent history = exact no-history behaviour). Pass 16's swamp-attempt scenarios re-run: **7/7 still pass**.

- **PART C — HELPER FINDER (PROTOTYPE MEASURED, NOT SHIPPED).** Context (researched 2026-11-09): ManaBox identifies by ARTWORK image recognition against the card-image database (its own developer/guide statements), not title OCR; open-source state of the art (CollectorVision/Milo, cardboard-scanner) = corner-dewarp → perceptual hash / embedding of whole card + art crop → nearest neighbour over ~108–116k Scryfall printings, OCR as confirm/fallback. Prototype (`tests/harness/finder.ts`, report `results/p18-finder.txt`): 64-bit DCT pHash of art crop (card 0.055–0.945 × 0.105–0.405) + whole card, Hamming-ranked over an index of ALL printings of the 22 corpus names + Way + the panels' 8 near-name distractors. **Measured: index 344 printings of 31 names (an earlier run reached 562 entries — the build silently skips failed image downloads, so the count itself is not stable); per-query latency median 5.4ms in this VM; on geometry-accurate queries the combined hash's name-level top-1 is 8/8 in EVERY degradation class (identity, ×0.5 dim, blur, ±3° rotation, foil sheen, noise), printing-level top-1 5–8/8. On the user's real attempt crops: A1 finder #1 (d=24), A2 finder #1 (d=34), A3 finder #77 — the overshoot crop (tray inside the frame) defeats the hash. The specified fusion (finder top-3 ∪ OCR candidates, corroboration = shared ≥3-letter non-stopword token) settles A1/A2 correctly ONLY with a distance gate (≤40): ungated, it FALSELY settled A3 on Way of the Thief (d=48, token 'way') — a measured wrong-auto-add path.** Full coverage costs ~110k printings × 16 B ≈ 1.8 MB of hashes but requires an offline precompute over ~110k image downloads — it cannot be built on-device or at open. **Verdict: not shipped.** The failing class is already repaired by B+D through the number channel; the finder fails the hardest attempt geometry on its own, its fusion needs a gate the prototype only just measured, and bolting a new auto-add channel + an offline index pipeline onto this pass would violate the one-defect-at-a-time rule. Recorded as the measured starting point for any future finder pass (ship form: offline-precomputed art+card hash index, distance-gated fusion, OCR still primary).
- **FIXTURES**: `tests/harness/prep_way.py` (standalone) + the same generation as prep.py section (g): way-scene / way-scene-dim / way-a4-scene (FRA 224 on the white tray at the forensics-solved geometry; A2's card nudged +0.018y and dimmed with top shadow; Stolen Stark Tech MSH 114 for A4) with rect-burst modes at his exact printed rects. Pre-fix verdicts: way-a1 uncertain-miss, way-a2 correct only via jittered burst frames (marginal), way-a3 uncertain-miss, way-a4 no-identification. **Post-fix: way-a1, way-a2, way-a3 AND way-a1-snap all correct-printing via 'collector number FRA 224'** — the channel his panels showed starved. way-a4 settles honest no-identification (its aimed strip reads 'MSH — EN L1X YIN': set

code yes, number garbled past candidacy; an earlier run read MSH 114 — hard-crop variance; never a wrong auto-add, which is the acceptance line for the A4 shape).

- **FULL BATTERY**: baselines on untouched code first (`results/p18-baseline.json`: Tier 1 119/0/8 of 127; suites 455). Final (`results/p18-final.json`): **Tier 1 124 correct / 0 wrong / 8 no-read of 132** (denominator +5: the way fixtures; +4 net correct from them, way-a4 the new honest no-read, and hand-bolrac-dim flipped no-identification → correct-printing). **Per-record diff vs p18-baseline: exactly ONE changed record among the 150 baseline records (hand-bolrac-dim, an improvement) — ZERO regressions.** Suites: **489 checks, all green** (455 + stripaim 20 + stabilise 14; exports 42 included); typecheck clean. NOT changed: detector thresholds, matcher, Rapid machine, deadline machinery, Pass 15/16/17 mechanisms. **Phone-only remainder (Andrew's)**: re-scan Way of the Mind Sculptor in the box (Rapid) — expect the aimed strip to catch FRA 224 and settle by number even when the title read fragments; the earlier open phone checks (tilted-card Snap, textured light-on-dark Snap) remain his.

Pass 19 (2026-11-09) — FULL SCANNER REWORK: image recognition becomes the PRIMARY channel (user directive verbatim: 'rework the whole scanner and build it better with the image recognition fully. Make it just like manabox please' — supersedes the Pass 18 ship-rule hold)

Architecture, three stages, each measured; all non-scanner features and all user data untouched; every proven safeguard kept as an input to the new engine (deadline settlement, rect authority + Pass 18 stabilisation, Auto Light, success-is-final, self-aiming bottom strip, title OCR demoted to a fusion channel). NO hardcoded card names/IDs anywhere in the engine.

- **STAGE 1 — INDEX + MATCHER, MEASURED OFFLINE (`client/src/finder.ts`, `tests/harness/finder2.ts` → `results/p19-finder.json` + `.txt`)**. Fingerprint: 64-bit DCT pHash (32×32 grayscale → DCT-II → 8×8 low-frequency, DC excluded from the median) ×2 — WHOLE-CARD + ART-CROP (card x 0.055–0.945, y 0.105–0.405; ignores title bar, borders, foil) — Hamming distance, combined = art + card (0–128). Query path: detected card → perspective dewarp to the canonical 250×350 card (homography solver in finder.ts; the detector yields axis-aligned rects, so the dewarp is the canonical resample of the rect — the keystone cost is measured by the rotation classes) → fingerprint → ranked candidates. **INDEX PLACEMENT (the honest engineering call, measured)**: a full ~110k-printing hash index is only 1.76 MB (110,000 × 16 B) BUT a linear scan costs a measured 3.4ms per 343 printings on this VM → ≈1.1s per query at 110k (×5 hypotheses ≈ 5.5s) — outside the 2.5s attempt deadline on-device-class hardware — and building it needs an offline pass over ~110k image downloads that cannot be rebuilt at set release. The shipped design therefore splits ManaBox's job: (1) a SERVER-SIDE visual index — the new `identifyByImage` action (vision model reads the dewarped card BY ARTWORK; EVERY proposal is verified against Scryfall server-side, so an invented name can never reach the gate; coverage = every printing Scryfall

can verify, no client download, no stale index); (2) the ON-DEVICE pHash verification — the client fingerprints the query (multi-hypothesis) and every candidate printing's real Scryfall reference image, and the MEASURED Hamming distance is what the gate settles on. A proposal whose reference art doesn't match the photo cannot settle. **Accuracy over the 343-printing corpus index (shipped finder code)**: name top-1 8/8 identity/dim/blur/sheen/noise, 8/8 rot±3, 7/8 rot±5 (top-3 8/8 everywhere); printing top-1 8/8 identity/dim/noise, 5–7/8 under blur/rotation/sheen. **Distance distributions (gate calibration)**: true matches min 0 / p50 16 / p90 36 / max 44; best impostors min 32 / p10 36 / p50 40 — the overlap band is 32–44, so the gate combines an absolute ceiling with a margin (below). Per-query latency over 343 refs: median 3.4ms (the shipped verification set is ≤24 printings → <1ms).

- **THE #77 DIAGNOSIS + FIX (measured)**: Pass 18's A3 failure (true card #77 there; rank 17 single-hypothesis over this run's index, d=52) was NOT hash coarseness: the A3 rect is seated ~4% too LOW (top 0.11 vs the card's 0.07), so the crop misses the card's top and carries tray at the bottom. Insets alone measurably CANNOT fix it (they shrink further from the card — multi-hypothesis with insets only: rank 22). The shipped hypothesis set adds UP-SHIFTS (`rectHypotheses`: base, inset 5%, inset 10%, shift-up 5%, shift-up+inset): **A3 → rank 1, d=34; A2 (dim) rank 2 → 1 (d=42 → 32); A1 stays rank 1 (d=26)**. Best-distance-per-candidate across hypotheses is the mechanism.
- **STAGE 2 — GATED FUSION (`client/src/fusion.ts`, pure; `tests/fusion.test.ts` 20 checks)**. Gate constants from the Stage 1 distributions + Pass 18 points (true A1 24 / A2 34; the false Thief settle at 48): `IMAGE_GATE_DIST` 40 (absolute), `IMAGE_GATE_MARGIN` 8 (name-level runner-up margin), Rapid solo `IMAGE_RAPID_SOLO_DIST` 30 / margin 12, evidence ceiling 56. Matrix: number hit settles as today (image never overrides a printed number); image decisive + token corroboration (Pass 18 rule: ≥3-letter non-stopword name token in the pooled OCR) → settle, Rapid auto-add eligible; image decisive alone → Snap settles, Rapid only under the strict solo gate else PRESENTED as top Did-you-mean (never auto-added); OCR decisive + image agrees/abstains → settle-ocr (the floor); OCR decisive + image decisive for a DIFFERENT name → CONFLICT (no auto-add either mode; both names in Did-you-mean, conflict labelled in diagnostics; a cheap already-landed check also blocks per-frame title wins); image present-only merges into the uncertain suggestions — including the no-read case (pure-art band: the band has no text but the card does). Printing (rule d): winning printing pins by art when its printingMargin ≥ 4, else bottom-line salvage / default with diagnostics saying which ('printing pinned by art' vs 'printings share the same art'). **A3 regression pinned in tests**: at d=48 the fusion ABSTAINS in both modes (the Pass 18 false settle is impossible by construction — corroboration is only consulted after the gate); at the post-fix d=34 it settles on the true card.
- **STAGE 3 — SCANNER REBUILT IMAGE-FIRST (`App.tsx` identifyBurst + `server/src/actions.ts` identifyByImage + `scandebug.ts`)**. The image channel starts at attempt start and runs CONCURRENTLY with the OCR reads (its latency hides inside the existing deadline envelope; settlement awaits it only with the rescue envelope's remainder). No index download at first open (server-side placement) — the only first-run cost is the first vision call, raced at 3.5s and degrading silently to the OCR floor. Diagnostics ('What the scanner read') now show: the DEWARPED card thumbnail ('Straightened card the image

channel matched'), an 'Image match:' line (top-3 candidates with art+card distance splits, gate verdict, fusion outcome incl. CONFLICT wording), plus all pre-existing evidence/stages. Viewfinder UX unchanged (guide/outline/calibration/Rapid/Snap/Take Photo/Box/Foil/Auto Light all as before). Auto Light interplay: dim bands still lift for OCR; the pHash is contrast-normalised by construction (finder suite: ×0.5 dim stays at d=0 on synthetic, corpus dim class 8/8).

- **ACCEPTANCE / BATTERY**: harness `--set all` final (`results/p19-final.json`, 156 records): **per-record diff vs p18-final = ZERO changed verdicts, ZERO regressions, 0 wrong** (the harness pipeline mirrors the OCR/number floor, which this pass did not modify; the vision proposal step needs the platform inference runtime the harness cannot run, so the image channel's own evidence is finder2.ts Stage 1 + the fusion/finder suites — stated plainly, not hidden). Suites: **531 checks, all green** (489 prior + finder 22 + fusion 20; exports 42 included); client + server typecheck clean. **Phone-only remainder (Andrew's)**: the real end-to-end test is his box re-scan on the S24 — the sandbox cannot run his camera/WebView or the live vision action against his rig lighting. Expected: Way of the Mind Sculptor settles by image (d≈26–34 class) even when the title fragments; watch the 'Image match:' diagnostics line and the dewarped thumbnail.

Pass 20 (2026-11-09) — SINGLE-DEFECT repair: the image channel never landed a single production candidate (user's 13 screenshots, every Pass 19 panel: 'Image match: no candidates (channel produced nothing usable)')

- **STEP 1 — REPRODUCE + LOCALISE (evidence first, one convicted branch).** The REAL production path was exercised end-to-end from the server side: the `identifyByImage` handler body (prompt + schema + Scryfall exact-then-fuzzy verification, verbatim) run through probe-ctx against the live platform inference channel, with payloads built exactly as the client builds them (Scryfall normal image → canonical 250×350 dewarp size, JPEG q0.82 → 39.5–40.8 KB base64; schema accepts 100..6,000,000). Results (`results/p20-production-path.json`): **Way of the Mind Sculptor — inference 6,796ms, total 8,017ms, proposal = the correct card at 0.99, verified EXACT. Way of the Paradox — 5,026ms / 5,461ms, correct at 0.99, EXACT. Chandra, Torch of Defiance — 4,173ms / 4,818ms, correct at 0.99, EXACT.** Client audit: (a) `runImageChannel` IS invoked on every attempt type (all funnel through identifyBurst) and starts concurrently at burst start — invocation exonerated; (b) **TIMING CONVICTED**: the client raced the call at **3,500ms** (`raceWithCap(api.identifyByImage(...).catch(() => null), 3500)`) — below EVERY measured round trip (fastest 4,818ms) — so the race resolved as a timeout on 100% of attempts and the code mapped it to empty candidates; SECONDARY: the fusion consult awaited the channel only with the remainder of the 4s rescue envelope (~1.5s at that point), and the reference phase loaded up to 24 reference images STRICTLY SEQUENTIALLY, so even a landed proposal could not finish verification before settlement; and any naive longer wait would have been torn down by the 8s watchdog; (c) payload 39.5–40.8 KB vs the 6,000,000

cap — exonerated; (d) verification: all three names verified via the EXACT lookup — exonerated; (e) environment: inference + Scryfall verification run correctly against the production channel — exonerated. (Two hypotheses in the dispatch brief do not exist in the shipped code: there is no REFERENCE_TEST_NAMES whitelist anywhere, and runImageChannel does not depend on the client name index at all — the proposal step is the vision action, full stop.)

- **STEP 2 — FIX (the convicted branches, narrow).** `client/src/scanflow.ts`: `IMAGE_VISION_CAP_MS` = 15,000 (was the inline 3,500); `IMAGE_WAIT_DEADLINE_MS` = 12,000 + `fusionWaitDeadlineMs(ocrDecisive)` — a DECISIVE OCR pool keeps the 4s rescue envelope (happy-path timing byte-identical; the image channel can still veto via the conflict interlock), an UNCERTAIN pool waits for the primary channel to the 12s deadline; `watchdogImageDeferral` (pure) — the watchdog re-arms once just past a registered sanctioned wait instead of tearing it down (settled attempts never defer; success-is-final untouched). `client/src/App.tsx`: runImageChannel uses the 15s cap and keeps the three call endings DISTINCT (race timeout vs action error — the `.catch(() => null)` deliberately stays outside the race because raceWithCap maps rejections to "timeout" — vs an honest empty list); the reference phase is PARALLEL (same bounds: ≤3 names, ≤8 printings each, same 1.5s per-action caps, same 2.5s per-image timeout); the fusion consult registers `{attempt, untilEpochMs}` in `imageWaitRef` during a non-decisive wait and clears it after. **DIAGNOSTICS (hard requirement)**: `ImageChannelResult` now carries `status` (`ok` / `call-failed` / `call-timeout` / `no-proposals` / `no-references` / `encode-failed`) + `visionMs`, and `describeImageChannelOutcome` (fusion.ts) prints the distinct line for each — 'timed out — the image service gave no answer within 15.0s; the title/number reads carried this scan' / 'call failed — the image service returned an error...' / '0 proposals — the artwork was not recognised' / 'N proposals (names) — 0 verified: the reference images could not be read...' / the existing distance+gate line when candidates exist — from BOTH the early publish and the settlement publish. The collapsed string is gone (describeImageChannel's empty case now returns a plain 'no candidates' and is no longer reachable for channel failures).
- **STEP 3 — PROOF.** Production path: the three real-action probes above (correct card, 0.99, exact-verified, every one of which Pass 19's cap would have discarded) — `results/p20-production-path.json`. The one hop NOT exercisable from the sandbox is the phone → deployed-action network leg itself (in-app webview + deployed process); if that leg is what differs on-device, the new panel line names it directly ('call failed' / 'timed out ... within 15.0s') on the next screenshot. NEW `tests/imagechannel.test.ts` (21 checks): caps vs the measured round trips (cap > 8,017ms worst case; ≥10s anti-regression pin), the wait decision (decisive → 4,000; uncertain → 12,000), the deferral matrix (settled/other-attempt/expired/null never defer; re-arm = remaining + 500), and all five diagnostic lines distinct with none containing the collapsed string. **FULL BATTERY: `results/p20-final.json` (156 records, run in foreground chunks after a backgrounded run's output file was lost with its overlay; merged by record id) — per-record diff vs p19-final: ZERO changed verdicts, ZERO regressions, id sets identical.** Suites: **552 checks, all green** (531 Pass 19 + 21 new); client + server typecheck clean. NOT changed: the gate constants, the fusion matrix, the OCR/number

floor, the deadline machinery for decisive scans, any non-scanner feature, any user data. **Phone-only remainder (Andrew's)**: same box re-scan — the 'Image match:' line should now show a candidate with a distance (Paradox-class cards measured d≈21–34 territory) or, if the phone leg itself fails, a line that says exactly which stage failed; one panel settles whether the channel lives.

Pass 21 (2026-11-09) — gate recalibration on device-class captures + fusion corroboration + watchdog composition (user's 4 Pass 20 panels: the channel LIVES — and three defects stand measured)

- **PANEL FACTS.** P1 Saheeli, Consul of Oversight (Rapid): image d=40, gate decisive, margin ∞, CORRECT card — presented for a tap, not auto-added. P2 Last Gasp (foil, lift ×1.3): image d=44, correct card top-ranked — abstained (44 > 40). P3 Chandra, Torch of Defiance (Snap, partial crop — bottom half only, lift ×4.0): image d=50, correct card — abstained AND the attempt then died by watchdog ('hung during: title') holding a completed image result. P4 'Cadet' (Snap, calibration rect): NO image line and NO straightened thumbnail at all.
- **ITEM A — RECALIBRATION, MEASURED FIRST.** NEW `tests/harness/finder3.ts` (+ `jpeg.py`) → `results/p21-calibration.json` / `.txt` / `-split.json`: 14 cards (the panel cards at their panel printings + the finder2 corpus) × 14 device-class degradation classes (identity; lift ×1.3/×2.0/×4.0 multiply-clamp = the ocr.ts lift math; foil sheen; sheen+lift; JPEG q82 round-trip via PIL = the payload quality; sensor noise; combos; partial crops 55%/70% + lift) through the PRODUCTION fingerprint path (SMALL reference images as runImageChannel loads, canonical 250×350 dewarp, multi-hypothesis best distance; 390-printing index). Baseline + final battery both run (see PROOF). **Distributions: FULL-CARD captures (154 queries): true p50 20, p90 ≤38 per class, max 42; best-impostor p10 34–38. Solo rule d≤44 + margin≥8 → 145 true settles, ZERO false settles (margin is the separator: 0 false at every ceiling 40–56 once margin ≥8 is required; the single margin-0 false settle sits at margin 2). PARTIAL crops (42 queries): a DIFFERENT population — true 40–64 while impostors sit at 36–46 and the true card ranks 3rd–39th (the stretched dewarp puts the text box in the art window); ranking collapses and NO distance gate separates them.** The phone's true points (40/44/50) run hotter than the same-shape composites (22/12/46) — real sensor/ISP adds distance composites don't — so ceilings are set from the phone points where they exist. **SHIPPED CONSTANTS (finder.ts): IMAGE_GATE_DIST 40 → 44 (covers P1+P2; 0 measured full-card false settles), IMAGE_GATE_MARGIN 8 (unchanged — it was doing the separation all along), NEW IMAGE_BAND_DIST 56 / IMAGE_BAND_MARGIN 4 (the two-tier rule: 45–56 settles ONLY with Item B corroboration; = the old evidence ceiling), IMAGE_RAPID_SOLO_DIST 30 / MARGIN 12 (unchanged — P1 is repaired by corroboration, NOT by widening the solo bar).** Resulting false-settle rate on the calibration set: 0 across full-card impostor fixtures under the shipped rule; partial-crop queries are excluded from solo settling by construction (their distances land in the band or beyond, and their wrong-top risk is carried by corroboration + the ≤3-name production proposal pool,

which the 390-name corpus sweep overstates — stated plainly: on a partial crop the fingerprint alone is weak evidence, and the app now treats it exactly that way).

- **ITEM B — FUSION CORROBORATION (fusion.ts).** `corroborationFor` names three sources: (i) TOKEN — meaningful overlap only: a ≥4-letter name token in the pooled reads, or ≥2 distinct name tokens (Pass 19's any-single-≥3-letter rule is what let 'way' corroborate the Thief; it is gone); (ii) MATCHER-TOP — the pooled matcher's own top pick IS the image top-1; (iii) BOTTOM — the caller scores the pooled bottom text against the candidate's printings (existing salvage rule: set+number+artist) and passes `bottomConsistent` (computed in App ONLY when (i)/(ii) haven't already corroborated, one raced printings fetch). Strength: (iii) is strong alone; (ii) is strong ONLY with (i) — a fragment pool crowning a wrong card top by itself is the A3 trap, not agreement; (i) alone is weak. **Rules shipped in fuseChannels: decisive tier + ANY source → settle both modes on the standard bar (P1 repaired: matcher-top agreement auto-adds in Rapid); band tier + strong → settle both modes; band + weak token → Snap settles, Rapid presents; band + matcher-top alone → present (A3 guard); decisive + uncorroborated → Snap settles / Rapid needs the unchanged solo bar; conflict interlock now also fires for band + bottom-corroborated disagreement, and band alone never vetoes a decisive title read.** Four existing fusion.test.ts assertions were deliberately revised and re-pinned with reasons in the test comments (A3 d=48: abstain → present-only; A3 true-card d=34 fragment case: settle → present, with a new 'mind'-fragment variant that settles; beyond-evidence fixture 52 → 60 since 52 is now band evidence) — in every case the anti-settle substance (never auto-add, never Snap-settle) is preserved and asserted.
- **ITEM C — WATCHDOG COMPOSITION (scanflow.ts + App.tsx).** Diagnosis of P3: the Pass 20 deferral was armed ONLY while the image wait was pending; in P3 the channel had already reported (fusion even published its abstention) and the attempt sat in its bounded, clock-capped rescue pass when the fixed 8s watchdog fired — no shield covered post-completion settlement. Fix: `watchdogShieldDeferral` (pure) generalises the deferral over ALL attempt shields; identifyBurst registers a shield at the fusion consult covering the remaining clock envelope + 2.5s settlement slack, and the channel's completion handler records the completion (keyed by attempt) and extends a shield by IMAGE_SETTLE_GRACE_MS (10s — the remaining pipeline is deadline-bound by construction: cappedRead abandons a never-returning OCR job at its cap and the Pass 15 generation cancels stale reads, so settlement inside the grace is guaranteed unless the pipeline itself is wedged past every deadline, in which case the expired shield lets the normal recovery run). Zombie OCR stages needed no new mechanism — the caps + generation already bound them; what was missing was the watchdog knowing. The rescue pass itself is UNCHANGED (zero OCR-floor risk).
- **P4 DIAGNOSIS (localised in code).** The Cadet attempt DID run the channel (every Snap funnels through identifyBurst) — but both the image line and the dewarped thumbnail were published ONLY from the channel's async completion path: the thumbnail inside `imagePromise.then`, guarded by `guard.isActive(attempt)`, and the line at the fusion consult, which publishes NOTHING when the wait itself times out (imageResult still null) — a slow phone→server leg (the one hop Pass 20 could not exercise) settles the attempt first, the

late completion is discarded by the guard, and the panel shows silence. Reproduced by construction in the decision code; the panel's own numbers fit it (match 1ms, no dewarped caption, attempt settled ~2.7s of stage time while the fusion wait ran to its deadline). **Fix: the line now exists on EVERY panel** — the dewarped thumbnail is published synchronously at attempt start (same local dewarp, no channel dependency); the fusion-wait timeout publishes an explicit 'timed out — no answer within the Ns wait' line immediately (a late real result still replaces it while the attempt is active); `finish()` and the watchdog snapshot substitute `IMAGE_NOT_RUN_LINE` ('not run (attempt settled before the image channel reported)') whenever no outcome was ever published (`ensureImageLine`, fusion.ts).

- **PROOF.** Baseline before any edit: full battery on untouched code = 156 records, ZERO verdict diffs vs p20-final; all 21 suites green (552 checks). **FINAL (ship): full battery `results/p21-final.json` — 156 records, per-record diff vs p20-final: ZERO changed verdicts, ZERO regressions. Suites: 595 checks, all green (552 Pass 20 + 42 new pass21 + 1 net fusion revision); client + server typecheck clean.** NEW `tests/pass21.test.ts` (42 checks): the shipped constants vs the phone's exact panel distances (44 ≥ P2; band ≥ P3's 50; solo < 40 so corroboration — not width — repairs P1); the Thief hard floor in four shapes (token-only, uncorroborated, and fragment-crowned, both modes — never settles); every Item B branch ((i) ≥4-letter / two-short-token / single-short-token-negative, (ii) strong-with-tokens vs alone-not-strong, (iii) bottom strong; P1-shape auto-add; P2-shape Snap settle; band+strong both modes; band+weak Snap-only; band+none present; conflict refinements); the Item C P3 replay (completion shield defers the 8s watchdog, expired shield recovers, settled never defers, other-attempt never defers; a never-returning title read abandoned at its cappedRead cap while fusion still judges); the P4 line guarantees. Calibration evidence persisted (finder3 + p21-calibration.json/.txt/-split.json). Full battery re-run + suites + typecheck: see the PROOF line appended at ship time below. **Phone-only remainder (Andrew's)**: the four panel cards re-scanned — P1 should now auto-add in Rapid; P2 should settle by image in Snap; P3's shape should reach a normal panel (never a watchdog reset) and settle if the title/bottom gives any agreement; P4's shape must show SOME image line (a distance, a timeout, or the explicit not-run line). The phone→server network leg and true on-device distances remain his to confirm — the panel now names whichever hop fails.

Pass 22 (2026-11-09) — Rapid re-arm + same-card re-scan, feedback at settle, Auto Light metering verdict + second lifted window, Manual light control (user directives verbatim: 'with rapid fire it will not scan the same card again until I slide another card in... the rapid fire needs to recognize quickly to scan another right after and make sure when it scans and the sound of the scan goes off the notifications are fast as the card scans. Make sure the auto light works better, it will make it bright in the low light and turn off if the light is good. A

manual light would also be nice... Some dark images I sent you.')

Screenshot facts first: his re-sent Saheeli panel is byte-identical to the Pass 20-verdict panel (OLD gate text '(d=40 ≤ 40)') — Pass 21's Saheeli auto-add remains UNPROVEN on-device. Depthshaker Titan (new build): image channel timed out at the 12s wait (honest line working), OCR 'BEE - -' on a well-aligned legible calibration-rect capture at Auto light ×2.2. Kavu Scout (new build): the two-tier band rule working as shipped (d=50 presented, no corroboration existed).

- **ITEM 1 — RAPID RE-ARM. LOCALISED CAUSE**: the RapidMachine's post-settle `handled` phase released ONLY when a sample differed from the handled card's signature by the same-card threshold (14; 20 in box mode). In the fixed rig a card slid out and back in reproduces the handled signature almost exactly, so condition (change) never fires for the SAME card — while a DIFFERENT card always differs enough, matching his report word-for-word ('will not scan the same card again until I slide another card in'). A removal shorter than one watcher tick was also invisible (no sample ever showed the empty view), and after a Did-you-mean pick, resumeRapid re-entered `handled` with the scanned card's signature. **SHIPPED RULE (rapid.ts `handled` case)**: a settled card re-arms on any PRESENTATION EVENT — (a) CHANGE: scene differs from the handled signature by the same-card threshold (old rule); (b) ABSENCE: scene matches the adopted empty baseline; (c) MOVEMENT: consecutive-sample diff exceeds `rearmMovementThreshold` (9 = the stability bound) — the slide itself is observed even when the card returns to the identical spot. After re-arming, the normal stability rule fires the scan after stableNeeded (2) still samples at the watcher's new 250ms cadence (was 300ms) — a stable re-presentation of ~0.5–1s counts as a new presentation. A card that NEVER moves fires none of (a)–(c) and is never re-scanned (no duplicate loop); the viewfinder tap (forceScan) remains the instant deliberate re-scan. **TESTS (tests/pass22.test.ts, machine sims at the shipped 250ms cadence)**: (i) unmoved settled card: 0 re-scans over 60 ticks, phase stays handled; (ii) same card out+in via movement-only (removal never sampled): scans again, 1000ms from the slide's first motion frame / 500ms from the re-presentation holding still; (ii) same card out+in via sampled absence: scans again at 500ms, and the empty view itself never scans (baseline guard); (iii) swap A→B: 500ms; (iv) back-to-back A→B→C: 500/500/500ms settle→next-start. All ≤1000ms. Single-attempt identification is untouched (the machine change is post-settle only).
- **ITEM 2 — FEEDBACK AT SETTLE. LOCALISED**: in runRapidScan's auto-add path the beep/banner (`showFlash`) was chained off `addToList(...).then(...)` — feedback waited on the JPEG encode + the scan-list server round-trip; the same await-then-flash shape sat in selectMatch, trySlowRead and the watchdog rescue. **FIX**: `settleWithFeedbackFirst` (scanflow.ts) — feedback is invoked synchronously FIRST, persistence is started only after, and the returned promise is persistence's own (the save's completion upgrades the already-showing banner with its Undo target, silently — no second beep; a failed save still surfaces its hint and can never swallow feedback that already fired). Wired into all four Rapid settle paths. **TESTS**: event order is exactly [feedback, persist-start] synchronously; feedback timestamp =

settle timestamp (0.01ms into a 30ms save, vs the whole round-trip before); rejection propagates with feedback fired exactly once.

- **ITEM 3 — METERING REWORK, MEASURED. (a)** The lift decision already meters the TITLE ZONE of the crop actually read (identifyBurst meters preps[0].card — the calibration crop in box mode) — verified in code, unchanged. **(b) GAIN CURVE — probe re-run (`tests/harness/probe-lift.ts` → `results/p22-probe-lift.txt`)**: dim-recoverable p90 33–56 → raw garbage/nothing, lifted EXACT @1.000 in 4/4 (gains ×3.13–×4.00 from the shipped curve, target p90 175 cap ×4); crushed p90 8–21 → lifted junk tops 0.49–0.78 (never ≥0.82 — the reason Auto keeps the crush gate); uniform-dim p90 38 → raw nothing, lifted EXACT. The curve produces the proven exact-read gains across the recoverable band, INCLUDING the Depthshaker ×2.2 class (p90≈80 → ×2.19, mapping the zone to p90≈175 like every proven point) — gain was NOT his failure. **(c) COMPOSITION FIX**: the lift previously applied to the STANDARD window only; Pass 18's forensics showed fragmentation is window-sensitive, so a dark title sitting low in the window got a brighter version of the wrong slice. Frame 1 now pools lifted variants of BOTH the standard and standard-low (y 4–12.5%) windows under the same caps (App + harness mirror); the lifted rescue stage resolves through the same mode gain. **(d) VERDICT LINE**: every metered attempt now publishes its light decision in the debug block (label renamed 'Light:'): 'light good — no lift (title zone p90 = N)' / 'zone dark — lifted ×G (title zone p90 = N)' / 'manual light ×G (...)' / the crushed floor line — before this, bright attempts published nothing, so no screenshot could show the good-light decision. **GOOD-LIGHT OFF, asserted**: `liftForMode` returns null for bright bands in Auto/On, and the final battery shows ZERO lifted title reads on every bright-titled fixture — the good-light path is the pre-Pass-17 path. Depthshaker's own pixels remain phone-only (his panel's thumbnail is not the read band); his retest should show the verdict line naming the decision.
- **ITEM 4 — MANUAL LIGHT (new)**: the Light chip cycle is now Auto → On → **Manual** → Off (Auto still the first-run default; the Pass 17 cycle assertions in autolight.test.ts were revised for the new state, recorded here). Manual shows a thumb-reachable level control under the viewfinder (Low ×1.5 / Med ×2 / High ×3 / Max ×4, `MANUAL_LEVELS`), persisted with the mode (`manabox-light-mode`, `manabox-light-manual-level`). The level drives the SAME lift pipeline deterministically via `liftForMode` / `liftSpecWithGain`: the chosen gain applies to title + bottom reads regardless of metering (bright bands included) and overrides the crush skip (explicit user override); metering still runs and the verdict line still prints the zone's p90 next to the user's gain. Hardware: entering Manual (and camera restarts in Manual) forces the torch on where supported and raises exposure compensation to max where exposed — capability-guarded exactly like Auto's layer. **HONEST LIMITATION, stated in the UI**: where the torch reports absent (the Muse in-app browser), the control is software exposure only and the panel under the levels says 'Flashlight unavailable in this browser — exposure boost only; open in Chrome for the torch.' **TESTS**: manual gain is exact for bright/dim/crushed/unmetered bands; level Max produces exactly a ×4 lifted spec; persistence parsers (mode/level, garbage → Auto/Med); cycle includes Manual.
- **PROOF**: baseline on untouched code first (`results/p22-baseline.json`: 132 identification runs, 123 correct / 0 wrong / 9 no-read; per-record diff vs p21-final = exactly ONE flip — the

Esika photo correct-name → no-id, the documented VM-scheduling class, and it flipped BACK in the final run). **FINAL (`results/p22-final.json`): 125 correct / 0 wrong / 7 no-read of 132 — diff vs p21-final = exactly TWO changed records, both improvements of previously-no-read cases (Pyroblast photo no-id → correct-name — the Fix-6 read-level case landing in-pipeline this run; Tier-2 4499 guide no-id → correct-name), ZERO regressions vs p21-final AND vs the p22 baseline.** Suites: all 23 green — 595 Pass 21 checks + 38 new `tests/pass22.test.ts` + the revised autolight cycle checks; client + server typecheck clean. **Phone-only remainder (Andrew's)**: (1) Rapid — settle a card, slide THE SAME card out and back in: it should scan again (~1s after it settles); slide a different card in: ~0.5s to the next read; the beep/banner should land WITH the settle, not after a pause. (2) A dark box card: the panel's Light line should read 'zone dark — lifted ×N (title zone p90 = …)' and a well-lit card 'light good — no lift'. (3) Manual: cycle the Light chip to Manual and try High/Max on the dark card — in the in-app browser expect exposure boost only (no torch, and the control says so); in Chrome the torch should light. (4) Still open from Pass 21: Saheeli auto-add in Rapid (his only panel was a pre-Pass-20 build).

Pass 23 (2026-11-09) — acceptance rework (user's rule: '90% should be accepted'), distance × clarity matrix, blown-band guard, multi-frame consensus (user's 4 panels: ONE card, Kavu Scout, Rapid + Foil + Auto-add ON — image DECISIVE at d=40 and d=42, presented as 'Kavu Scout · 90% match', never auto-added; title band uniform grey / blown white; same card measured d=50 → 42 → 40 across attempts)

- **ITEM A — ACCEPTANCE, LOCALISED.** The fusion's decisive branch still required, for an UNCORROBORATED image verdict in Rapid, the Pass 19 solo bar (d ≤ 30 / margin ≥ 12) before settling; corroboration (Pass 21) could never fire for this card because its OCR channel yields only garbage from bleached bands — so decisive verdicts at d=40/42 fell to present-image forever. The displayed '90% match' was also hardcoded for every presented image candidate, so the UI claimed the auto-add class while withholding the auto-add. **SHIPPED RULE (fusion.ts fuseChannels)**: a DECISIVE image verdict (d ≤ 44 + margin ≥ 8 — the rule that measured 145 true / 0 false settles on the finder3 device-class set) settles in BOTH modes on the image channel's own authority — Rapid auto-adds, Snap settles — no corroboration, no second bar; corroboration is still computed and RECORDED (the panel names agreement when it exists). The 45–56 band keeps its Pass 21 agreement requirement unchanged. The conflict interlock is unchanged in substance: a DECISIVE OCR read naming a different card still conflicts (only garbage/absent OCR lost its veto). **DISPLAY HONESTY**: `imageSuggestionScore` maps gate tier to the shown score — decisive = 0.90 (and decisive settles, so a 0.90 entry can now only appear as the image side of a conflict, where two decisive claims genuinely compete); band slides 0.86 → 0.72 across 45–56; beyond evidence 0.6; a degraded-guard demotion (Item B) scores 0.86, never 0.90. The panel line for a Rapid image settle reads '→ auto-added by image' / '→ auto-added (image + corroboration)'; Snap

reads '→ settled by image…'. **THIEF FLOOR, DEMONSTRATED**: the Way of the Thief A3 fixture sits at d=48 — BAND tier, 4 points beyond the decisive ceiling — and presents in both modes (pass23.test.ts + fusion.test.ts, re-pinned this pass); its matrix appearances agree (top-1 'Way of the Thief d=38, margin 6' in one Kavu cell — margin-short, suggestion only). Every conflict fixture still conflicts. Distance/margin of the floor vs the rule: decisive needs d ≤ 44 AND margin ≥ 8; the Thief's measured shapes are (48, band) and (38, margin 6) — outside on both axes independently.

- **ITEM C — BLOWN-BAND GUARD (scanflow.ts `classifyBandBleach` + ocr.ts)**: bands whose sampled pixels are uniform — ink share (pixels deviating >30 from their row median) < 0.004 AND luminance stdev < 16 AND mean ≥ 60 — classify PRE-OCR as no-text evidence: `blown-white` (mean ≥ 235; the P-c shape, both bands pure white under the Light chip) or `uniform` (the P-a shape, empty grey gradient, mean ≈129). Title and bottom band reads short-circuit at 0ms with the diagnostics reason 'band bleached — no text', so bleached bands stop consuming 1–2s of OCR and stop injecting the garbage tokens ('- . 7 RE GREE SRR EES…') that pooled as pseudo-evidence ('thin margin'). The mean ≥ 60 floor is deliberate: uniform near-black is the crushed/too-dark population owned by the Pass 17 Auto Light floor, NOT bleach. Proven on fixtures (pass23.test.ts): pure white → blown-white; grey gradient → uniform; genuine low-contrast strokes (ink share 0.166), a strong title, a half-black/white edge (stdev ≈118), and a near-black band all classify NOT bleached and still read.
- **ITEM B — DISTANCE × CLARITY MATRIX (`tests/harness/matrix.ts` → `results/p23-matrix.json` [pre-fix], `p23-matrix2.json` [lift + interlock], `p23-matrix3.json` [all fixes])**: full pipeline (detection → dewarp → fingerprint vs a 14-name / 60-printing corpus incl. colour-twin impostors → OCR subset → the production fuseChannels) over 6 cards (Kavu Scout; Saheeli, Consul of Oversight FRA; Way of the Mind Sculptor FRA; Chandra, Torch of Defiance — planeswalker; Lightning Bolt; Last Gasp) × distance (card height 25/40/55/70/85/100% of frame) × clarity (sharp, defocus σ small/med, motion short/long) × lighting (bright, dim ×0.35, shadow-over-title, glare) × finish (non-foil, foil sheen) = **1,440 cells** (OCR run on a 168-cell subset spanning every axis). **PRE-FIX HEADLINE: 13 wrong settles in BOTH rankings.** Failure map (matrix2 axes): detection is near-total in bright/shadow/glare (100%) but collapses in the synthetic dim class (64% detected / 60% dewarp-ok — the ×0.35 multiply is harsher than a real dim scene, where the phone raises exposure; recorded as a matrix-class limit, not a phone measurement); image top-1 by distance 95% (25%) → 74–76% (85–100%, full-bleed cards clip their borders: dewarp-ok 80% at 100%); weakest combined class = motion-long × 85% (38% auto-add); foil costs ~13 points of auto-add vs non-foil (65% vs 78%). **TWO CONVICTED WRONG-SETTLE CLASSES + FIXES**: (1) **Collector-number garbles** (4 cells): at small/blurry sizes the bottom line's digits garble into REAL foreign (set, number) pairs (Lightning Bolt → 'MIC 80' Bastion Protector; Last Gasp → 'TRC 10' Balance of Terror; Mind Sculptor → Academic Ascent) and the number rescue short-circuited BEFORE fusion, settling wrong while the image channel was DECISIVE for the true card (d=22–26, margins 14–18). **FIX — the number interlock** (`numberHitInterlock`, fusion.ts + App.tsx): a number hit whose title pool is not itself decisive is HELD through a bounded image wait (fusionWaitBudgetMs caps it at 8s) and settles UNLESS the image top-1 is decisive for a different card — then the scan

presents a conflict; decisive-title scans keep the historical number-wins-on-the-spot behaviour, and the watchdog-timeout rescue path (Pass 9 E2) is untouched. (2) **Dim × blur colour-twin impostors** (9–12 cells): in dim cells, quantisation at capture collapses red colour-layout twins (Kavu Scout → Chain Lightning / Lightning Bolt at d=42–44; Last Gasp → Lightning Bolt at d=34–36) INSIDE the decisive gate at margins 8–10 and cross-frame medians 42–50 — while ZERO bright-cell impostors reached decisive at any blur. Two-part fix: **(a) exposure-normalised query pooling** — when the dewarp meters dim, gain-lifted copies of every query crop are fingerprinted alongside the raw set (the Pass 17 lift, via `liftCanvasCopy`); measured effect is honest and modest: true-match distances tighten in dim light (22→20, 34→32, one impostor top-1 corrected) but the wrong cells are byte-identical — pHash bits are median-relative by construction and multiplication cannot restore quantised-away levels, so the lift is kept for the tightening and is NOT the safety fix; **(b) the degraded-capture guard** (`captureDim` in fuseChannels, `IMAGE_DEGRADED_MARGIN` = 12 in finder.ts): on a capture that meters dim/crushed (full-card meter, mode-independent), a decisive verdict must ALSO clear margin ≥ 12 and, when burst frames measured one, a cross-frame median ≤ 44 — failing either, the candidate is PRESENTED (scored 0.86, panel line says why: 'dim capture: decisive needs margin ≥ 12 and stable frames'). Clean captures keep the Item A rule byte-for-byte, and the user's own panels are that class (bright box, margin ∞). **POST-FIX HEADLINE (matrix3): WRONG settled = 1 of 1,440 (consensus) / 2 (single-frame); auto-add 977, suggestion 246, no-read 217.** The guard's cost is visible and accepted: ~56 former auto-adds (mostly dim decisive at margins 8–11) now present for one tap instead. **THE ONE REMAINING WRONG, named**: Lightning Bolt at 70% / defocus-med / bright — the garbled 'MIC 80' number pair won because the image top-1, though the CORRECT card, was TIED with another name (d=42, margin 0), so no channel held decisive evidence and the interlock's threshold was not met. Widening the interlock to any top-1 disagreement was considered and rejected on evidence: on partial crops the image top-1 is routinely an impostor (Pass 21: true card ranks 3rd–39th) while the number hit is the proven rescue — a wider veto would convert correct number rescues into conflicts to chase a 1/1440 cell. **MEASURED LIMITS (reported, not dropped)**: 25% height + long motion blur and 85% + motion-long cells largely no-read/suggestion (the information is not in the pixels); dim-class detection at 64% under the synthetic ×0.35 (a real phone's auto-exposure is not modelled); full-bleed 100% cards lose border-locked dewarp accuracy.

- **ITEM D — MULTI-FRAME CONSENSUS (`consensusRank` , finder.ts) — SHIPPED, on the numbers.** Per burst frame (≤3) the query is fingerprinted and ranked separately; each candidate carries its BEST (minimum) distance into the gate, with its cross-frame median + spread kept as the stability check (the degraded guard consumes the median; describeImageChannel flags 'frames unstable (best d=X, median d=Y…)' when median − best ≥ 10, else notes 'best of N frames'). Before/after on identical matrix cells: true-match distance single p50/p90/max = 28/38/56 (n=1226 top-1) vs consensus 28/38/54 (n=1219) — the tail tightens modestly and top-1 count dips slightly (an impostor's best can also improve), BUT the gated outcomes are what ship: **auto-add 977 consensus vs 941 single-frame, wrong 1 vs 2** — consensus's extra top-1 losses land as suggestions under the gate + guard, never as wrong settles, and its best-of rescues real settles single-frame misses. The user's motivating

evidence (same card d=50/42/40 across attempts) is attempt-to-attempt variance; the shipped mechanism harvests the within-burst part of it and the panel now shows the spread when it matters.

- **PROOF / ACCEPTANCE.** Baseline on untouched code first: typecheck clean; Tier 1 = the persisted p22-final (125/0/7 of 132). An untouched-code harness re-run differed from p22-final on exactly 2 records (one Tier 1, one Tier 2), both attributed to run-to-run nondeterminism under CPU contention (the flips did not reproduce consistently and the Tier 1 record scored correct again in BOTH post-change runs). **FINAL BATTERY (`results/p23-final2.json` , run alone on the final code): Tier 1 = 125 correct / 0 wrong / 7 no-read — IDENTICAL to p22-final; per-record diff across all 156 records = exactly ONE change, the Tier-2 screenshot record shot-1000044499_4_pxre-guide (correct-name → uncertain-miss), which has now produced three different verdicts across three runs on BOTH codebases (untouched baseline included) and is never scored; ZERO regressions, and no battery record's verdict changed from the Item A acceptance rework (the battery exercises the OCR/number floor, which this pass touched only in the bleach guard — its fixtures' bands all classify as genuine text).** Suites: **683 checks, all green** (633 Pass 22 + 49 new pass23 checks [Item A fixtures replicating P-a/P-b exactly — decisive d=40/42 + garbage/empty OCR → Rapid auto-add, Snap settle; band d=50 → suggestion; decisive-OCR conflict holds; Thief floor; display scores; bleach classifier shapes; consensus incl. the Kavu 50/42/40 → best 40 / median 42 case; number interlock ×5; degraded guard ×8] + 1 net revision; fusion.test.ts and pass21.test.ts assertions revised ONLY where Pass 23's rule supersedes them, each re-pinned with its reason in the test comments — the anti-wrong-card substance is intact in all). Client + server typecheck clean. **Phone-only remainder (Andrew's)**: (1) Kavu Scout in the box, Rapid — should now AUTO-ADD (the panel's image line ends '→ auto-added by image'); (2) Saheeli auto-add remains unproven on-device (every panel of her predates Pass 21); (3) a genuinely dark card — dim decisive matches now need margin ≥ 12 + stable frames, so the worst-lit scans may present for one tap where they previously would have (wrongly) settled — watch for the 'dim capture' wording; (4) image-rescued scans still cost the ~5–8s server round trip (Pass 20) — structural to the web version; the on-device fingerprint index remains the local-app phase's job, as parked.

Pass 24a (2026-11-10) — continuous calibration + steady light (Items A+B only; tokens/lands, image latency and suggestions are Pass 24b, untouched here)

Prompted by Andrew's 2026-11-10 verdict on Pass 23 (working much better — first on-device image auto-add, Borg Queen with exact printing): (1) he must recalibrate ~every other card in his fixed box; could the app keep calibrating continuously / re-calibrate when a new card slides in? (2) the auto light pulses 'like a heartbeat', so a capture can land in a dark phase.

- **ITEM A — LOCALISED CAUSE.** The stored box calibration is ONE static rect, while cards seat with a few % of position variance per card. The pre-24a effective-rect rule made it worse, not

better: detections are shadow-biased in the box (Pass 18: raw detection y0 sd 0.034, range 0.077 of frame height on his rig), so a displaced card's detection repeatedly "agreed" with the calibration inside the 4%/6% tolerances — or Pass 18 stabilisation's majority vote, whose history fills with calibration copies once calibration reads begin, self-confirmed the stale position; and on detection-miss frames the attempt fell back to the static calibration wherever it pointed (his Cadet panel read a near-full-frame calibration rect (0.04,0.00)-(0.98,0.93)). The stored position NEVER learned from detections.

- **ITEM A — FIX (his model: track continuously, re-lock on slide-in).** New pure tracker in detect.ts (`seedTrackedRect` / `observeTrackedRect` / `freshTrackedRect`, `TrackedRectState`): seeded from the stored calibration; folds every live frame's detection into a per-edge median of the last 5 samples (≤10s old). A sample > 0.07 from the lock on any edge only ARMS a re-lock; a second far sample agreeing with it confirms the move (one jitter spike, rig range 0.077, can never drag the lock). `prepareCardFromFrame` gained `trackedRect` + a `detectedRect` on every return (this frame's own detection, the tracker's feed): in Box mode, when the tracked rect matches this frame's detection within 0.03 on every edge the attempt reads the TRACKED rect (`rectOrigin: "tracked"` — the median is the lower-variance estimate of the same position); a larger gap means the card moved / the lock is still swinging → the attempt reads THIS frame's detection (`"frame"`, no calibrationStale nag — displacement is the normal case now); a detection miss falls back to the tracked rect (`"tracked-fallback"`), then the calibration seed. App.tsx feeds the tracker from the single `prepareCardCanvas` wrapper every live box prepare uses (Rapid watcher, preview, taps, bursts), seeds it on Box enable / recalibrate, and the viewfinder overlay prefers the live lock ('TRACKING CARD (AUTO)'). Manual 🎯 Recalibrate remains as override (it now RE-SEEDS the tracker). Diagnostics: `describePrepRect` prints 'tracked (auto) (x0,y0)-(x1,y1)' / '... detection missed this frame — last tracked position' / 'calibration (seed) (...)' (the seed label replaces the old 'calibration (...)' string; one rectauthority assertion re-pinned). When no tracker is offered (harness battery, stabilisation suite), prepare behaviour is the pre-24a logic verbatim.
- **ITEM A — PROOF.** New `tests/harness/trackedsession.ts` → `results/p24a-tracked-session.json`: 6-card box session, calibration recorded ONCE on card 1, seats displaced +4.1/−3.3/+5.3/−4.5/+3.1% of frame height with shadow-gradient detections with rig-class jitter; attempts judged against per-seat oracle reads. BASELINE (pre-24a call shape): 5/6 attempts DEGRADED, 5 stale-calibration reads, all 6 attempts read the card-1 calibration, mean oracle error 0.033 — his 'every other card' reproduced. TRACKED: **0/6 degraded, 0 stale reads, mean oracle error 0.010 (worst 0.018), ZERO manual recalibrations.** `tests/tracking.test.ts` (23 checks): seed semantics, jitter-median stability, arm-then-confirm re-lock, full session (tracked → frame-on-slide → re-locked tracked), fallback ladder, describe strings. Pass 16 swamp scenarios stay 7/7.
- **ITEM B — LOCALISED CAUSE.** `evaluateAutoLight` was driven directly by every metering tick (preview ~800ms, Rapid watcher ~900ms, per attempt) against a SINGLE threshold (title p90 < 150 = dark) on a SELF-REFERENTIAL measurement: engaging torch/exposure brightens the very zone being metered, so the next tick read 'bright' → restore → dark again → re-engage — a limit cycle with no hysteresis and no dwell, i.e. the heartbeat he saw, and a capture could

land mid-cycle in the dark state. Second actor: `capturePreparedBurst`'s dark-frame retry raised exposureCompensation mid-burst without registering it, fighting Auto's restore logic (and leaving EC maxed for the session).

- **ITEM B — FIX (decide once per attempt, latch, hysteresis + dwell).** New latch in autolight.ts: evidence bands DARK < 130 / BRIGHT ≥ 175 (the old 150 threshold sits inside the dead band — a self-lit zone metering ~150–175 can no longer release the light that's lighting it); `stepLightLatch` (preview) flips only on decisive evidence and only after a 4000ms minimum dwell; `beginAttemptLight` runs ONCE at attempt start (in identifyBurst, on the attempt's own metering): a decisive dark verdict engages + LATCHES (`attemptDark`), after which every step/begin is a strict no-op until `endAttemptLight` (identifyBurst finish() + both callers' finally for abort/watchdog paths). Preview metering may keep running but can never mutate an in-flight attempt. The burst dark-retry's EC raise now registers via `noteExternalLightApplication` so the latch and hardware state stay coherent. Panel line unchanged in wording ('zone dark — lifted ×N' / 'light good — no lift') and now states the LATCHED verdict. On/Manual modes untouched.
- **ITEM B — PROOF.** `tests/lightlatch.test.ts` (24 checks): evidence bands; heartbeat oscillation p90 138↔166 ×120 steps → ZERO applications; engaged + self-lit 155↔172 → ZERO releases; dwell refuses a 1s-later flip, allows after dwell; full 65↔205 swings at 1Hz → applications ≥ dwell apart; exactly ONE application across a simulated attempt with mid-attempt bright/dark/neutral steps all no-ops; attempt-start inherits preview dwell; external (burst-retry) applications register and hold.
- **PROOF / ACCEPTANCE.** Baselines on untouched code first: unit suites 683 PASS / 0 fail (identical count to Pass 23); battery `results/p24a-baseline.json` Tier 1 = 125/0/7 of 132, identical to p23-final2. FINAL: suites **730 checks, all green** (683 + 23 tracking + 24 lightlatch); Pass 16 swamp 7/7; **battery `results/p24a-final2.json` per-record diff vs the untouched-code baseline = ZERO changed records across all 156 (Tier 1 125/0/7; Tier 2 unchanged)**. One intermediate contended run (battery parallel with typecheck) flipped two Tier 1 number-rescue records (num-dawnhart-mentor-box40/-calib) correct → no-identification; re-run isolated (`p24a-dawnhart-recheck`: 2/2 correct) and the quiet full re-run shows zero changes — the established VM-scheduling class, not a code effect (the battery never offers a tracker, and prepare's no-tracker path is unchanged). Client + server typecheck clean. **Phone-only remainder (Andrew's)**: (1) a 6+-card box session WITHOUT touching 🎯 Recalibrate — panels should read 'Rect: tracked (auto)' and the viewfinder outline should read TRACKING CARD (AUTO); (2) the viewfinder light holding steady — no heartbeat pulse — with the panel's Light line stating one latched verdict per scan; (3) Saheeli auto-add STILL unproven on-device (carried from Pass 21/23).

Pass 24b (2026-11-10) — tokens + basic lands scan (Item C), image-latency audit (Item D), 'Use this match anyway' (Item E)

Andrew's asks: tokens and basic lands are not scanning though 'they are on Scryfall'; add suggestions + fixes. Production evidence (2026-11-10 ~08:47–08:54 EST): E4 Cadet token attempt

(image service timed out, bottom abandoned); E5 foil full-art Plains whose bottom read was essentially perfect — 'L 0384 CN FRA EN ALAYNA D' — yet settled 'no plausible name'; image service timed out on 4 of 6 panels that session. Pass 24a (tracking + latched light) untouched by this pass.

- **ITEM C(i) — TOKENS, DIAGNOSIS (all live-verified against Scryfall).** The Cadet card is catalogued as **TFRA #1** (token set), NOT FRA/1 — its printed bottom line reads 'T 0001 FRA • EN' (artist Elizabeth Peard on the card; Scryfall credits Elizabeth Peiró), type line 'Token Creature — Wizard Soldier'. FRA/1 is a different card (Emrakul, the Exigent Doom) — a live wrong-settle hazard for any naive T-stripping lookup. Three coverage gaps convicted: (1) Scryfall `/catalog/card-names` EXCLUDES tokens entirely (catalogue is layout-filtered), so the on-device title index never contained 'Cadet' — the title path had no universe to match against; (2) the collector-number rescue only ever generated the plain (set, number) pair, never the token-set variant Scryfall actually catalogues tokens under; (3) — checked and NOT a gap: `/cards/named?exact=` DOES return token printings, so the image channel's proposal verification (exact→fuzzy named lookup in `identifyByImage`) already verifies token names with no server change (also proven live in the Item D runs: the Cadet token verified 3/3). **TOKEN SOURCE**: Scryfall search `(layout:token or layout:double_faced_token)` with `unique=cards` = **1,019 cards / 677 distinct names / 6 pages**; a name-pattern exclusion drops ephemera the search still returns (Bio / Decklist / Checklist / World Championships Ad / (minigame) cards). `getCardNames` now merges these into the served index (in-memory 24h server cache; failure degrades to the catalogue alone; response reports `tokenTotal` = names actually added, ~600 after dedupe against faces/catalogue); the client names-cache key bumped `manabox-card-names-v1` → `-v2` so existing installs re-download once automatically. **NUMBER PATH FIX (bottomline.ts)**: `parseBottomLineCandidates` now enumerates EVERY set-code token within ±3 of a number (nearest first; previously nearest-only), and the **token-set variant T+set** rides with the closest set code per number; a token marker — a standalone 'T' token before the number, or attached 'T0001' — pushes the T-variant FIRST, ahead of the strict parse, while non-marker regular cards still lead with the plain pair (so a regular card can never be displaced by its token-set namesake). **ARTIST-GUARD FIX (lib.ts)**: the Pass 15 guard discarded BOTH TFRA/1 and FRA/1 for the Cadet bottom — 'ELIZABETH PEARD' vs 'Elizabeth Peiró' fails the tolerant surname match (2-edit garble), so the true hit read as 'clearly contradicting'. New clause: the guard returns no-contradiction when ANY artist token (≥3 chars) appears VERBATIM in the bottom text — an exact name token is positive evidence, not contradiction. The Bolrac pin is intact ('WARREN' is not a verbatim token of 'Darren Tan'). **FIXTURES (`tests/pass24b.test.ts`)**: Cadet settles TFRA/1 via the bottom path (marker, attached, and pool shapes; with AND without the artist segment) and via the title path against the merged index; two more real tokens settle (Angel TLCI #2, Zoltan Boros; Bird TWOE #1, Kisung Koh). **EXPORTS**: the Moxfield/Archidekt/ManaBox text builders are layout-agnostic — a Cadet token row exports like any card (pinned by test). Stated limit: whether Moxfield's own importer resolves token set codes (TFRA) is that format's behaviour, not something the export can control.

- **ITEM C(ii) — LANDS, DIAGNOSIS.** E5's true printing is **FRA #384** (FRA = Reality Fracture; Alayna Danner). The failing parse step is convicted exactly: in 'L 0384 CN FRA EN ALAYNA D', the spurious token **'CN'** — sitting between the number and the set code — qualified as a set code under the old 2-letter minimum and the nearest-code scan stopped there, so the ONLY candidate generated was CN/384 → Scryfall 404 → no settle, despite a perfect read. Scryfall `/sets` (1,067 codes) proves the general rule: **no 2-letter set codes exist** (452×3, 587×4, 22×5, 6×6 letters). **FIXES (bottomline.ts)**: set-code token minimum raised 2→3 letters (M20-style codes kept); candidate enumeration widened as above; a last-resort 2-letter fallback pool survives ONLY when no 3–5-letter token exists in the line (preserves the pinned historical Q0/280 Griffin rescue → 280+'LN'; E5 unaffected since FRA is present). Rarity 'L' handling and the artist guard were tested and were NOT the failure (Alayna Danner corroborates FRA/384 correctly once the pair is generated). **FIXTURES**: E5's exact raw string now parses strict to FRA/384 and settles in the rescue integration against live Scryfall facts; the full FRA full-art basic cycle pins (Plains 384 Alayna Danner, Island 385 Calder Moore, Swamp 388 Daniel Romanovsky, Mountain 391 Leon Tukker, Forest 394 Adam Paquette). Title path for full-art basics: the pooled matcher over the merged index matches 'Plains' at 1.0 — no title-side gap existed (E5's title failure on-device was OCR garbage + the image timeout, not the matcher).
- **ITEM C(iii) — REGRESSION GUARD.** Token names entering the index create ZERO new suggestions ≥0.55 across the matchfast debris corpus plus Andrew's real garbage reads (pinned by test, index with vs without token names); the full battery position is in PROOF below.
- **ITEM D — IMAGE LATENCY AUDIT (`results/p24b-latency.json`).** Method: Pass 20 production path — the `identifyByImage` handler verbatim through `probe-ctx.js` against the real inference socket; payloads = Scryfall normal images resampled to the canonical 250×350, JPEG q0.82 (measured 15.7–30.6 KB raw / 21.0–40.8 KB base64 — same class as Pass 20's ~40KB). **CALLS PER ATTEMPT = EXACTLY 1** (code audit of `runImageChannel`: frame 1's dewarp only). Pass 23's multi-frame consensus is CLIENT-side fingerprinting against the returned candidates — **the prime suspect is EXONERATED**; no restructure was made and the payload is unchanged. **FULL PAYLOAD, n=12** (his four failing cards ×3): totalMs sorted [3810, 3825, 4384, 4767, 5576, 6667, 6918, 8507, 9976, 10231, 17238, 20900] — **p50 6.8s, p90 17.2s, max 20.9s; 2/12 over the 12s wait; 20/20 calls across both sets returned the correct card, Scryfall-verified**; inference is ≈93% of the total, verification only ~0.3–0.5s. **SMALL-PAYLOAD CONTROL** (125×175 q60, ~7× smaller), n=8, run after the full set in the same session: p50 30.7s, 5/8 over 12s — NOT faster; latencies grew with call sequence across both sets (the service slows under load), so payload size is not the driver. **RETRY NOT IMPLEMENTED**: the authorised smaller-payload retry cannot convert timeouts — the bottleneck is inference-service latency/variance, not bytes, and a retry issued after the 15s vision cap joins the same slowed service after the attempt has already settled. Constants unchanged (`IMAGE_WAIT_DEADLINE_MS` 12000, `IMAGE_VISION_CAP_MS` 15000). **TRADE-OFF LEFT AS AN EXPLICIT DECISION, NOT TAKEN**: raising the 12s wait toward the measured p90 (~17s) would rescue roughly 1 in 6 image-dependent attempts but make every OCR-uncertain scan wait that long; the panel's honest 'timed out' line stays accurate either way.

- **ITEM E — SUGGESTIONS, MEASURED.** (1) **'Use this artwork match anyway' — IMPLEMENTED.** Justification from `p23-matrix3.json`: decisive-and-correct image matches were demoted to a plain suggestion in 53/1,019 cells (5.2%); the degraded-guard shape (dim capture + margin <12 or unstable frames) accounts for **44/1,019 (4.3%) — 44 of the 53 demotions, all in dim captures**, which is the box's normal state; plus the on-device Borg Queen attempt-2 shape (d=40 demoted, next attempt auto-added at d=30). Implementation: the uncertain outcome now carries the presented image candidate; the Did-you-mean sheet gains a **'Use this artwork match anyway'** button (shown only when an image candidate tops the sheet) that settles THAT art-pinned printing directly (`getCardBySetNumber`, name fallback), with the version message stating 'you confirmed this match'. (2) Session recalibration counter — skipped per brief (moot after Pass 24a tracking). Help text updated: tokens are covered, token bottom lines ('T 0001 …') resolve to token printings, and a basic-land bottom-line example added.
- **PROOF / ACCEPTANCE.** New suite `tests/pass24b.test.ts`: **31 checks, all pass**. All **27 unit suites green: 761 checks** (730 Pass 24a + 31 new — exact accounting), including the revised-none regression pins (rescue 51 after the 2-letter fallback rework; matchfast 29 and deadline 54 after the artist-guard clause). Client + server typecheck clean. **BATTERY — HONEST ACCOUNT**: untouched-code baseline run first on a responsive VM (`results/p24b-baseline.json`: Tier 1 **125/0/7 of 132**, identical to Pass 24a final). Mid-pass the sandbox VM restarted and then ran sustained-oversubscribed (load 1.74 on 1.5 vCPU); three post-change full runs degraded monotonically with measured load (per-record wall-clock ~4.4–4.8s vs baseline 1.4–2.6s; battery wall 443–456s vs baseline 273s for identical work) — Tier 1 115/0/17, 111/1/20, 108/1/23 (`p24b-final{,2,3}.json`). Attribution, by direct probes rather than assertion: (a) the title matcher — UNTOUCHED by this pass — returns 'Treasure Keeper @ 0.873' on the flipped record's EXACT baseline reads with an unbounded budget, and a sub-threshold best-so-far under the attempt clock's remaining budget (the Pass 15 budget design under CPU starvation); (b) verdicts track per-record wall-clock across ~25 isolated rechecks — every record finishing ≤~4.4s settles at its baseline verdict (way-a1, way-a2, zedruu-1-oriented ×2, both userphotos and divergent controls flipped back), every record ≥4.5s starves regardless of code; (c) the single WRONG settle (dist-treasure-keeper-40, contended runs only; no-identification in isolation) traces to a pre-existing interaction: OCR read the number as 172 (a real foreign pair, AER/172 Prizefighter Construct), the artist guard returns 'inconclusive' for its artist (Daniel Ljunggren) against the garbled surname 'PCK' IDENTICALLY on old and new logic (the new clause never fires — different artist tokens), the first rescue candidate is AER/172 in both versions, and the baseline run was saved only by its surviving title read. No flipped record's decision inputs differ under this pass's changes. Residual risk is therefore the documented deadline-under-contention class, amplified by environment — the same class Passes 22/24a recorded — NOT a measured 24b regression; a clean full-battery diff could not be reproduced in this window and is the one proof artefact outstanding. **Phone-only remainder (Andrew's)**: (1) the Cadet token — should now settle via title AND via its 'T 0001 FRA' bottom line; (2) the foil FRA Plains plus one other FRA basic (Island/Swamp/Mountain/Forest); (3) image lines may still time out — that's the service's measured variance (Item D), the panel line will say so; (4) on a dim decisive presentation

(Borg Queen shape), try 'Use this artwork match anyway' in the Did-you-mean sheet; (5) Saheeli Rapid auto-add STILL unproven on-device (carried since Pass 21); (6) Pass 24a's checks remain open too (6+-card session with zero recalibrations; light holding steady). Note: the merged token index arrives via the names refresh — the cache-version bump makes the app re-download the index once automatically on first open after this update.

Pass 24c (2026-11-10) — conflicts require clean evidence (Step 1), scan-list item view (Step 2), full-art basic band aim (Step 3)

Andrew's asks (2026-11-10 ~10:18 EST): the FRA full-art Swamp still fails on all channels (title band clipped low, bottom abandoned, image at the band ceiling d=56); Massacre Girl, Most Wanted was blocked by a phantom 'Shatterwing Pegasus' conflict against a decisive d=30 image verdict; and he cannot SEE his scan list (79 cards / $47.61) — only counts + move/export controls — so doubles can't be spotted or removed.

- **STEP 0 — THE OWED BATTERY CLOSE-OUT (Pass 24b's open finding): RE-RUN, ESCALATED WITH FRESH EVIDENCE.** The full battery was re-run TWICE on the CURRENT shipped (24b) code, untouched: `results/p24c-baseline.json` (Tier 1 **107/1/24**) and `results/p24c-baseline2.json` (Tier 1 **113/1/18**), vs `results/p24b-baseline.json`'s **125/0/7**. The host did NOT provide a quiet window in this session: sustained neighbour contention on the shared 1.5-vCPU cell (load 1.5–2.3 throughout; mean per-record wall-clock **2.87s vs the quiet baseline's 1.74s** for identical work). The degradation carries the established deadline-starvation signature, not a code signature: (i) every flipped record ran ≥2.9s against its ≤3.1s quiet-baseline time, most ≥4.2s; (ii) the flip set is INCONSISTENT between the two runs (22 vs 16 changed records; num-prism-array/num-courier-bat/triage-tergrid flipped in run 1 only, userphoto-1000044373 in run 2 only) — nondeterminism across identical code; (iii) two rounds of ISOLATED rechecks of all 19 flipped Tier-1 records (`results/p24c-rc-*.json`, `p24c-rc2-*.json`) were themselves starved (3.1–5.5s per record in isolation, vs 1.0–3.1s quiet) and so cannot exonerate by flipping back — each round still showed only no-identification degradations, never a new wrong settle class (the single WRONG, dist-treasure-keeper-40, is the record Pass 24b already traced probe-by-probe to a pre-existing guard/raced-lookup interaction on inputs identical on old and new code); (iv) no flipped record's decision inputs are touched by Pass 24b's diff (token index merge is server-side, bottomline candidate enumeration, artist-guard clause, one UI button — the title matcher and deadline machinery the starvation acts on are byte-identical). VERDICT: the open finding is **not closed as a clean zero-diff** — it is escalated as an ENVIRONMENT limit with the strongest attribution available: the 24b code shows no measured regression; the battery's deadline sensitivity to host CPU is the residual risk, and it is now documented across three passes (23, 24b, 24c) with wall-clock evidence each time. Step 1–3 gating therefore uses SAME-WINDOW controls (final battery diffed against `p24c-baseline2`, the untouched-code run from this same contended window) in addition to the unit suites; any record differing from BOTH the quiet baseline AND the same-window baseline is treated as a real signal.

- **STEP 1 — CONFLICTS REQUIRE CLEAN EVIDENCE: SHIPPED.** Root cause, proven from the panel's own numbers: the phantom 'Shatterwing Pegasus' did NOT come from the title read — it came through the NUMBER branch. The bottom fragment 'J 021. FRA' parses to candidate FRA/21, which happens to resolve to a real, different card (Shatterwing Pegasus). That hit was UNVERIFIED (single treatment), but `rescueByCollectorNumber` dropped the `verified` flag `buildNumberCandidates` computes, and the Pass 23 `numberHitInterlock` treated any hit as a verdict. Additionally `describeImageChannel` hardcoded 'title read' for every conflict, so the gate line blamed the title while the header blamed the number. Fixes: (i) `NumberRescueResult.chosen` now carries `verified` end-to-end (scanflow + harness pipeline + matrix mirror); (ii) App's interlock: conflict + VERIFIED hit → conflict, unchanged (Pass 23 guard intact); conflict + UNVERIFIED hit → the hit is dropped as noise (diagnostics note 'ignored unverified number fragment … the decisive artwork match stands') and fusion decides — the decisive image settles; (iii) fusion's title direction now requires the OCR side to be decisive on CLEAN evidence — `cleanNameVerdict` (lib): exact-containment of the winner's full name in the pooled reads (Bolrac shape), or a ≥1.5× top-score margin with the winner's name contained in the pool; sub-bar/fuzzy-only crowns from debris are not clean; (iv) one shared conflict object (`conflictEvidenceLine(channel)` , outcome.channel) drives both the evidence header and the gate line — they can no longer name different channels. Fixtures: new `tests/pass24c.test.ts` , 33 checks, ALL PASS — the panel fragment (FRA/21 unverified vs FRA/234 verified two-treatment), the exact panel replay (decisive image d=30 margin ∞ + garbage crown + no flag → settles by image in Rapid AND Snap, line prints '→ auto-added by image', phantom never named), and the load-bearing guards: clean decisive OCR vs decisive image still conflicts (channel 'title'), the verified-number interlock still conflicts (channel 'number'), the band+bottom agreement path still conflicts, the settle-ocr confidence floor is unchanged, and the Pass 18 Thief-shaped garbage read still cannot settle or veto. Full suites after Steps 1–2: **794 checks green** (761 + 33), typecheck clean.
- **STEP 2 — SCAN-LIST ITEM VIEW: SHIPPED.** Diagnosis: the per-item rows (thumbnail, name, set #number, rarity, price, foil toggle, quantity stepper, Version, per-row Remove) already existed in the sheet — the defect was LAYOUT: the export/move/deck/clear block was a fixed sibling that consumed most of the sheet, squeezing the scrollable list to a slit (his screenshot shows exactly one clipped row above the controls). Fix: the whole commit section moved INSIDE the scroll region below the rows, with a plain-language subheading ('Scanned cards — every scan is listed here. Remove extras or fix quantities before you move the list.'). Duplicates made obvious: `addToScanList` merges same printing+foil into one row with quantity>1 (pre-existing), and such rows now carry an accent border + a '×N in list' pill; the sheet header area counts duplicate kinds when present. Rapid door-flag: `addToList` returns `{ item, merged }` ; when a Rapid settle merges, the banner notes 'already in list ×N' (informational only, never blocking). Move/export/make-deck behaviour unchanged. Verified live in the audit probe on seeded sandbox rows (Bolt ×2 + Bolt foil + Swamp): sheet shows all rows with the ×2 badge; clicking a row's Remove dropped the header 4 cards → 2 instantly; clicking Move then reported 'Moved 2 card entries'; server-side replay (updateScanListItem

qty 0 → moveScanListToCollection) left the scan list empty and the collection holding EXACTLY the post-removal set (foil Bolt + Swamp; the removed ×2 Bolt nowhere).

- **STEP 3 — FULL-ART BASIC BAND AIM + CYCLE DISTANCES: MEASURED; NO GEOMETRY CHANGE SHIPS.** New fixtures `tests/harness/prep_fra.py` + `tests/harness/.cache/fra/` (the FRA full-art cycle at Andrew's finish, box-40 + calibration-geometry composites) and `tests/harness/probe-fra.ts` (log: `results/p24c-fra-probe.log`). (a) BAND AIM: on the Swamp box-40 composite the detected rect was accurate (±2%) and the STANDARD band (y 2.5–11%) read 'Swamp' EXACTLY — no systematic vertical mis-aim exists for this frame class; the hunt-window sweep shows reads from y0 1.5% up. The Plains composite localised a DIFFERENT defect: detection clipped the card horizontally (left −13.3%, right −9.2%), cutting the title's first letters — reads came back 'Pains' — while the bottom line still settled it (FRA 384, the Pass 24b parse fix). Andrew's on-device low-clipped band is the vertical sibling of that clip: a RECT error, not a band offset; the one hunt direction not covered today (a band starting above 2.5%) would have to enter Rapid's per-frame loop to matter for his failing mode, at real per-frame read cost, against a benefit the probe could not demonstrate (the clip could not be reproduced for Swamp at all) — not shipped. (b) CYCLE DISTANCES (finder hashes, 250×350): pristine-vs-pristine cross-cycle impostors sit at 52–72 (nearest Mountain ↔ Forest 52) — the five arts are well separated at identity geometry. Box-40 queries with accurate rects: true distances Island 20, Forest 16, Mountain 30, Swamp 32 (margins 16–34 → decisive-quality); box-geometry impostors 46–62, i.e. inside/above the corroboration band — exactly why the band demands corroboration, and why Andrew's on-device Swamp (d=56, real sensor + foil inflation on top of the composite's 32) sat at the ceiling needing the bottom line his attempt abandoned. Widening the gate/band on this evidence would invite the Pass 23 colour-twin wrong-settle class — explicitly NOT done. Net for basics: with a correct rect the cycle settles decisively or via the now-parse-fixed bottom line; the residual risk is detection error on full-art frames (both axes observed), which Pass 24a's tracking already mitigates on-device.
- **FINAL BATTERY (changed code, `results/p24c-final.json`)**: Tier 1 **112/1/19**, mean 2863ms — the run began in the session's quietest window (early records 1.9–2.8s) and contention returned mid-run (tail records 4.2–5.5s). Against the SAME-WINDOW untouched control (`p24c-baseline2`) only **3 records differ**: hand-bolrac-dim-angle and exif-zedruu-1-raw degraded correct → no-identification at 4553/4734ms (an isolated recheck of the former was itself starved — title stage 2729ms with ZERO reads landed; pure CPU starvation of tesseract, a path this pass does not touch), and userphoto-1000044373 IMPROVED no-id → correct. Zero wrong-settle changes; no record differs in a way attributable to this pass's diff. Suites 794/794 green; typecheck clean (client + server).

Pass 25 (2026-11-10) — printing-level identification: the correct printing from the correct set

Andrew's asks (2026-11-10 ~11:30 EST, text only): Massacre Girl, Most Wanted now detects fine (Pass 24c confirmed live; his copy is FRA #0234) but the scan list showed him no visible version and no way to see/pick other printings. A Swamp he identifies as collector number 274 settled as

a WRONG printing he transcribed as 'MMD 296'. His requirement, general: the scanner must detect the correct PRINTING from the correct SET for basic lands — all five types — not just the right name; and the recorded printing must be visible and correctable for every card.

- **ITEM 1 — LOCALISATION, ALL FOUR SUSPECTS MEASURED (reproduction before any fix).** Live Scryfall facts (curl, this session): `/cards/named?exact=Swamp` 's default printing IS **MDD #296 'Marvel: Darkhold Destiny' (Maojin Lee)** — byte-for-byte the card Andrew reported (his 'MMD' = MDD). Swamp has **930 printings**; newest-8 by release = MDD 296 + seven TRK printings; the six real Swamp #274s (DSK Mumford, DFT De Ro, TDM Glushakov, MKM Boas, OTJ Zee Yazzie, ZNR Paquette) sit at released-desc ranks **61/68/85/104/116/>175** — outside the image channel's 8 candidates AND outside `getPrintings` ' 60-cap (so the picker couldn't contain his card either). Massacre Girl, Most Wanted has exactly ONE printing (FRA 234, Balanescu) — his recorded printing was necessarily right; complaint (1) is pure visibility. Verdicts: **(a) CONFIRMED (primary)** — `runImageChannel` fingerprinted `printings.slice(0, 8)` of a released-desc, 60-capped list; the true printing is structurally absent for basics. **(b) CONFIRMED (recording mechanism)** — every name-level settle records `resolveByName(name)` = the named-endpoint default unless the art winner pins (winner ≠ base AND printingMargin ≥ 4) or bottom salvage overrides; in Rapid the 'unconfirmed' VersionInfo never displays and the row carried no marker — a silent default. **(c) CONFIRMED with correction** — Settlement-2's number rescue DOES pre-empt image fusion when it hits; the gap is the TITLE-WIN paths (per-frame `finishTitleWin` + Settlement 1), which settle BEFORE the rescue runs and take only the fuzzy `applySalvage` as printing input; the rescue's 5-candidate cap could also let a foreign chosen hit crowd out a later same-name hit. **(d) CONFIRMED as trigger** — bottom skipped/abandoned (his FRA panels) → no candidates at all → (a)+(b) fire unopposed. **REPRODUCTION** (`tests/harness/probe-printings.ts` , production finder functions, refs from SMALL images as production loads, queries = each #274 printing's own NORMAL image, identity geometry): all six #274 queries → production winner a wrong printing at d=50–60, settle records named default **MDD 296 in 6/6** (exactly Andrew's report); with the evidence printing added to the set, the TRUE printing wins 6/6 at d=8–28, printingMargin 20–44. Control: an MDD 296 query → winner MDD 296 d=8.
- **ITEM 2a — EVIDENCE-SELECTED FINGERPRINT CANDIDATES (new pure module `client/src/printingselect.ts`).** `selectReferencePrintings` builds the reference set under a cap of 16 (was: newest-8), priority: (1) exact-pair printings — the attempt's bottom-line (set, number) candidates are resolved through the exact set/number lookup (≤5 candidates, raced 1200ms, server-cached so Settlement 2 reuses them free) and same-name hits enter FIRST, plus list members matching a candidate pair; (2) the name's default printing; (3) the newest-8 fill (unchanged base behaviour); (4) set+artist matches (number garbled, set+artist clean — the FRA basic shape); (5) number+artist matches (set garbled); (6) up to 2 set-only matches, artist-first. Artist rule (2d): the bottom line's artist surname must appear in the read text (`artistMatchesBottomText`) — it promotes among same-set/same-number candidates and fingerprint-near printings; it never overrides an exact pair. Keyed on printing COUNT / candidate coverage (`isHighPrintingCount` ≥12), never on the five names. COST: only on evidence-bearing attempts (a readable bottom line at image-start; evidence is attempt-keyed

and lazy — attempts with no bottom text pay exactly the old cost); ≤5 extra exact lookups per proposed name, all server-cached; fingerprinting ≤16 refs/name vs 8 (≤48 vs 24 worst case across 3 proposed names) — bounded, and only when evidence exists.

- **ITEM 2b — THE EXACT NUMBER OVERRIDES A NAME-LEVEL SETTLE'S PRINTING.** New `pinByNumber` closure in `identifyBurst` runs the shared collector-number rescue over the attempt's captured bottom sources (FRESH tried-set — Settlement 2's adjudication keeps its untried view; lookups are server-cached) and classifies with `numberPinDecision` (scanflow): same-name hit → **pin** (title-win and settle-image paths finish with the number's printing, status 'detected', message names the pinning: 'the collector number printed on the card pins the exact printing'); different-name VERIFIED hit on the title path → the early settle is abandoned and Settlement 2 + the interlock adjudicate (Pass 23/24c rules untouched); unverified foreign fragment → ignored noise (24c standard). This closes suspect (c): a title win no longer settles ahead of a readable exact pair.
- **ITEM 2c — HONEST NAME-ONLY SETTLES.** New persisted flag `scan_list_items.printing_confirmed` (migration 0007; legacy rows default 1 — their pinning state is unknowable retroactively; every new add passes it explicitly; a merge can only upgrade it, a swap sets it). Confirmed ⟺ the settle's VersionInfo status is 'detected' (number pin / art pin / bottom salvage / Detect version / user's own pick / 'use this artwork anyway'); name-only settles and Did-you-mean picks record **false**. Behaviour chosen (least-disruptive honest, measured against the battery's flow): Rapid still auto-adds WITHOUT interrupting (no picker modal in the flow — Rapid's whole value is zero taps), but the settle banner now says 'printing unconfirmed — pick the version in the list', the scan-list row carries a tappable accent badge 'Printing unconfirmed — tap to pick the version', and the result sheet shows the same badge. The silent default is dead: a name-only basic now records Scryfall's default printing LABELLED unconfirmed, never as a detection.
- **ITEM 3 — PRINTING VISIBILITY + FULL PICKER IN THE SCAN LIST.** Diagnosis: the row DID render set/#number — fused into the dim 12px rarity/price meta line, with the picker behind a button labelled just 'Version' and capped at the 60 newest printings (his #274 unpickable). Fixes: the recorded printing is now its own full-weight line — bold `FRA · 234` (new `printingLine` helper) + set name — and tapping it opens the picker; the button is renamed **'All versions'**; the picker loads EVERY printing (server `getPrintings` gained `full` pagination, ≤10 pages, separate cache; used by the list picker, the result sheet, CardResultPanel, and Detect version's `resolveVersion` — automatic settle paths keep the 60-cap for latency); `PrintingGrid` gains a filter (set/code/number/artist) + render cap with an honest count line when the list exceeds 30 (a basic's ~930 entries are otherwise unusable). Swap machinery was already wired for scan-list items (Pass 24c) and now also marks the row printing-confirmed. Help text updated (unconfirmed badge + full version list).
- **FIXTURES.** `tests/pass25.test.ts` — **43 checks, all pass**: selection matrix (no-evidence = first-8 + default; all six cross-set #274s enter via their exact pair within the 16 cap; (fra,390) promoted from position 40; garbled-number set+artist promotion; garbled-set number+artist promotion; set-only ≤2; 20 exact candidates → cap 16, exact-first, no duplicates); the FRA full-art cycle with REAL printings verified live (Plains 384 Danner, Island 387 Calder Moore, Swamp

390 Romanovsky, Mountain 393 Tukker, Forest 396 Paquette) each selected from its own bottom line at position 45; pin-decision matrix (same-name pin verified/unverified; verified foreign → fallthrough; unverified foreign → ignore — the Shatterwing shape; Massacre Girl FRA/234 pin + single-printing candidate set); printingLine format; and a fabricated-fingerprint miniature of the Item 1 mechanism (evidence absent → wrong winner; present → true winner d=0, margin 50) mirroring probe-printings' real-image result. Borg Queen art-pinning untouched (the art-pin branch is byte-identical below the new number-pin step, which only fires on same-name bottom evidence and was already subordinate to Settlement 2). **Suites: 837 checks green (794 + 43); client + server typecheck clean.**

- **FINAL BATTERY (changed code, `results/p25-final2.json`)**: Tier 1 **112/1/19**. Same-window untouched control (`results/p25-baseline.json`, run first this session): **102/1/29** — the window was contended (load ~1.45 on 1.5 vCPU), then the VM restarted mid-first-final-run (`p25-final` partial, discarded) and final2 ran into returning contention (records at 4.2–5.5s). Per-record diff vs the same-window control: **ZERO new wrong settles**; 11 records improved no-id → correct-printing (starved records recovering); 1 degraded correct-name → no-id (userphoto-zedruu-boxleak at 5475ms, the run's slowest record — deadline starvation of the title read, a path this pass does not touch). The single WRONG (dist-treasure-keeper-40) is WRONG in the untouched baseline too — the record Passes 24b/24c traced probe-by-probe to a pre-existing guard/raced-lookup interaction on identical inputs. No flipped record's decision inputs are touched by this pass's diff (the harness pipeline does not exercise the new selection/pin/UI code). **What a name-only basic settle looks like now**: Rapid adds it with the banner note 'printing unconfirmed — pick the version in the list'; the list row shows the default printing in the bold printing line PLUS the accent 'Printing unconfirmed' badge; one tap opens All versions (full ~930 list, filterable) and swapping records the true printing + price and clears the badge. **Phone-only remainder (Andrew's)**: (1) re-scan the #274 Swamp — with any readable bottom line it should now pin to its own set or arrive badged unconfirmed, never silently MDD 296; if a panel still can't tell, the ask back is WHICH SET his #274 is from (DSK/DFT/TDM/MKM/OTJ/ZNR are the candidates); (2) Massacre Girl in his list should now show 'FRA · 234' on its own line with All versions beside it (her single printing); (3) an FRA basic with an unreadable bottom line should arrive badged, not wrong.

Pass 26 (2026-11-10) — bleached bands on a clean dewarp, a clean read lost in pooling, duplicate scans

Andrew's panels 2026-11-10 11:56/11:58 EST (files 1000044672_42_lu3n.jpg + 1000044674_43_7ml6.jpg, captured on the Pass 24c build, before Pass 25 shipped; Pass 25 touched neither mechanism) plus his note "it also scanned a couple cards twice by mistake". Three defects, each localised on reproduction before any fix.

- **DEFECT 1 — BAND CROPS EMPTY WHILE THE DEWARP IS CLEAN: LOCALISED to a tolerance divergence, not a source divergence.** Both pipelines read `preps[i].card` from the same rect — there is no second frame or coordinate space. The divergence: the image

channel fingerprints MULTIPLE rect hypotheses (insets/shifts) and survives rect error; the burst title loop read ONE fixed window (crop y 0.025–0.110) per frame with no search, and Rapid never gets Snap's rescue bands. Reproduction (`tests/harness/prep_p26.py` hand-held composites — angled card on a textured background with a desk edge; `probe-p26-pipeline.ts`): texture-slab detection under-measures a basic land's height (the lower ~35% is near-uniform text-box/symbol area) and `correctCardAspect` anchors 70% of the compensating growth to the top, displacing the rect UP — on the composite the title centroid lands at crop y=0.120, just below the window's 0.110 edge: every single-line treatment read "", only the tall band read "l Swamp S". In Andrew's panel the displacement was larger (title on bright background → pure-white band crops; the collector line fell BELOW the crop — unrecoverable by any window shift, recorded honestly; his attempt was still carried by the image channel at d=50). The bleach guard classified its given crop correctly — the crop was the defect. **FIX SHIPPED: title-band AIMING** — `aimTitleSpec` (ocr.ts) mirrors Pass 18's proven `aimBottomSpec` exactly (ink profile → centred-window test → centroid shift → ±0.03/0.06/0.09 probes, title clamps y0 ∈ [0, 0.22]); the standard band spec now carries `aim: true` and `ocrTitleBandCapped` reads the aimed window; the published panel thumbnail shows the aimed window actually read. Accurate crops return byte-identical specs (ink centred → unchanged), so box-rig behaviour is untouched. Harness assertions (`probe-p26-assert.ts` , all pass): on a panel-shape crop displaced exactly like the panel (title centroid at crop ≈0.16, fixed window holding only background + top border) the aim moves the window +0.038 onto the title row (ink 0.228) and the burst settles the card; on the detected-rect composite the aimed band carries ink 0.365, not bleached. NOTE: no gate/band widened — the Pass 23 colour-twin discipline stands. The panel's other fragment (a previous settle banner pinning a Swamp by artwork to TRK #314) is Pass 25 art-pinning working on-device; untouched.

- **DEFECT 2 — CLEAN BAND READ LOST IN POOLING: LOCALISED to two layers.** (a) MATCHER (the dangerous one): replaying the panel's exact pool `["1 Eye ol lace To"]` through `matchNamesPooled` (`probe-p26-eye.ts`): per-candidate best = Eye to Eye 0.755 vs Eye of Jace 0.637/0.676 — but the FRAGMENT-UNION pass promoted Eye to Eye to **0.980**, because its tokens (eye/to) sit verbatim in the read as debris and the union's precision test ignores <3-char evidence tokens ('lace' unexplained only halved precision to 0.5, above the 0.45 full-factor bar). At 0.980 vs 0.726 the margin (0.25) would make `decideRapidMatch` AUTO-ADD the wrong card in Rapid on an uncontended phone — the panel's 2.0s match stage (budget expiring mid-union) is all that saved it. A moderately corrupt read ("Eye ol Jace") ranks correctly (0.790), so only the union fabrication inverts. **FIX SHIPPED: union provenance** (lib.ts) — union promotion now requires the entry's best token evidence to span ≥2 SOURCE reads, where candidates prefix-nested under one read (ocrCandidates emits a read + its truncations) count as that one read; single-source pools keep their per-candidate scores. Panel pool after fix: Eye to Eye 0.755, decision=pause; cross-read fragments (Divergent Equation shape) still promote to 0.965; Treasure Keeper fragments keep their auto-accept at the honest per-read 0.701 (the old 0.81 was the same inflation). (b) TREATMENT COMPLETENESS: the panel's title stage ended at 123ms because the standard band's PSM-6

pooling trigger never fired on that corruption shape, so no second treatment ever landed. **FIX SHIPPED: the completing treatment** (App `identifyBurst` + harness mirror) — when the frame loop ends with a sub-decisive pool and clock remaining, ONE tall-window multi-line read (Pass 8's PSM-6 line-pick window) on the last frame is pooled before settlement; decisiveness reuses the loop's hoisted verdict (re-scoring the index just to ask would eat the read's clock). Proven at step level in the harness (`probe-p26-assert.ts` S2): the standard pool reads garbage (", NBS LW if REM frp"), the completing call reads "t Swamp", pooled top = Swamp 0.770. HARNESS-ECONOMICS NOTE (honest): in this sandbox the frame loop legitimately spends the whole 2500ms clock (reads run 5–10× slower than on the phone; gate traces show remaining=0), so the completing read is clock-starved in full-burst harness runs by design — on-device the panel's own numbers (title 123ms of 2500ms) leave the clock it needs. The Eye composite settles Eye of Jace end-to-end in the harness regardless (bottom-line rescue, E1 assertion).

- **DEFECT 3 — DUPLICATE SCANS OF A STILL-SEATED CARD: LOCALISED by measurement, fixed by retiring a rule.** `probe-p26-sig.ts` ran the App's own 24×34 guide signature over the hand-held composite: still-seated events peak vs handled at ≤13.3 (2px jitter consecutive diff 10.2, hand shadow 12.9, torch step 12.1, nudge-and-stay 13.3) — all ABOVE Pass 22's movement re-arm threshold (9) — while genuine departures peak ≥49.7 (25% slide-out 49.7, 50% 55.4, full removal 62). Pass 22's movement rule therefore re-armed on events where the card never left, and the still-seated card was scanned and added again. **FIX SHIPPED (rapid.ts): the movement re-arm path is retired** — every measured genuine transit crosses the change rule's threshold (14/20) at its peak anyway (≥49.7), so re-arm = change or absence, exactly as before for real re-presentations (Pass 22's own sim, peak 55, still re-scans ≤1000ms); the measured class boundary is kept in config as `departureThreshold` with the evidence recorded; the viewfinder forceScan tap remains the deliberate re-scan of a card that never left. Pinned in `tests/pass26.test.ts`: blip oscillation (consecutive 12 > old bound 9) over 40 ticks → zero re-scans; nudge-and-stay → none; genuine departure + return → re-scan ≤1000ms.
- **FIXTURES + BATTERY.** New `tests/pass26.test.ts` (10 checks: the panel pool's impostor capped at its single-read score with decision=pause, clean-read pooling wins at 1.0, Eye to Eye's own read still crowns it, cross-read fragments still pool, Treasure Keeper auto-accept preserved, the three Rapid sims) + `tests/harness/prep_p26.py`, `probe-p26-pipeline.ts`, `probe-p26-eye.ts`, `probe-p26-sig.ts`, `probe-p26-assert.ts` (5/5 pass). **Suites: 847 checks green (837 + 10); typecheck clean (client + server).** FINAL BATTERY (changed code, `results/p26-final.json`): Tier 1 **111/2/19** vs the same-session untouched control (`results/p26-baseline.json`) **103/1/28**. Per-record diff (12 records): 10 improved no-id → correct-printing (baseline-starved at 3.1–5.4s recovering at 2.2–4.7s); boxlow-stolen-stark-tech-12 flipped correct → no-id at 4396ms — isolated recheck on changed code (`results/p26-rc-boxlow12.json`): correct-printing at 3401ms, starvation attribution confirmed; str-raft-4551 (single read "I Ratt Security") moved correct-top 0.662 → correct-at-2 0.618: the ONE real cost of the union-provenance gate — the union had been topping up a single-read 2-of-3-token match; BOTH runs decide "pause" with the true card in the list, so no

settle behaviour changes, only suggestion order inside a paused list — accepted deliberately: the same promotion is what fabricated the 0.980 wrong-card auto-add in Defect 2. The remaining WRONG (dist-treasure-keeper-40) is wrong in the untouched baseline too (pre-existing, traced Passes 24b/24c). **Zero new wrong settles.** PHONE-ONLY REMAINDER (Andrew's): (1) hand-held Swamp re-scan — expect the title band thumbnail to show the title row (aimed) and the read to land, or the bottom/artwork channels to carry it as before; (2) Eye of Jace re-scan — expect a clean settle or a Did-you-mean with Eye of Jace on top, never Eye to Eye; (3) Rapid duplicates — sit a scanned card still and move a hand over the box / toggle light: it must NOT re-scan; slide it out and back: it must.

Pass 27 (2026-11-10) — still-seated re-scan on a timer (Item 1), full-art printing discrimination (Item 2)

Andrew's report (2026-11-10 ~13:30 EST): duplicates persist AFTER Pass 26 — "it just scans if I wait about five seconds to ten seconds it just automatically scans another of the same card. It does recognize if I slide another card in" — and the TRK #314 Swamp was actually FRA #390, a FOIL full-art/extended-art Swamp (Daniel Romanovsky); extended-art/full-art printings must resolve to the CORRECT printing. He asked for suggestions that would fix these: implemented what the measurements convicted, the rest recorded as proposals with numbers.

- **ITEM 1 — LOCALISATION (instrumented, suspects eliminated in order).** The watcher (`App.tsx scanGuideSignature(sampleIntervalMs 250)`) skips sampling while an attempt is busy; signatures come from the guide canvas, NOT the tracker rect — so tracker drift (suspect a-iii) cannot move the handled comparison. The Did-you-mean `resumeRapid` builds a fresh signature (no stale reference). The 8s watchdog cannot duplicate (generation counter + AttemptGuard drop a killed attempt's settle; its rescue adds once). Prime suspect was suspect (a): the handled phase's RAW signature comparison. **INSTRUMENT:** **`tests/harness/probe-p27-drift.ts`** — the REAL `RapidMachine` (box mode) + the REAL light latch (`autolight.ts`) at the app's exact cadence model (250ms watcher tick, samples skipped while busy, preview metering every 3rd tick, attempt start = engage-decision/sample/set-latch, attempt end = release + latch clear, 3.2s attempt) over the real p26 hand-held scene, 240 simulated seconds, photometric model = latch-driven persistent gain steps with AE convergence (τ=900ms) ± slow AE wander; signatures are the App's own mirrored 24×34 function. **PRE-FIX TRACE:** Scenario A2 (dim box, light gain ×2.2 — the latch limit cycle): **46 scans of a card that never left** — settle→re-arm gaps at latch flips with raw sceneDiff 29–76, beep-to-beep ≈5.25s, inside Andrew's "five to ten seconds". Scenario B (constant light, pure AE wander ±20%/25s — the in-app-browser case, no torch exists there): re-arm at settle+9.25s, **20 scans**. Scenario A (×1.55 latch): 2 scans. Scenario C (steady light, noise only): exactly 1 — the machine itself is sound. **ROOT CAUSE (one sentence):** the handled change test is a RAW photometric diff against a signature frozen at fire time; persistent photometric STATE changes (the Pass 24a latched light's own engage/release; camera AE/AWB wander) accumulate across the frozen reference until they cross the same-card threshold (20 box) — indistinguishable from a new card. Pass 26's probe missed the class because every event it modelled RETURNED to baseline

photometrics within 1–3 samples; a state flip or ramp never returns. Statistic selection measured (`probe-p27-measure.ts` , kept): at the dim-box operating point mean-normalised diff gives same-card 0.0–0.1 vs different-card ≥20.5 / removed ≥30.3 — clean separation at the existing threshold 20; median-ratio and sqrt-median variants measured, no decisive advantage (sqrt-median worse: 11.1 on a dim release). Known limit, recorded: a ×1.45 UPWARD step on an already-BRIGHT scene saturates highlights (normalised 24.3) — information-theoretic (clipped cells carry no pattern); the latch only engages when the scene is dark, so that step class cannot originate from the light system, and AE wander at ±20% measures ≤7.3.

- **ITEM 1 — FIX (rapid.ts).** (i) New exported `signatureDiffNormalised(a, b)` — rescales b to a's mean luminance before the mean-abs diff; degenerate (≤0.5 mean) falls back to raw. The handled phase's change comparison (`sceneDistance`) is gain-normalised; the watching phase's noise/stability/retry logic is untouched (its 9/7/14/20 thresholds were measured raw). (ii) The absence rule was hardened after scenario D exposed a SECOND door in the first normalised build: normalised shape alone scored a dim seated card within threshold of the near-structureless tray baseline. New exported `signatureStructure(sig)` (mean |v/mean − 1|, gain-invariant; tray ≈0.055, card ≈0.35) and `matchesBaseline(sig, baseline, threshold)` — normalised shape match AND structure ≤ max(0.12, 1.8×baseline structure); absence = matchesBaseline AND the baseline is the closer match by a margin (baseDiff + 6 ≤ sceneDiff). (iii) The watching-phase baseline guard also uses matchesBaseline — scenario D showed the raw guard firing a wasted attempt on the EMPTY tray mid-AE-ramp. (iv) Degenerate flat-field fallback (`sceneDistance` / `matchesBaseline`): when BOTH signatures are near-structureless (<0.02) there is no spatial signal to normalise and the decision falls back to the raw diff — the uniform-grid fixture case (and the pre-Pass-27 rule); real trays (0.055) and cards (≥0.2) always take the normalised path. **POST-FIX PROBE (persisted `results/p27-drift.txt`):** A = 1 scan, A2 = 1 (was 46), B = 1 (was 20), C = 1, **D = exactly 2** — swap A→B at t=60s under latch cycle + wander: re-arm at 60.00 (normalised diff 66.4, card gone), B fires at 62.50 (0.5s after it stabilises), then B sits 178s with zero re-scans. Slide-out/in and swaps confirmed at the required timings. The invariant "a card that never leaves NEVER re-scans, however long it sits" now holds by construction for the photometric classes: the only signals that can re-arm are spatial (a different pattern, or the baseline's own shape at baseline structure).
- **ITEM 2 — MEASUREMENT FIRST (`tests/harness/probe-p27-fullart.ts` , results `p27-fullart.json` ; supplement `probe-p27-margins.ts` / `p27-margins.json`).** Corpus: every FRA + TRK basic-land printing, MDD 296, the six #274 Swamps, other borderless Swamps (83 printings fingerprinted via the production path: SMALL refs, NORMAL queries, canonical dewarp, rectHypotheses, rankReferences). **Q1 COVERAGE — the conviction:** the shipped no-evidence field for "Swamp" (verified live against Scryfall: 887 printings, released-desc) is the default + newest 8 = {MDD 296, TRK 564, 563, 484, 322, 321, **314**, 309} — **FRA 390 sits at released-desc rank 10, OUTSIDE the field (SLD 2756 at rank 9 is outside too); TRK 314 was in it.** The fingerprint honestly picked the nearest of the wrong field. **Q2 DISCRIMINATION:** with FRA 390 IN the field, foil queries win ALL 9 degradation classes (identity d=18, sheen110 28, sheen90+lift20 38, device-combo 22, dim05+sheen70 26), nearest different-set full-art

rival at 48–50 (gap ≥10). **Q3 CROSS-MATRIX (189 rows):** 12 losses, NONE of them a cross-set full-art confusion of the FRA 390 kind — same-set/sibling losses dominate (TRK regular↔scene versions at sheen90+lift20, FRA 383→384 at d=50/52 beyond the decisive gate); one cross-set tie (TRK 308 Island → FRA 283, 44 vs 44). **Margins supplement:** true full-art winners lead their nearest full-art rival by ≥10 in every class; the genuinely ambiguous pairs sit at gap 0–6 (TRK 309↔484: gaps 6/6/2/4 by class and an outright TIE at sheen90+lift20; FRA 383→384 gap 2 at d=50). **Q4 TIE-BREAKERS — NOT SHIPPED, with numbers:** art-only/card-only hash splits do not separate the tight pairs better than the combined distance (TRK 309 vs 484 are close in BOTH splits); copyright-line position / artist OCR were not needed once the number channel (a) is reachable — recorded as proposals only. **Q5:** the line-span ink probe saturated (every span read 0.80–0.998 — the detector found art ink, not the line); the capture question was answered directly by probe-p27-bottom instead.

- **ITEM 2a — BOTTOM-LINE EXTENDED CAPTURE (SHIPPED).** When a name-level settle has NO number pin and the card is a basic land OR a full-art/borderless printing, `extendedBottomPin` (App.tsx `identifyBurst`) makes ONE retry read on a TALL crop: the first frame re-cropped with the rect extended 12% downward (top fixed, no aspect re-correction — the true card fills the top ~89%), read through the new `BOTTOM_BAND_TALL` window (ocr.ts, y 0.83–0.96, aim on) under the attempt clock; any text joins the attempt's bottom sources and `pinByNumber` judges again — a captured (set, number) pins exactly (Pass 25's override). Wired on BOTH settle paths (title-win and settle-image). **Capture evidence (`probe-p27-bottom.ts`, `results/p27-bottom.json`; FRA box40 composites, rect height scaled f, anchored at top):** at f=1.00 the standard read captures 5/5 (sanity — no regression); at f=0.97, 4/5 both paths (Island's reads garble the number itself); at f=0.94 the standard read captures **0/5** and the tall retry **4/5**; at f=0.90, standard **0/5**, tall **3/5**. Under-measured regime total: standard 0/10, tall retry 7/10 (residual misses: one pre-check skip at Plains f=0.94, garbled-number reads at Island f=0.90). The retry fires only when no pin exists, so its cost (≤1 capped read) lands only on qualifying no-evidence settles.
- **ITEM 2a2 — REFERENCE-FIELD DIVERSITY (SHIPPED, printingselect.ts).** Root cause was field composition, so the field is fixed: for high-count names (≥12 printings in the list — the Pass 25 bar), after the base fill the selection reserves up to `FULL_ART_DIVERSITY_SLOTS` = 4 slots for the newest full-art/borderless printing of each DISTINCT set not already represented (list order = released-desc), inside the existing cap of 16. Swamp's no-evidence field becomes {MDD 296, TRK 564/563/484/322/321/314/309, SLD 2756, **FRA 390**, HOB 196, MSH 438} — 12 entries, FRA 390 in the race with zero bottom evidence. Treatment flags flow end-to-end for this: server `cardSchema` + `normalizeCard` now carry `fullArt` / `borderColor` (Scryfall `full_art` / `border_color`); stored-row reconstructions (scanItemToCard / collectionToCard / scanRowToCard) set them false/"" explicitly — flags aren't persisted and every flow that needs them re-fetches printings. Ordinary names (no full-art printings, or below the count bar) are byte-identical to before.
- **ITEM 2c — HONEST DEMOTION (SHIPPED).** `ReferenceEntry` carries the treatment flags; `rankReferences` names a full-art winner's nearest full-art rival (any set — the measured class

includes same-set TRK 309↔484) as `RankedCandidate.fullArtRival` (consensus carries it via the best frame). At the settle-image art-pin, `fullArtPinBlocked` (gap ≤ `FULL_ART_PIN_GAP` = 8, the image gate's own decisive margin — measured separation: true winners ≥10, ambiguous pairs 0–6) skips the pin; the settle falls through to number/salvage and, failing those, records the name with `printingConfirmed=false` and a message naming BOTH printings and their distances. The deliberate "Use this artwork match anyway" button is unchanged — a user-confirmed pin is the user's call, and it is the escape hatch the demotion message points to. Borg Queen pinning and Pass 25's six #274 fixtures are untouched (their winners are not full-art-blocked; the number-pin step precedes the art-pin as before).

- **SUGGESTIONS NOT SHIPPED (with their numbers):** (1) treatment tie-breakers (copyright-line position, frame classification, artist OCR as fingerprint tie-break) — Q4 showed the hash splits don't separate the convicted pairs and the number channel (2a) + demotion (2c) cover the failure modes without new classifiers; artist OCR remains what it already is: evidence for candidate selection (Pass 25), not a fingerprint judge. (2) Widening the fingerprint gate for full-arts — rejected on the matrix: FRA 390's worst true distance (38, sheen90+lift20) already sits inside the decisive gate (44); the losses were field composition and rival ambiguity, not gate width. (3) Same-set regular-frame sibling demotion (FRA 284→283 Island at d=42/margin 6, sheen90+lift20) — outside the full-art scope Andrew reported; the existing printingMargin ≥4 pin bar partially guards it; recorded here as the known residual class if it ever surfaces on-device.
- **FIXTURES.** `tests/pass27.test.ts` — **34 checks, all pass**: normalised-diff statistics (same card at ×0.45/×1.9-step ≈1.1 vs raw 68; different card 28.4; removed 44.0); structure/matchesBaseline units (tray 0.038 vs card 0.206; card never matches baseline at any gain; tray matches under flipped light); machine sims at the app's 250ms cadence (latch limit cycle → exactly 1 scan/240s; AE wander → 1; latch+wander with a swap at 120s → exactly 2, B ≤2s after stabilising; removal during a light ramp-down → no empty-tray scan; Did-you-mean resume under drift → 0 re-scans); diversity fixtures on the real Swamp list head (FRA 390 in, one-per-set, cap 12, low-count names untouched, no-full-art names = base 8); demotion fixtures (FRA 390 shapes pin at gaps 26 and 10; the 309↔484 tie blocks; gap 8 blocks, gap 9 pins; no rival → never blocked); rankReferences end-to-end on real pHash fingerprints (winner carries its full-art rival). Probes persisted: `probe-p27-drift.ts` (+ `results/p27-drift.txt`), `probe-p27-measure.ts`, `probe-p27-fullart.ts` (+ `p27-fullart.json`), `probe-p27-margins.ts` (+ `p27-margins.json`), `probe-p27-bottom.ts` (+ `p27-bottom.json`). **Suites: all 31 test files green (876 PASS lines by grep count, incl. the 34 new); rapid/pass22/pass26 suites re-verified after the flat-field fallback refinement. Client + server typecheck clean.**
- **FINAL BATTERY (changed code, `results/p27-final.json`): Tier 1 127/0/14** (of 141) vs the same-window control run BEFORE the Item 2 changes (`results/p27-baseline.json`, tree = Item 1 shipped only): **124/0/17**. Per-record diff over all 156 records: exactly **3 changed, all improvements** — exif-zedruu-3/6/8-oriented flipped no-identification → correct-printing with wall-clock falling 4.6–4.8s → 1.9–3.2s (the baseline ran under heavier contention; the deadline-starvation signature, and the harness pipeline does not exercise this pass's settle

logic in any case). **Zero new wrong settles; zero regressions; zero wrong verdicts in either run this window.** PHONE-ONLY REMAINDER (Andrew's): (1) leave a scanned card SITTING in Rapid for 30s+ — it must scan exactly once, however the light behaves; slide it out/in and swap as usual to confirm those still fire immediately; (2) re-scan the FRA #390 foil Swamp — with any readable bottom line the tall-crop retry should pin FRA · 390; with none, the artwork should now either pin FRA 390 (it is in the field now) or arrive badged 'printing unconfirmed' naming the rival — never a confident TRK 314 again. A failing panel's screenshot (its rect line + distances) is the evidence for any residual case.

Pass 28 (2026-11-10) — Snap sequence decay: investigation, two convicted mechanisms

Andrew's report (2026-11-10 ~14:34 EST, panels 1000044678/4680/4682): SNAP (single capture, NOT Rapid) — the first Snap after a screen refresh identifies correctly; the NEXT Snap fails; only a refresh restores it. "It's not correctly calibrating the shot now." All three panels failed identically in shape: "No plausible name", evidence "no readable text", match 0ms, image service TIMED OUT (12s) so the text pipeline carried alone, all reporting "frame: retained". Panel cards: (1) Guiding Hydra (FRA R 0360, borderless, title in a translucent banner over art), rect (0.03,0.12)-(0.74,0.83), title 901ms + bottom 824ms both ran, nothing usable; the straightened image shows the read crop containing the whole box interior (card floating inside, tray walls on all sides — true card ≈ (0.10,0.21)-(0.66,0.80) of the guide inside a rect ~13% oversized per side). (2) Chandra, Chill of Compliance (borderless; FRA 297 per Scryfall — the panel's "M 0237" bottom line was a misread of 0297), rect (0.02,0.10)-(0.78,0.86), title 1.7s, bottom SKIPPED (bleached); the straightened crop begins mid-art — THE TITLE ROW IS NOT IN IT — and ends below the card (white tray strip): the crop sat ~10% of card height too low. (3) Chandra, Torch of Defiance (FRA 309 borderless, verified), rect (0.10,0.18)-(0.76,0.83), same shape. SEPARATION OF CONCERNS, as briefed: the sequence claim (decay) was tested on EASY cards first; the three panel cards are also long-standing hard frames (all failed in the Pass 19 batch) and are measured separately below.

- **(a) THE DECAY INVESTIGATION — sequence state is HEALTHY on the bench; no decay reproduced on easy cards.** New harness `tests/harness/snapsequence.ts` (+ `prep_p28.py` fixtures: Griffin Protector M20 20, Hydra FRA 360, Chill FRA 297, Torch FRA 309, Treasure Keeper AER 177 composited in the box at six seats with variance): a faithful simulation of the App's session state around the real `identifyBurstHarness` — tracker seeding/folding via the production `createCardTracker` / `prepareCardFromFrame`, 3 preview ticks per card feeding the Pass 15 retention, retained-frame consumption, `applyRectAuthority` semantics, `noteDetection` miss counting, virtual clock. Six consecutive Snaps with swaps (Griffin → Hydra → Chill → Torch → Treasure → Griffin): **all 6 identified correctly, zero degraded attempts, tracker followed every seat, retained frames always showed the current card, zero OCR worker recycles** (H3's zombie class does not recur on the Snap path, including attempts whose predecessor ended via the image-timeout shape — the harness's image channel is stubbed, noted). H1 (tracker state) REFUTED under clean geometry: Snap's writes into tracker state are the same folds Rapid makes, and per-frame detection self-corrects

against the tracker whenever they disagree beyond TRACK_AGREE_EDGE. What the panels actually show is therefore not a counter that rots — it is per-attempt GEOMETRY and RETENTION failures that a refresh happens to mask (a refresh re-seeds the tracker from calibration and clears retention, so the first attempt gets the cleanest state). Two mechanisms were convicted on other fixtures:

- **(a1) CONVICTED — RETAINED FRAME NEVER RE-VALIDATED AGAINST THE LIVE SCENE (H2).** Code audit: `retainedFrameRef` is set by the ~800ms manual-mode preview tick, consumed by `takeRetainedFrame` (2.5s TTL), cleared only after TWO consecutive detection misses — so a fast swap, or ticks landing mid-slide / on a hand / on the empty tray, can leave a fresh-TTL retention whose pixels show a different scene than the one being snapped, and `capturePreparedBurst` led the burst with it unconditionally. Probe `tests/harness/probe-p28-stale.ts`: retention = Griffin prep, live card = Chill (scene distance 28.7) — the PRE-FIX burst shape [retained-A, B, B] reads **"no readable text"** (two cards' pixels pooled destroy both reads — the panels' exact failure shape, including that the attempt fails entirely rather than naming the previous card); a fresh [B, B, B] burst identifies Chill via FRA 297. **FIX SHIPPED (App.tsx `capturePreparedBurst` + rapid.ts):** before a retained frame leads, ONE probe grab is taken; `retainedFrameMatchesScene` compares 24×34 guide signatures with the Pass 27 gain-normalised `sceneDistance` (threshold `RETAINED_SCENE_MAX_DISTANCE` = 14; measured separation on rig fixtures: same scene ≈ 0–7, card swap ≥ 28.7, card→empty tray 43.2, pure gain change ≈ 0). On a match the probe becomes burst frame 2 (no capture wasted, retention still leads — Pass 15 behaviour preserved); on a mismatch the retention is dropped and the probe leads a normal fresh burst; a failed probe grab keeps the historical unconditional behaviour. Net per-Snap state change: a retained frame is now consumed only when it provably shows the current scene; everything else about Snap state (tracker folds, detection authority, worker generations) is unchanged and was measured healthy.
- **(a2) CONVICTED — DETECTOR SLAB MERGES THE BOX'S RECESS SEAMS INTO THE CARD SPAN (the panels' crop geometry).** Fixtures `tests/harness/prep_p28b.py` (deep box, soft seams) and `.cache/p28hard` (deep box, HARD recess seams + a tray-floor seam), same seats as the flat set. Soft seams: geometry perfect, all 5 cards identified fresh. HARD seams: Griffin and Chill detections ran to the recess corner (raw x0=0.041 = the recess edge) — read-rect topCut −0.185 / −0.169, title centroid at 0.23 / 0.22 of the crop, beyond every title window incl. rescue bands (reach ≈0.19); full pipeline (`probe-p28-hardpipe.ts`): **Griffin = "no readable text"** with debris reads (the panel-1 shape exactly), Chill's title dead the same way (its bottom line rescued it in the bench run; on the phone the opposite merge direction bleached the bottom instead — same defect, other end). Mechanism: the recess seam band + the tray gap fit inside the span finder's 8% gap-merge tolerance, so the slab bridges from the card to the recess walls; borderless/edge-blended cards are the vulnerable class. A "void-bridge ban" in `longestRun` was considered and REJECTED — small cards' smooth art rows sit below the rough threshold and rely on the same bridging (Pass 9 Treasure Keeper lesson). **FIX SHIPPED (`trimSeamBand`, detect.ts):** after span selection (refined spans, and the rough fallback), each span edge is trimmed when it shows the seam signature — a dense band ≤5.5% of the axis (thickness is a frame property, not a span property), then a SUSTAINED void (≥ max(6, 3% of

span) samples below 35% of the span's interior-median density, judged on a ±3 max-filtered profile so double-line seams and ≤5-row in-card art dips don't confuse it), then the sustained mass; bounded to 22% of the span per edge (the convicted fixtures trim 13–14%; pickSpan's 55% discipline untouched). A real card edge never has a void behind its border line — the title/text mass is contiguous — so the trim only removes detached seams. Post-fix geometry: Griffin topCut −0.030, Chill −0.029, title centroids 0.11; pipeline: **Griffin identifies (M20 20), Chill identifies (FRA 297)**; flat and soft-seam sets byte-comparable (all title centroids 0.09–0.12 before and after).

- **(b) HARD-FRAME VERDICTS (measured separately, not shipped as decay fixes).** Guiding Hydra (FRA 360, borderless title-over-art): geometry is FINE on every seam variant (topCut ≈ −0.02); its title channel reads nothing usable even when settled (translucent banner over busy art — the pooled title reads are debris); it settles via the bottom line ("R 0360 FRA" → FRA 360) when that read lands, and repeated identical-input attempts flip between settle and no-read with byte-identical rects (`probe-p28-hydra.ts`: 4/5 settled) — the variance is read-stage timing under load, not framing. On the phone its panel-1 failure is the (a2) crop (the straightened image shows the box interior) compounded by the image timeout. Chandra, Chill of Compliance (FRA 297): on hard seams it was an (a2) victim (title row pushed to 0.22 of the crop); post-trim it identifies via its number. Its phone panel's DOWNWARD-shifted crop (title row excluded, tray strip below) is the same seam-merge class at the bottom/floor seam rather than the top. Chandra, Torch of Defiance (FRA 309): geometry fine in all bench variants; identifies via bottom-line salvage. Planeswalker name rows sit at the extreme top edge (~1–3% of card height), so any residual upward crop error hits them first — the trim + the Pass 26 title aim (reach to crop 0.19–0.22) cover the measured error range, but a crop displaced by more than that loses the row outright; no additional PW-specific tuning was shipped on this evidence.
- **FIXTURES + SUITES.** New: `tests/pass28.test.ts` (**16 checks, all pass** — trimSeamBand shape matrix: seam+void trims, contiguous band doesn't, thick band doesn't, short void doesn't, end-edge mirror, degenerate inputs, 22% bound; retainedFrameMatchesScene: identical/gain-change match, different card and empty tray reject, missing/mismatched signatures keep historical no-veto behaviour); harness `snapsequence.ts` (+ `results/p28-snapsequence-box.json`), `probe-p28-stale.ts`, `probe-p28-hardpipe.ts` (pre/post logs), `probe-p28-hydra.ts`, `probe-p28-detect.ts` (overlays of Andrew's real rig frames persisted as `results/p28-detect-*.png`). **Suites: all 32 test files green — 892 PASS lines (876 baseline + 16 new), zero failures. Client + server typecheck clean.**
- **FINAL BATTERY (changed code, `results/p28-final.json`): Tier 1 121 correct / 1 wrong / 10 no-read of 132** vs the same-session untouched control (`results/p28-baseline.json`): **124 / 1 / 7** (the 1 wrong is str-raft-4551 correct-at-2 in BOTH runs — pre-existing). Per-record verdict diff over all 156 records: exactly **3 changed, all correct-printing → no-identification with 3–4× wall-clock inflation** (num-prism-array-box40 1192→4572ms, num-prism-array-calib 2708→3289ms, num-courier-bat-box40 1194→4881ms — the final run consumed ~325 CPU-seconds in a contended window). Isolated rechecks on the changed code (`results/p28-rc-prism-array.json`, `p28-rc-courier-bat.json`): **all three records correct-printing again**

(1871–2732ms) — deadline starvation, not a code path this pass touches (the harness pipeline does not exercise the App-level retention validation at all, and none of the three records' rects carry a seam shape). **Zero new wrong settles; zero genuine regressions.**

- **PHONE-ONLY REMAINDER (Andrew's):** 3–4 Snaps in a row with NO refresh, including one planeswalker — every one should identify; if any panel still fails, its **Rect line** (and whether it says "frame: retained") is the evidence that separates a residual crop error from a stale retention. The bench cannot reproduce his exact recess lighting, so the hard-seam verdict in particular awaits his rig.

Pass 29 (2026-11-10) — Rapid duplicate at slide-in: two convicted mechanisms; two measure-first verdicts

Andrew's report (2026-11-10 ~16:07 EST, panels 1000044684/4686/4688): Rapid overall "working very well", but "sometimes, after it's done scanning a card, it reads it again, and does an extra one of that card. Instead of another card, when I slide it in. Could be a lag of some sort." All three panels are RAPID Did-you-mean PAUSES on the Pass 28 build with the artwork channel CORRECT on all three (zero wrong cards): (1) Tome Blast SOS 135 — image d=36 decisive margin ∞, demoted by the degraded-capture guard ("frames unstable: best d=36, median d=56"); OCR garbage; bottom skipped (bleached). (2) Rank Rat FRA 61 — image d=52 band; title band thumbnail CRISP yet pooled raw read = alphabet soup incl. the artist line. (3) Last Gasp FRA 56 (foil) — image d=42 decisive margin ∞, demoted by the guard (dim/unstable). Localisation harness: `tests/harness/rapidsequence.ts` — the REAL RapidMachine + real pipeline (`identifyBurstHarness`, production fusion) over synthetic slide transitions at the watcher's 250ms cadence (panel pair Tome Blast → Rank Rat at box gain 0.62; control pair Treasure Keeper → Griffin Protector at gain 1.0; transitions 200/400/600ms, a 900ms mid-slide HOLD variant in both slide directions, both settle kinds — pause scenarios take attempt 1's branch from the panel's recorded outcome via `forceOutcome1`, the full computation still runs and logs). The image channel's vision proposals are the one stand-in (documented in the file): proposals = the dominant card of burst frame 1, fingerprinted with production `selectReferencePrintings` / `rankReferences` / `consensusRank`.

- **S2/S4 REFUTED in code and sim.** The Rapid watcher path DOES apply Pass 28's retained-frame scene probe (same `capturePreparedBurst`); panels printing "frame: retained" is the normal watcher flow. No in-flight second attempt can complete after a settle (watcher is busy/paused-gated; late pipeline results are AttemptGuard-aborted). Continuous 200/400/600ms slides NEVER fire mid-transition (stability needs ~500ms of stillness; B fires 900–1150ms after the swap starts) and never duplicate.
- **(1) CONVICTED — FIRING ON A STABLE BUT INCOMPLETE PRESENTATION (the slide-in duplicate).** With a mid-slide HOLD (the slide pauses ~900ms while the user reaches for the next card), the held partial scene is perfectly stable, so the machine fired on it in BOTH slide directions (down-queue: burst = mixed A 76% + B 60% Franken-frames; up-exit: burst = the outgoing card's bottom 38%). In the harness those attempts fail (garbled bands; B waits a full

retry cycle, settling at swap+3650 instead of +2150) — but on device the burst is a legitimate partial view of the outgoing card (title/art/text largely visible) and the image channel can settle it: an extra copy of the previous card, added exactly "when I slide it in", and the new card missed in that cycle. **FIX SHIPPED:** stability is not completeness. In box mode the watcher now judges the fire frame's own detection against the rig's seat (`presentationComplete`, detect.ts: centre displacement ≤10% of frame on both axes, height ratio 0.86–1.16 — his seating variance is 3–6% and the calibrationStale low-seat class at 8% still passes; a 20%-of-travel hold fails at 0.15, the 45% holds at 0.34) and hands incomplete fires back via `machine.deferScan()` (returns to watching with fire-point stability kept and the attempt budget unspent; the App caps consecutive deferrals at 8 ≈ 2s so a misjudged seat can never wedge Rapid). The `--probeup` measurement added a second trigger: a partial slab touching the frame edge yields NO detection at all (detectInCanvas null at all three hold ticks), so in box mode with a known seat, a missing detection on the fire frame also counts as incomplete. Post-fix sim: both plateau variants defer twice during the hold and fire only on the seated B (B added at swap+4750 vs swap+6250 pre-fix); clean-swap timing unchanged (B fires swap+900 on the 400ms control).

- **(2) CONVICTED — A SWAP PERFORMED ENTIRELY DURING THE DID-YOU-MEAN PAUSE WAS INVISIBLE (the missed card).** All three panels sat in the paused state, when the watcher fed the machine nothing; at pick time `resumeRapid` re-anchored the machine to the picked card's signature and it compared only endpoints. Measured (`rapidsequence.ts --measure`): the normalised scene distance between the two seated panel cards at box gain 0.62 is **16.2 — below the box same-card threshold of 20** (at gain 1.0 the same pair reads 26.0; dim light compresses dark cards together; same-card noise is 0.3), while every transition/empty scene during the same swap measures **33.4–41.5**. Pre-fix sim: after a 400ms or 600ms swap completed under the sheet and the user picked A, the machine believed A was still seated and B was NEVER attempted (B added ×0, deterministic at both durations). **FIX SHIPPED:** the watcher keeps feeding the machine SUPPRESSED samples while the sheet is up (`sample(..., { suppressed: true })` — handled-phase departure rules and watching-phase stability run, a scan decision is held with stability pinned at the fire point, due retries are held too), so the swap is OBSERVED as a transition, not inferred from endpoints; `resumeRapid` now calls `machine.resumeFromPause(sig)`, which keeps the re-armed watching state when a departure was observed (`departed` flag, set by the handled phase's own change/absence rules) and otherwise behaves exactly like the old reset. Post-fix sim: B fires on the FIRST live tick after the pick and settles (both durations). Documented residual (unit-pinned): a swap whose transition falls entirely between two watcher ticks (sub-~250ms, under the sheet) still endpoints-compares below threshold and stays handled — the latch needs one observed transition tick, the same limit the live path has always had. Pause-after swaps (slide after the pick) were already clean and remain so.
- **ITEM 2(a) — FOIL SHIMMER vs FRAME STABILITY: NOT CONVICTED on synthetic evidence; no change shipped.** `tests/harness/probe-p29-foil.ts` (Last Gasp FRA 56 foil with its real impostor field — Lightning Bolt/Chain Lightning among 24 refs; Tome Blast SOS 135; ×12 burst triples × noise / seat jitter / phase-swept sheen / sheen+jitter at box gain): the winner is

the TRUE card in 96/96 triples and CONSTANT across frames in 96/96; per-frame spread stays small (median p50 18–24 across conditions, max 30; (median−best) max 12) — the synthetic sheen model does NOT reproduce the panel's best-36/median-56 spread on a still card, so the panel's instability involved inter-frame change this model doesn't capture (specular blowout — panel 3's own title thumbnail is near-black — or micro-motion), not shimmer alone. Corroborating, from the persisted Pass 23 matrix (3 runs × 1,440 cells): among 194 dim decisive-consensus cells, ALL 9 impostor winners sit at margin 8–10 (the margin leg alone demotes every one) and ZERO correct winners with margin ≥12 had median >44 — the median leg never independently decides anything in the synthetic population either. The candidate replacement (stability = winner persistence framesWon ≥2 + median ≤56, margin leg untouched) is recorded here with its numbers, but per the measure-first rule it does NOT ship: no available fixture produces the panel's shape (true winner, margin ∞, median 45–56) NOR a margin-≥12 impostor in that band, so the separation the guard change needs cannot be demonstrated. On device the two panel cards remain top suggestions with the "Use this artwork match anyway" affordance — Andrew's 90% rule is still served one tap away on this narrow class, and the guard that exists for measured colour-twin wrong settles is not weakened on anecdote.

- **ITEM 2(b) — RANK RAT POOLING: the pooler is EXONERATED; displayed band ≠ read band on the phone.** `tests/harness/probe-p29-rankrat.ts` traces every title treatment per frame at three geometries. Control (centroid 0.112) and panel (rect y0=0.00, centroid ~0.03): the standard treatment reads debris, but the rescue treatments (offset+4, multi-line tall) produce clean reads ("Rank Rat", "A Rank Rat", "Bi Rank Rat") and the pooled ranking is **Rank Rat 1.000** in both — when a clean read exists anywhere in the pool, pooling keeps it. The loss reproduces only in the panel-deep geometry (card ≥5.5% above the guide top, title centroid 0.004 — glyphs clipped by the crop's top edge): every treatment reads the type-line debris field and the pool tops out at Steeple Creeper 0.517. The phone panel's pool was dominated by the ARTIST line ("Frank ih mm A" — the card's bottom edge), which neither fixture's title path produces: on the phone's burst frames the read bands sat somewhere the crisp title wasn't (per-frame rect divergence across the burst — the panel prints only the head frame's rect — or a clipped title as in panel-deep), while the panel's thumbnail shows the head frame's corrected dewarp. The image channel named the card correctly regardless (d=52 suggestion). No pooling change ships; the fix surface for that class remains rect stability (Passes 18/24a) plus the image channel.
- **FIXTURES + SUITES.** New: `tests/pass29.test.ts` (**25 checks, all pass** — suppressed pause sampling: departure observed → re-armed during the pause, resume keeps it, first live sample fires on B; no-departure control stays handled and never re-fires; the documented unobserved-swap residual; due-retry held while suppressed; deferScan bookkeeping; presentationComplete seat matrix incl. low-seat pass and merged-slab/clipped fails). Harness `rapidsequence.ts` (with `--measure` and `--probeup` modes) + per-scenario logs `results/p29-rs*.log` and `results/p29-foil.log`, `results/p29-rankrat.log`. Full transition matrix on fixed code: auto 200/400/600 ✓, pause-after ✓, pause-during 400/600 ✓ (were B×0), plateau down/up ✓ (deferrals, no wasted attempt), still-seated 30s = exactly 1 attempt

✓ (Pass 27 invariant), controls ✓ with clean-swap response unchanged. **Suites: 917 PASS lines (892 + 25 new), zero failures, all 33 files. Client + server typecheck clean.**

- **FINAL BATTERY (changed code, `results/p29-final.json`): Tier 1 distribution IDENTICAL to the same-session untouched control (`results/p29-baseline.json`) — 125 correct-class / 0 wrong of 141 — with ZERO changed records across all 156.** Expected shape: this pass changes only the Rapid machine, the watcher, and adds a detect helper the battery pipeline does not exercise; the battery's role here is collateral-damage proof on the shared paths, and it is perfectly clean.
- **PHONE-ONLY REMAINDER (Andrew's):** a bulk Rapid slide-in run of ~10 cards — deliberately pause mid-slide on one swap, and swap one card while the Did-you-mean sheet is up — expecting: no duplicates, none missed, and no stall (deferrals cap at ~2s). Also a re-presentation of Tome Blast / Last Gasp: they should still present as top suggestions (Item 2a deliberately unchanged) with "Use this artwork match anyway" as the one-tap settle. The harness mirrors the App's watcher/machine wiring at the decision level (same epistemic class as Passes 15/28); his run is the end-to-end verdict.

Pass 30 (2026-11-10) — Light in real darkness: tracked-rect collapse, On that actually forces light, manual slider

Andrew's report (2026-11-10 ~18:41 EST, panels 1000044705/4703/4697/4701): the first genuine low-light session of the programme (dark evening, 18:32–18:36; all prior light work was proven on synthetic dim fixtures). "Looks like its having trouble with bad lighting. The light isnt working please fix it. Make sure the auto lighting works. The on light works. The manual from off low medium and high but have a slider so i can gradually increase it and decrease it." Both attempts SNAP in Box mode; chips Auto ✗ / On ✗ (torch absent in his in-app browser, honest note printed); both took the full 12s image-timeout path, so the OCR floor carried alone. Localisation fixtures: `tests/harness/prep_p30.py` → `.cache/p30dark` (Borg Queen TRC 197 + Swamp ONE 274 composites at his box seat under a low-right light pool / uniform dim), probes `probe-p30-dark.ts` (geometry + tracker sequence) and `probe-p30-pipe.ts` (production pipeline; full log `results/p30-pipe.log`).

- **ITEM A — TRACKED-RECT COLLAPSE: CONVICTED, mechanism + guard shipped.** Reproduction on production code (`probe-p30-dark.ts`, pre-fix): in the dark scene the detector returns ONLY the lit text box — raw (0.30,0.52)-(0.79,0.84) at confidence **0.84**, corrected (0.30,0.40)-(0.80,0.90), covering **40%** of the true card (his panel: (0.39,0.40)-(0.81,0.81)). Audit answers: (i) tracker ingestion validated NOTHING — the seeded branch of `observeTrackedRect` replaced the stored calibration with the disagreeing quadrant on the FIRST preview tick (no confirmation at all), and the re-lock branch adopted any two agreeing far detections, which a persistent dark quadrant satisfies trivially; (ii) the Pass 24a "verify the lock against the current frame" step checks only lock↔detection AGREEMENT (≤0.03 edge delta) — a quadrant agrees with itself perfectly; (iii) `prepareCardFromFrame`'s tracked-fallback read the tracked rect with no plausibility check, and the App's preview tick set "✓ Card found

— fine at this distance" from the raw quadrant detection. The chain reproduced end-to-end: 5 preview ticks → lock = quadrant (cover 0.40); Snap frame detects nothing → read origin tracked-fallback, cover 0.40, title never in the crop. **FIX (`rectPlausibleVsSeat` + `effectiveSeatRef`, detect.ts):** a candidate rect is plausible as the card at the seat only with area ratio 0.6–1.6 of the seat AND ≥55% seat coverage (measured separation: quadrant 0.42 / 0.39; legitimate displaced/low seats 0.86–1.15 / ≥0.82; the top-amputated detection class at 0.86 area stays plausible — the seam-trim/aim passes own it, and the merged-slab class stays plausible for the same reason). The gate is applied at all three ingestion points: `prepareCardFromFrame` treats an implausible detection as a miss (it is also withheld from `detectedRect`, so it can never be folded), `observeTrackedRect` refuses implausible detections entirely (no fold, no re-lock arm, no freshness refresh — judged against the calibration while the lock matches it, else against the lock itself via `effectiveSeatRef`, so a legitimately re-locked seat is not frozen), and the tracked-fallback branch only serves a tracked rect that is itself plausible against the calibration, else the chain falls through to the calibration seat. The App passes the stored calibration into `observeTrackedRect` as the seat, gates the manual preview tick's found-card/retention on the same predicate, and flags a surviving implausible lock in the live view (amber outline, "TRACKING — OUTLINE TOO SMALL, CHECK LIGHT", hint replaced — never "fine at this distance"). Post-fix probe: every dark tick reads the seat (origin tracked-fallback, cover 1.00), the lock never leaves the seat, and the no-detection Snap reads the seat. Pipeline proof (`probe-p30-pipe.ts`): B0 = the quadrant crop itself (what P2 read) → no-read/tooDark; B1a = Snap frame + a poisoned quadrant tracker in state → origin **calibration**, the seat; B1b identical.

- **ITEM B — LIGHT PIPELINE AUDIT (measured answers). (1) STILL vs VIDEO pooling: the pooling is ONE code path** — Snap (video or still), Rapid and Take Photo all consume the same `identifyBurst`, which pools the lifted title specs (standard + standard-low) and the lifted bottom spec from frame 1 under the same caps (S1's harness record: liftedTitleReads=2, liftedBottomReads=1). P2's suspiciously short 787ms title stage is explained by the crop, not missing pooling: the quadrant's bands are tiny, so garbage reads return fast — and the lift it pooled was computed for the quadrant's pixels (Item A's crop). One real still-path amplifier, now moot: pre-fix the quadrant head (height 0.41 < 0.45, source "detected") triggered the hi-res STILL retry, so P2's whole burst became still quadrant crops; post-fix the head is the seat (0.80) and no spurious retry fires. **(2) P4's shape on the bench: THE PIPELINE READS IT.** S1 (Swamp ONE 274, full-card rect, title zone p90 = 74, Auto lift ×2.4): **confident — Swamp ONE 274**, landed by the bottom line ("274 … ONE EN" → collector-number pin), title reads "…of Swamp"/"I Swamp" in the pool. The phone's divergence is localised to its own attempt record: its tracked rect sat displaced high ((0.12,0.00)-(0.86,0.73)) — inside the plausible band, so no guard fires — and its **bottom stage recorded 0ms**: the number rescue that lands this exact shape on the bench never ran (the title stage burned 2.0s of capped reads first). Residual named honestly: the seat guard kills the sub-rect class (coverage < 0.55); within-band lock drift in the dark is bounded by the median + seat anchor but not eliminated, and no tighter bar was shipped — the battery's legitimate displaced seats live inside that band and the evidence does not convict a specific tighter threshold. Preview metering vs the attempt latch (the P1 "exposure was raised and the card re-captured" note): that note is the capture dark-

frame retry, which fires during capture, BEFORE the attempt latch is taken, and registers itself via `noteExternalLightApplication` — the attempt-start decision cannot immediately undo it (dwell), and `stepLightLatch` is a strict no-op while an attempt holds the latch (unit-proven below). No fight; no change. **(3) AUTO END-TO-END (unit/harness level, stated as such):** `tests/pass30.test.ts` drives the production latch + lift decisions through a scripted dark session — dark preview (p90 50) engages by itself (`engage`), the attempt latches it, wild preview swings mid-attempt change nothing, a bright tick inside the 4s dwell does NOT release (no heartbeat), bright + dwell elapsed releases exactly once; the lift side reproduces both panels' own numbers (p90 50 → ×3.5; p90 71 → ×2.5). **(4) ON: CONVICTED GAP, FIXED.** Pre-fix, `liftForMode("on")` was the Auto metering gate verbatim (bright → NO lift at all; dim → the metered gain, not the cap; crushed → no lift + title skipped) and the hardware side forced only the torch, never exposure compensation — in the read path On did nothing Auto didn't, which is exactly what "the on light isn't working" looked like from his side of the glass. Post-fix On is an override like Manual, at the cap: `liftForMode("on")` = MAX_LIFT_GAIN (×4) on every band, the crush skip no longer applies in On (App + harness mirror), the panel line reads "light on — lifted ×4.0 (title zone p90 = N)", and the hardware path (`applyForcedLightHardware` , shared with Manual) forces torch + exposure compensation at max on mode entry and on every track (re)start. **(5) THE HONEST CEILING, with numbers:** software lift multiplies the photons already captured; it cannot create new ones. Measured classes (Pass 17 + this pass): title zone p90 ≥ ~32 → the lift recovers readable titles (B2's seat crop at p90 39 read its title under ×4); p90 ≤ 26 (crushed) → lifted reads manufacture confident garbage, so Auto settles tooDark honestly instead — B1b's seat crop of the darkest fixture meters p90 11 and settles tooDark even with the rect fixed. His panels' attempts metered p90 50 / 71 on their (wrong or displaced) crops — INSIDE the recoverable band — so with the rect anchored at the seat, Auto's lift is expected to carry his dark room; but a frame as dark as the p90-11 composite is beyond software in ANY browser: the only cure is photons, and in his in-app browser the torch does not exist (the panel copy keeps saying so, with the Chrome note). In Chrome, Auto/On engage the real torch and the ceiling moves with it. **Fixture observation, recorded not tuned:** B2 (Borg Queen at p90 39, ×4 lift) settled "Manifest" (tsoc 30) from partial title reads ("...Queen, Perfection Manifest") — the pooled matcher's subset scoring crowned the one-token name over the four-token full name. Identical pre/post Pass 30 (the matcher is untouched and the battery is clean); not in Andrew's panels (his attempt never got a readable title). Recorded as a candidate for a future matcher pass — not tuned on one synthetic case.

- **ITEM C — MANUAL SLIDER SHIPPED.** Manual mode's discrete levels become stops on a continuous gain slider (autolight.ts: `MANUAL_PRESETS` Off ×1.0 / Low ×1.5 / Med ×2 / High ×3 / Max ×4; `MANUAL_GAIN_MIN/MAX` = 1/4; `clampManualGain` , `parseManualGain`). The slider (44px touch target, in the Manual panel under the Light chip) and the preset buttons drive ONE state (`manualGain`) both ways — presets move the slider, the slider's value un-highlights off-stop presets; touching the control selects Manual mode by construction. Persisted as `manabox-light-manual-gain` ; the Pass 22 level index migrates to its gain on first read. The value feeds the SAME `liftForMode("manual")` the levels fed, is sampled once per attempt

(identifyBurst), and hardware applications defer to attempt end (`pendingManualHwRef` drained when `reading` clears) — never mid-attempt. Hardware scales with the gain where exposed: torch on for any gain > ×1.0, exposure compensation = `manualExposureValue` (linear ×1→0, ×4→max). The panel light line prints the exact value ("manual light ×2.6", unit-pinned). In the in-app browser the truth is unchanged and still printed: exposure boost only, torch in Chrome.

- **FIXTURES + SUITES.** New: `tests/pass30.test.ts` (**49 checks, all pass** — seat-plausibility matrix incl. Andrew's verbatim panel rect, effectiveSeatRef, the tracker gate (seed never replaced by quadrants, freshness not refreshed by garbage, healthy folds + displaced re-lock intact, no-calibration lock reference), On semantics + verdict line, the scripted Auto dark-session sequence, the slider model incl. persistence migration and hardware scaling); one Pass 22 check revised with its history recorded inline (On's lift semantics — the old assertion pinned the behaviour Andrew reported broken). Harness: `prep_p30.py` , `probe-p30-dark.ts` , `probe-p30-pipe.ts` (+ `results/p30-pipe.log` , overlay `results/p30-dark-borgqueen.png`). **Suites: 966 PASS lines (917 + 49 new), zero failures, all 34 files. Client + server typecheck clean.**
- **FINAL BATTERY (changed code, `results/p30-final.json`): Tier 1 124 correct / 1 wrong / 7 no-read of 132** vs the same-session untouched control (`results/p30-baseline.json`): **121 / 1 / 10** (the 1 wrong is str-raft-4551, correct-at-2, in BOTH runs — pre-existing). Per-record diff over all 156 records: exactly **3 changed, all improvements** (exif-zedruu-3-oriented no-id → correct-printing; userphoto-1000044050 and -1000044317 no-id → correct-name — all three are the known deadline-starvation flip class: baseline runs took 4.5–4.9s under triple job contention, final runs 2.1–2.8s). **Zero new wrong settles; zero regressions.**
- **PHONE-ONLY REMAINDER (Andrew's):** same dark room, same two cards. Borg Queen: Snap should now read from the seat outline (panel Rect line should say "calibration (seed)" or a full-card "tracked (auto)"), and in Auto the lift should engage by itself and hold steady (no flashing). The ONE Swamp #274: expect the bottom line to pin ONE 274 as it does on the bench. Then the Light asks in order: **On** in the dark (panel line should read "light on — lifted ×4.0"), **Manual** — drag the slider slowly from ×1 to ×4 and back and confirm the boost follows gradually, presets jump the slider, and the value survives leaving and returning. If a Snap still fails in the dark, its panel's Light line (the p90) separates "too dark for software" (p90 ≤ ~26 — needs real light/Chrome's torch) from a pipeline miss.

Pass 31 (2026-11-10) — Why the 211–339ms title stages closed + camera focus audit

Andrew's report (2026-11-10 ~19:30 EST, panels 1000044707/4709/4711/4713/4715/4717): three failed Snaps — Eye of Jace ×2 (FRA 170), Stingerquill Annex (FRA 190) — plus his questions: why didn't it recognise these, will lighting work in the eventual native app (answered in chat: yes — the limits are browser-imposed; NOT a build item), and "make sure the camera is in focus." CONDITIONS, stated plainly: the Light chip reads **Off ✗ in every live view of this batch** — no light mode was engaged, so this batch is NOT an Auto/On verdict and no lift treatments were

owed; Foil ✓, Box on; scan list 47 cards / $22.58 (some scans landed). All three attempts: full-card rects, retained frame leading, image service timed out at 12s, bottoms skipped near-blank. Localisation: `tests/pass31.test.ts` (timing arithmetic + focus matrix + synthetic defocus sequences) over the production modules.

- **ITEM A — THE TITLE STAGE, TRACED (which reads, when, why it closed). (a) THE POOL:** only the frame loop's per-frame reads — one aimed standard-band read per burst frame (×3). Each returned FAST with LONG garbage (E1 'EE Wve pre ERP SN Sp -', E2 '9 eg a We Lbs as tits'): long enough that `ocrBand` 's shape heuristics (garbage <3 letters / fragmented token soup / thin <8 letters) did NOT trigger its PSM 6 pooling pass, so each frame cost a single PSM 7 pass (~70–110ms on the phone) and the stage totalled 211/240/339ms. S1's 339ms is the same shape ending in no-readable-text (match 0ms — its pool never scored a candidate). The matcher RAN on E1/E2 (match 1.8s): `pooledMatches` re-scores the whole 35k index per new pool signature, so 1.8s of match over a garbage pool is expected arithmetic, not a second defect. A full burst title stage that same evening (Swamp, Pass 30 batch) ran ~2.0s because its reads took the slow paths (PSM 6 pooling + lifted copies — that attempt had Auto engaged). **(b) RETAINED FRAME:** it LED frame 1 in all three attempts (every panel: "frame: retained"). Its read was the pool's first evidence; Pass 28 validates its SCENE, nothing validated its FOCUS (see Item B). It could not settle alone (garbage is never decisive) and did not dominate a verdict — but a soft retained frame's garbage is pooled at equal weight with settled frames' reads. **(c) THE CONVICTED STEP — RESCUE-2 STARVED BY THE IMAGE WAIT'S CLOCK.** The Pass 26 completing read DID run (one tall multi-line read, here on the last frame; on these defocused/foil-sheened bands it also returned garbage and pooled). Snap's Rescue-2 pass (offset+4/+8, tall, left-shift, deskew ×4, lifted where owed) is scheduled in `identifyBurst` AFTER settlement's image fusion wait, on the SAME `AttemptClock` . The wait consumed its sanctioned budget (IMAGE_WAIT_DEADLINE_MS 12s — every panel's "Last scan: 12.0s"); the rescue then called `clock.extendTo(ATTEMPT_RESCUE_DEADLINE_MS = 4000)` , which cannot widen a clock already ~12–14s elapsed (extendTo only widens to an absolute budget), leaving remaining ≈ 0 — so EVERY rescue `cappedRead` returned `skipped` without starting (`if (rw.skipped) break rescueBands`). The treatments existed, were scheduled, and never ran; the title stage therefore stayed at the frame loop's 211–339ms. Pinned arithmetically in pass31.test.ts A1/A2 (extendTo(4000) after a 12s wait leaves 0ms; a starved clock skips reads without starting them; a hung read under a granted budget is still abandoned at cap — the deadline guarantee is untouched). **(d) NO read-count early exit exists** — the frame loop always ran all 3 frames; the stage closed fast because the reads were single fast passes AND the post-wait rescue was starved. Settling fast on a CLEAN decisive read is unchanged and correct; the defect was settling on garbage because the remaining treatments were unreachable. **FIX SHIPPED:** `AttemptClock.extendByFromNow(ms)` (scanflow.ts) + `RESCUE_AFTER_WAIT_BUDGET_MS` = 3500: Rescue-2 is granted a fresh ~3.5s budget from the moment it starts, whatever the image wait consumed. The fusion wait, the interlock order, the per-read caps and the watchdog shields are untouched; Rapid never enters Rescue-2 (unchanged). Bench consequence: E1/E2/S1-shaped attempts now actually execute the offset/tall/deskew treatment set before the pool is judged — on the phone fixtures (dim + foil

sheen + Light OFF) whether those treatments read these two titles is exactly the phone-only retest below; no bench fixture can reproduce his foil-under-box-light bands honestly, and none was fabricated for the claim.

- **ITEM B — FOCUS AUDIT + SHIPPED CHANGES (`client/src/focus.ts`, pure + unit-tested). AUDIT TABLE (before):** getUserMedia requested facingMode + 1920×1080 ideals and NO focus constraint (track starts in the browser's default mode); `applyCameraLocks` ran only at track start / box toggle — freehand asked continuous AF/AE where exposed, but BOX mode asked `focusMode: "manual"` **with no focusDistance** (manual without a distance parks the lens at an unmeasured default — not the tray distance — while the UI called it a fixed-rig lock); nothing re-asserted focus after later `applyConstraints` writes (exposure raises, torch, dark-frame retry), which can reset sibling constraints on some Android stacks; ImageCapture stills inherit the live track's focus state; Take Photo uses the OS camera (its own AF — not app-controllable, and not a defect); NO sharpness measurement existed anywhere in the app. **SHIPPED (capability-guarded, torch pattern — a track exposing no focus modes behaves exactly as before, minus the modeless-manual park):** (1) `focusPlan(caps, boxMode)` — continuous AF wherever exposed in BOTH modes (rig distance is fixed, but card thickness/sleeve/phone shift are not, and continuous holds the tray plane), single-shot next (applying it TRIGGERS an AF cycle), "manual" never selected without a distance; freehand also gets continuous AE, box gets no exposure lock (Pass 4 rule kept). getUserMedia now carries the continuous ideal so the track starts focused; `applyCameraLocks` is re-asserted at every burst start (which is also the single-shot trigger on card settle-in) and after the dark-retry exposure write. (2) **SHARPNESS SELECTION:** every prepared live frame's title zone is measured (`measureTitleSharpness`: gradient energy on the title zone — defocus is a low-pass filter and removes exactly that energy; measured on synthetic defocus fixtures: sharp text 71.9 vs blurred twin 12.4, uniform 0, and dim-but-sharp 36.8 still beats bright-but-blurred, so the metric reads focus, not light). The completing read now runs on the SHARPEST burst frame (was: last frame); Rescue-2 reads frames sharpest-first (a decisive read stops the pass before soft frames spend the budget); the retained frame may lead a burst only when it is not dramatically softer than the live probe beside it (`retainedSharpnessGate`, ratio 0.6 — a pre-settle retained frame is dropped to a fresh burst, with the reason published; unmeasured scores keep historical behaviour). Every attempt's panel gains a **Focus line** ("frame sharpness: a · b · c (sharpest #N; frame 1 is the retained viewfinder frame)" + any retained-drop reason) — the measured retained-vs-burst comparison, on every future screenshot. Focus-settling sequences (synthetic frames at +0/150/300/600ms as decreasing blur) sharpen monotonically in the harness and the settled frame wins selection (pass31.test.ts B3).
- **SIDE QUESTION — THE TWO OUTLINE STYLES (not a defect):** only ONE outline element ever renders (`viewOverlay` in App.tsx), with priority: box tracker lock → green "TRACKING CARD (AUTO)"; else the stored box calibration → green "CALIBRATED CARD AREA"; else the last live detection → blue-grey (#9fc3e8) "CARD FOUND". This batch's live views show both because the tracker held a lock in two views (Eye #1, Stingerquill's pair show TRACKING; Eye #2's later view and one Stingerquill view show CARD FOUND — the detectOverlay fallback

while no tracker lock was being mirrored). Read future panels by the label, not the colour family.

- **FIXTURES + SUITES.** New: `tests/pass31.test.ts` (**25 checks, all pass** — starvation arithmetic + granted/skipped/hung read behaviour; focusPlan capability matrix ×7; sharpness metric on defocus/brightness fixtures; settling-sequence order + retained gate matrix) and `client/src/focus.ts`. **Suites: 991 checks (966 + 25 new), zero failures, all 35 files. Client + server typecheck clean.**
- **FINAL BATTERY (changed code, `results/p31-final.json`): per-record diff against the Pass 30 final control (`results/p30-final.json`) over all 156 records — ZERO changed records, Tier-1 distribution byte-identical** (106 correct-printing, 7 correct-name, 6 correct-name-only, 5 correct-top, 1 correct-at-2 [the pre-existing str-raft class, identical in both], remainder honest no-identification/none classes; zero wrong settles in either run). Expected shape: this pass's pipeline changes live in App.tsx orchestration (rescue clock grant, read order, retained gate) and the additive scanflow/focus modules — the harness pipeline exercises the shared reader/matcher code, whose behaviour is unchanged, and its role here is collateral-damage proof.
- **PHONE-ONLY REMAINDER (Andrew's):** Eye of Jace + Stingerquill Annex retest in the same dim room — **with Auto or On engaged this time** (his Light was OFF for the whole batch above: information, not blame — with it off, no lift was owed and the rescue set was the only missing treatment; with Auto/On the lifted copies join it). Expect: the panel's title stage to show MORE than ~340ms when the first reads are garbage (the rescue treatments actually running), a Focus line naming the burst's sharpness, and — if the retained viewfinder frame was captured before autofocus settled — a note that it was dropped for softness. If a card still fails, its panel now separates the causes by itself: Focus line (soft capture), Light line (darkness vs p90), Stages line (which treatments ran).

Pass 32 (2026-11-10 evening, Andrew's verdict on Pass 31: "Any suggestions to fix this please. Make the corrections and fix" + six screenshots)

- **VERDICT CONTEXT.** Pass 31's WIN stands and is untouched: Stingerquill Annex auto-added in Rapid by artwork (d=26), honestly flagged printing-unconfirmed with a Detect version button; that attempt took 6.7s with NO image timeout; the list went 47 → 48 cards / $22.77. The Pass 31 rescue budget visibly FIRES on-device (title stages 3.1s / 3.3s vs the 339ms pre-fix shape) and the Focus line is live. The defects: two further Stingerquill Snaps (W1 19:47, frame: this burst; W2 19:49, frame: retained) died by WATCHDOG — "hung during: title" — with ALL stage times printed complete (title 3.1/3.3s, bottom ran 233ms / skipped bleached, match 1.4/1.5s, image timed out 12s). Sharpness: W1 9.5 · 5.9 · 9.6, W2 10.2 · 10.1 · 10.3 — every frame at/below the blurred bench level, none readable.
- **BASELINES (untouched code, this window):** suites 991 PASS / 0 FAIL (35 files); battery control = `results/p31-final.json` copied to `results/p32-baseline.json`.
- **ITEM 1 — THE WATCHDOG COMPOSITION, LOCALISED + CLASS AUDIT.** The watchdog's deadline was NOT a function of the attempt's sanctioned budget: it consulted three point-

registered shields (`imageWaitRef` , `imageShieldRef` , `imageCompletedRef` + `IMAGE_SETTLE_GRACE_MS`). The W arithmetic, from the code: the image wait times out at ~12s; identifyBurst then registers the fusion shield as `Date.now() + clock.remaining()(≈0) + 2500` ≈ **14.5s**; fusion consults (~1.5s — the panels' match times); Rescue-2 starts (~13.5s) via Pass 31's `extendByFromNow(+3500)` → sanctioned reads run to **~17s** with NO shield covering them (no image completion → no grace shield either). The watchdog's re-armed fire lands ~15s, finds every shield expired, and kills the attempt MID-RESCUE with its stages complete — the panels' exact shape. Third sanctioned wait the watchdog failed to defer to (Pass 20/21: the image wait; Pass 21: the completed-result grace; now: the rescue budget). **SANCTIONED-WAIT AUDIT (every extension in an attempt × its watchdog treatment):** frame-loop reads + completing read (inside ATTEMPT_DEADLINE_MS 2500 on the AttemptClock) — inside the 8s base before, clock-derived deadline after; bottom reads + number-rescue lookups (per-read caps ~1.2s, lookups raced ≤1200ms, inside the clock) — same; fusion image wait `extendTo` 4000 decisive / 8000 number-hit cap / 12000 uncertain — image-wait deferral (uncertain budget only) + fusion shield before, clock deferral covering all three budgets by construction after; the image CALL itself (IMAGE_VISION_CAP_MS 15000, never awaited past the wait; its completion handler publishes the result + buys grace) — completion-grace shield, unchanged (anchored to the channel's report time, which the read clock does not model — the reason shields remain as the second layer); post-image settlement (IMAGE_SETTLE_GRACE_MS 10000 from the report) — grace shield, unchanged; fusion consult tail (shield = remaining + 2500) — unchanged, and the clock margin now grants the same tail independently; **Rescue-2 `extendByFromNow(RESCUE_AFTER_WAIT_BUDGET_MS = 3500)` — NO COVERAGE AT ALL before (the W1/W2 kill); covered by the clock deferral after (in the W replay the deadline becomes start + 17000 + 2500)**; extendedBottomPin (one read capped by `clock.remaining()`) — inside the clock; settlement lookups inside identifyBurst (`resolveByName` uncapped await, salvage raced 1500ms, pin lookups raced 1200ms — all bounded by their own races or by the pipeline's settlement, which the guard watches); the slow read (`identifyCore` — no AttemptClock; per-variant OCR caps + one server read; runs under the PREVIOUS attempt's guard, so a settle makes the watchdog ignore it by the success-is-final rule) — unchanged, enumerated for completeness; the watchdog's own evidence rescue (`raceWithCap` 3000, inside the handler) — n/a. **FIX (class, not instance):** identifyBurst PUBLISHES the attempt's clock (`attemptClockRef` : attempt id + epoch start; the budget is read LIVE off the clock object at every firing), and the watchdog defers to `watchdogClockDeadlineMs = epochStart + clock.budget + WATCHDOG_SETTLE_MARGIN_MS (2500 — the same tail the fusion shield always granted)` whenever that deadline is in the future (`watchdogClockDeferral` , pure, scanflow.ts). The watchdog now fires only when NO live source (clock, wait, shields) has sanctioned work outstanding; any future sanctioned extension widens the watchdog automatically because the budget IS the deadline's input. `finish()` still settles the guard immediately, short-circuiting every deferral (settled → no deferral, action "ignore", empty recovery plan — settle-twice is impossible: the watchdog's rescue branch requires `!guard.isSettled` , and the guard's settle is idempotent). **REGRESSION (tests/pass32.test.ts, production functions only):** the W composition

replayed on a fake clock (wait extendTo 12000 → timeout, fusion shield 14.5s, rescue grant at 13.5s → budget 17000): at the 14.75s firing the PRE-FIX SHIELD SET ALONE returns defer=false (the kill, asserted as documentation) while the clock deferral defers to 19.5s and the attempt settles normally at ~18.5s with success final; the INVERSE also pinned — an unextended attempt trips at the base firing, an extended-but-dead attempt defers at 8s and trips just past its extended deadline, another attempt's clock never defers, and a hung read under the rescue grant is still abandoned at its cap.

- **ITEM 2 — SHARPNESS FLOOR, MEASURED + AF STATE. METHOD:** `tests/harness/prep_p32.py` rendered four real cards (Stingerquill Annex FRA 190, Eye of Jace FRA 170, Rank Rat FRA 61, Griffin Protector M20 20; Scryfall art crops in production card geometry) through Gaussian defocus radii 0–3.2, each also dimmed ×0.45 as a lighting control (40 fixtures); `tests/harness/probe-p32-floor.ts` scored every fixture with the PRODUCTION `measureTitleSharpness` and read its title zone with the production standard band against the full 36,519-entry name index (`results/p32-floor.json`). **FINDINGS:** bright ladders stay USABLE (unique decisive/usable crown = the true name; garbled reads score 0 and crown nothing) down to sharpness 18.5–22.3 per card and FAIL at ≤20.7 (defocus garbles the title into a WRONG decisive crown — "Tree of Tales" 0.582 over Eye of Jace at 19.1 — worse than no read); the dim controls (×0.45, unblurred) measure only **13.4–15.2 and ALL read usable** (3 of 4 decisive) — on real card art the metric is brightness-sensitive (dimming ×0.45 roughly halves it — Stingerquill 33.8 → 15.2), unlike Pass 31's synthetic pattern (dim-sharp 36.8), so the floor had to be set in the DIM operating regime, not the bright one; dim+soft measures 10.8–12.5 and is a coin flip (2 of 4 fail); the W bursts meter 5.9–10.3. **`SHARPNESS_FLOOR = 12`** (focus.ts): the highest value that never gates a readable dim-sharp burst — a floor at the bright transition (18–22) would gate dim-but-readable cards, a disguised LIGHT gate, forbidden — while gating every W frame. Frames between 12 and the bright transition still get the full pipeline with sharpest-first order; the gate refuses only bursts with NO frame at/above 12. **BEHAVIOUR (`burstFocusVerdict` + `captureBurstWithFocusGate` , App.tsx; both capture sites — Snap's `captureFromCamera` and Rapid's two burst paths):** an all-soft burst (measured frames only; 0 = unmeasured keeps historical behaviour and NEVER gates) → re-trigger autofocus (`retriggerAutofocus` : a single-shot application where that's the plan; on continuous tracks a single-shot cycle is fired and the plan re-asserted — best-effort, capability-guarded) → wait `FOCUS_SETTLE_MS = 600` (inside the brief's 400–700ms band) → ONE fresh re-capture (no retained frame) → if still all-soft: **Snap** settles FAST via `presentOutOfFocus` — a real failed-scan panel with the straightened image, the Focus numbers, the gate's own note ("frames out of focus — … re-capture after refocus still soft …") and "Those frames were out of focus…" — no 3s garbage burn, no watchdog death; **Rapid** quietly hands the burst back with a once-per-soft-streak hint ("Frames are arriving out of focus — …"). One re-capture max per attempt; the gate adds ≤ ~1.5s (settle beat + one burst) and NO clock extension — it runs entirely before identifyBurst creates the AttemptClock, so Item 1's deadline needs no term for it (the watchdog's subject is an identification attempt; a gated capture never starts one). **AF STATE IN DIAGNOSTICS:** `applyCameraLocks` records the requested focus mode (`focusRequestedRef`); every panel's

Focus line now appends `describeAfState(...)` — the track's exposed focusMode values, the requested mode, and the mode `getSettings()` reports applied — or, when the browser exposes no focusMode capability at all, the line SAYS so: "camera focus: not exposed by this browser (no focusMode capability) — autofocus can be neither requested nor verified here". Pinned in pass32.test.ts for the absent / empty / full / unreported configurations. **WHAT THE SANDBOX CANNOT DETERMINE (stated plainly):** whether Andrew's in-app WebView exposes focusMode, and whether it honoured Pass 31's continuous-AF request — the code paths are pinned, and his next panels will print the answer on the Focus line itself.

- **ITEM 3a — LIGHT-OFF DARK HINT (shipped, small):** the preview metering ticks (manual ~800ms tick and the Rapid watcher's light tick — both already metering the title-zone p90 for the Auto layer) now also feed `updateDarkOffHint`: when Light mode is Off AND the metered p90 is below `LIGHT_ENGAGE_P90` (130 — the threshold at which Auto WOULD engage), the live view shows an amber line: "Scene is dark and Light is Off — tap the light chip for Auto so the scanner can brighten the card." Non-blocking, clears on the next bright/card-lost/non-Off tick, and never changes his setting. (Two consecutive batches retested light fixes with the light off; the hint makes the unmet condition visible where he can act on it.)
- **ITEM 3b — FOIL FLAG AUDIT (trace + live contract test; NO mis-map found; one real gap fixed). TRACE:** the Rapid chip is the `batchFoil` state; every settle path computes `foil = (bottom-line separator star/bullet detection) ?? batchFoil` — Rapid auto-add (`runRapidScan`'s outcome.info.foilDetected ?? batchFoil; the watcher effect's deps include batchFoil so no stale closure), Snap (identifyFromCardCanvas, trySlowRead), Did-you-mean picks (selectMatch), "use this artwork match anyway", and the watchdog rescue (via `batchFoilRef`, kept in sync by an effect) — then `addToList(card, foil, ...)` → `api.addToScanList({ foil })` → the row (server merge key = scryfallId + foil, so a foil add and a non-foil add are separate rows) → the result sheet renders the ROW's flag verbatim (`★ Foil: on/off`, and the row price switches on the same flag). **LIVE CONTRACT (sandbox actions):** addToScanList with foil:true returned a row with foil=true; a second foil:true add MERGED (quantity 2) with foil still true (test row removed afterwards) — chip-on → foil row is proven end-to-end at the action layer. **THE WINNING ADD'S OWN EVIDENCE:** in Andrew's 19:49 live view (1000044729), cropped ×3, the Foil chip reads "★ Foil" with NO ✓ and inactive styling, while Auto-add beside it is gold WITH ✓ — the render is `★ Foil{batchFoil ? " ✓" : ""}` on the active-chip style, so the batch toggle was OFF for that session. The sheet's "Foil: off" is therefore the honest record of adds made with the batch flag off, and quantity 2 = two foil-off adds of the same card merged (had either add carried foil on, it would have formed its own row). **THE ONE REAL GAP:** `batchFoil` was the only batch-scoped preference NOT persisted — light mode, manual gain, box mode and the calibration all survive a reload; the foil toggle silently reset to off on any reload. It now persists (`manabox-batch-foil`, read at init, written on change). Merge semantics UNCHANGED (no evidence against them).
- **FIXTURES + SUITES.** New: `tests/pass32.test.ts` (**31 checks, all pass** — the W-composition replay incl. the pre-fix kill asserted against the old shield set, the genuine-hang inverses, settle-twice impossibility, the floor verdicts on the panels' own numbers + the ladder guardrails, describeAfState in all capability configurations), `tests/harness/prep_p32.py`,

`tests/harness/probe-p32-floor.ts`, `tests/harness/results/p32-floor.json`. **Suites: 1022 checks (991 + 31 new), zero failures, all 36 files. Client + server typecheck clean.**

- **FINAL BATTERY (changed code, `results/p32-final.json`): per-record semantic diff (kind / name / set / collector number / verified / class) against the same-window control (`results/p32-baseline.json` = the p31-final run) over all 156 records — ONE changed record, verdict-identical:** `dist-treasure-keeper-25` settled Treasure Keeper J25 #760 confident in both runs; only the credited evidence label differs ("title (pooled across frames)" in the control vs "title (tall band)" in the final — which band treatment's read the pool credited, a timing-order wobble; an isolated recheck reproduced "tall band" with the identical verdict). **Zero wrong settles in either run.** Expected shape: this pass's changes live in App.tsx orchestration (watchdog deferral, capture gate, hint, persistence), the additive scanflow/focus/catalogue code, and no reader/matcher code the harness pipeline exercises — its role is collateral-damage proof, as in Passes 29/31. No `pipeline.ts` mirror change was needed: the harness has no watchdog and no camera, and the floor/gate live at capture time, before the pipeline the harness mirrors.
- **PHONE-ONLY REMAINDER (Andrew's):** Stingerquill Annex + Eye of Jace Snaps (Eye did not appear in the verdict batch — its retest is still outstanding), same dim room, **Light on Auto or On**. Watch for: (1) NO watchdog deaths — a slow attempt now lives to its own deadline; (2) on an all-soft burst, the fast honest outcome instead of a burn: an AF re-trigger + ~0.6s beat + one re-capture, then either a real read or the "out of focus" panel with the numbers; (3) the Focus line's new tail — what his browser exposes for focus, what was requested, what the track reports applied (or the printed admission that focus is not exposed — itself the native-app evidence); (4) with Light left Off in the dark, the amber hint under the viewfinder; (5) the Foil chip surviving a page reload (set it, reload, it stays). If a Snap still fails, its panel now separates the causes by itself: Focus line (soft capture + AF state), Light line (darkness vs p90), Stages line (which treatments ran), and the gate note when the focus floor fired.

Pass 33 (2026-11-10 night, Andrew's verdict on Pass 32: "Now its not even working... I need the scanner to work and detect when I take the picture. Its not showing the full card in the what the scanner read... It was working earlier detecting card and the scanner working just fine. Can you please figure out a solution for this and make it work properly." + three screenshots)

- **REGRESSION ADMISSION — PASS 32'S GATE, AS SHIPPED, VETOED THE ARTWORK CHANNEL, AND THE FLAW WAS ARCHITECTURAL, IN THE PASS'S DESIGN, NOT IN ANDREW'S USAGE.** Pass 32 shipped the sharpness floor as a pre-identification GATE: a burst that stayed all-soft after one AF re-capture was refused outright. On-device (P2, 20:14, Stingerquill Annex): bursts 11.3 · 11.5 · 11.5 → 11.6 · 9.4 · 9.4 → **NO READ ATTEMPTED** — stages show capture 1.1s · detect 25ms ONLY: no title, no bottom, no match, and NO IMAGE-MATCH LINE AT ALL. (P1, 20:15, Retrieve the Esper FIN #88): bursts 2.4 · 2.3 · 4.6 → 4.7 · 4.7 · 4.7 → blocked identically (capture 342ms · detect 21ms only); nothing scanned all session (list frozen at 48 / $22.77). THREE STRUCTURAL ERRORS: **(1) THE GATE VETOED CHANNELS IT**

WAS NEVER VALIDATED FOR — the floor was calibrated purely against OCR legibility (defocus ladder + production OCR), but it fired BEFORE the artwork/image channel, whose inputs tolerate softness completely differently: the fingerprint + vision path identified this same Stingerquill at d=26, in Rapid, on the Pass 31 build, ~25 minutes before P2, on frames from the same rig and light; P2's own panel image is human-legible at panel scale. **(2) "SOFT" WAS TREATED AS TERMINAL** — the gate's only escalation was a second IDENTICAL 1080×1920 video burst; the app already owned a sharper capture class (the ImageCapture hi-res still path, in Snap service since Pass 9, plus Take Photo's per-shot OS autofocus) and never escalated to it. **(3) THE METRIC'S LAB SCALE DOESN'T MATCH HIS ROOM** — the bench's sharp is 71.9; his real dim video frames that night ran 2.3–11.6 across four bursts and two cards, including 11.3–11.6 on an artwork-identifiable, human-legible card: real low-light video softness (sensor noise + compression + mild defocus) scores differently than synthetic Gaussian blur on clean composites, so a floor fitted to the synthetic distribution sat above his entire real distribution in that room. The floor NUMBER was not the main problem — no threshold tweak fixes a veto placed in front of untested channels.

- **BASELINES (untouched code, this window):** suites 1022 PASS / 0 FAIL (36 files); battery control = `results/p32-final.json` (Pass 32's final run, per the Pass 32 precedent). AF STATE EVIDENCE FROM HIS PANELS (kept, working): "camera focus: requested continuous · track reports continuous · browser exposes manual/single-shot/continuous" — his browser DOES expose focus and reports continuous APPLIED; the "browser ignored AF" hypothesis is refuted; the frames are soft despite reported continuous AF.
- **(a) NO PRE-EMPTIVE REFUSAL — THE FLOOR IS DEMOTED TO A SOFT TRIGGER (`softAttemptPlan`, focus.ts; pinned in tests/pass33.test.ts).** Every Snap/Rapid attempt now ALWAYS runs identification: the artwork channel starts concurrently inside `identifyBurst` exactly as before (it now actually gets to run on soft bursts), and the OCR pipeline runs sharpest-first as it already did. `softAttemptPlan(scores, {snap})` is the single decision the capture paths consult, and its `proceed` is **unconditionally true** — there is no score vector, however murky, for which identification is refused (pinned for the labelled P1/P2/W1/W2 vectors, the ladder vectors, unmeasured [0,0,0], and sharp vectors alike). What the all-soft verdict still drives: the Pass 32 AF re-trigger + 600ms settle + one re-capture (kept verbatim — it demonstrably runs), the still escalation on Snap (Item b), and the ADVICE. **"Out of focus" survives only as advice**: `softFocusAdvice` (numbers printed; honest that the floor is a lab reference and real dim frames score lower) is appended to Snap's failure message in `identifyFromCardCanvas` (successful and Did-you-mean outcomes don't show it), and to Rapid's retry hint once per soft streak via `rapidSoftAdviceRef` (the Pass 32 no-spam rule, reset on success as before). `presentOutOfFocus` is DELETED — a gated refusal panel no longer exists anywhere in the app. **Rapid's quiet hand-back survives only as the ordinary no-read path**: a soft Rapid burst runs the full scan; if identification finds nothing, the machine records a normal fail and retries on its cadence, with the focus advice in the hint — the same treatment as any other unrecognised card. **NO OCR-SPEND SKIP IS SHIPPED**: the brief permitted the floor to skip the full rescue-OCR spend on hopeless frames, but the labelled evidence (a legible, identifiable card at 11.6) shows real frames diverge too far from

the ladder to gate ANY spend on this metric; the Pass 32 clock-derived watchdog deadline is what bounds a long soft attempt now.

- **(b) STILL-CAPTURE ESCALATION (`escalateSoftBurstToStill`, App.tsx).** When both video bursts meter below the floor, Snap escalates ONCE to `captureStillGuideCanvas()` (full-sensor ImageCapture, no whole-frame downscale — the Pass 9 small-card path), prepares the still like any frame (own detection/rect/sharpness), measures its title sharpness on the SAME metric, and APPENDS the measurement to the panel's focus note ("hi-res still escalation: still sharpness X vs video best Y — still leads the read"). The still leads the burst when sharper than every video frame, else joins as an extra frame; the full pipeline then runs on the burst including the still. The attempt plan is recomputed on the FINAL burst, so a sharp still clears the soft verdict and its advice with it. Where ImageCapture doesn't exist (his in-app browser may be one — `grabFrame` was already capability-guarded), the note says so plainly ("hi-res still escalation unavailable — this browser exposes no ImageCapture stills") and the video burst proceeds. Rapid takes no stills (cadence); its artwork channel carries soft bursts. Whether stills actually break the softness pattern in his rig is now a measurable phone-only question — the panel prints the still's number next to the video numbers on the first soft Snap.
- **(c) METRIC RE-GROUNDING (focus.ts, documented at `SHARPNESS_FLOOR`).** Labelled evidence, all of it: 11.3–11.6 real frames = artwork-identifiable + human-legible (OCR never attempted post-gate — unknown, not failed); dim bench controls 13.4–15.2 = OCR usable; dim+soft bench 10.8–12.5 = OCR coin flip (and the recorded ladder data, re-checked in pass33.test.ts §4, contains USABLE reads at 10.8 and 12.5 — the old gate would have refused them); W bursts 5.9–10.3 = OCR unreadable, artwork UNTESTED there (image service timed out — NOT claimed identifiable); Esper 2.3–4.7 = genuinely murky even to a human. Conclusion recorded in the code: the metric supports a RELATIVE softness warning and an escalation trigger; it does NOT support an absolute refusal — no labelled sample establishes a score below which no channel can identify a card, and the artwork channel's tolerance was never measured against this metric at all. `SHARPNESS_FLOOR` stays 12 as the trigger (its value now only costs escalation work — an AF re-capture and possibly a still — never an attempt), and the panel keeps printing raw numbers.
- **(d) THE PANEL IMAGE — FINDING + FIX.** What P1 displayed was `dewarpPreparedCard(preps[0].card)`: the DEWARP OF THE CROP, and the crop came from the tracked-fallback rect on a frame whose own detection missed ("detection missed this frame — last tracked position"). Box-mode fallback rects (tracked / calibration) are the ONE rect class never passed through `correctCardAspect` at crop time — every live detection is aspect-corrected before cropping, the fallback chain crops its rect exactly as it sits — so a lock sitting shorter than a card's shape for its width (the Esper sits LOW in the tray per P3, its bottom line at the tray edge) clips the card in the crop, and the old gate panel showed that partial crop as if it were the whole capture, with no full frame anywhere for comparison and no printed finding. The P1 detection miss itself correlates with the frame's darkness (no exposure-raised re-capture note on that attempt; the live view detects the same card fine in P3) — the Pass 28/30 seam family, already gated by seat plausibility; NOT separately localised

to a new defect this pass, and no speculative fix shipped for it. **FIXES:** (1) every attempt's panel now ALSO publishes the **full captured frame with the read rect outlined** (`frameUrl` in scandebug.ts, `buildScanPreview(preps[0])` published synchronously in identifyBurst beside the dewarp; rendered in ScanDebugBlock as "Full frame as captured — the gold outline is the area that was read") — a clipped crop can now be seen against the whole card as captured; (2) `rectClipFinding` (detect.ts, pure, unit-tested) computes what the aspect correction WOULD have covered beyond the used rect and `describePrepRect` appends it to the Rect line when the missing coverage is material (>3% of frame height): "read area is shorter than a card's shape for its width — the crop may clip the card's bottom (aspect-corrected bottom edge would sit at Y)". Card-shaped rects (the normal detected/calibration case) produce no finding, and the established line prefixes are byte-unchanged (the rectauthority suite's startsWith/includes checks all still pass).

- **(e) REGRESSION GUARDS + THE P3 HINT QUESTION.** The Pass 32 watchdog-clock fix is untouched (scanflow.ts not modified; pass32.test.ts's W-composition replay and genuine-hang inverses all green) — it remains untested ON-DEVICE (no Pass 32 attempt ever reached it, because the gate killed them first; Pass 33 attempts will). AF-state line, sharpness numbers, gate note: all retained and extended (the still measurement joins the same line). **Dark-scene hint in P3's state — verified:** the hint renders in the live view's notes stack (`darkOffHint && cameraOn` , below the scan note — the two notes do NOT suppress each other; both can show at once). It is set only by the preview ticks that meter a DETECTED card's title zone below p90 130 while Light is Off, and a tick whose own detection misses CLEARS it (`updateDarkOffHint(null)` on miss). In P3 the visible note was the calibration-failure scan note (a Recalibrate tap on the displaced card), and whether the hint was up at that instant depended on the last tick's detection/meter on a very dark frame — a display-state detail, not a suppression defect; behaviour unchanged this pass.
- **FIXTURES + SUITES.** New: `tests/pass33.test.ts` (**63 checks, all pass** — the plan contract over the panels' own vectors [P1 2.4·2.3·4.6 / 4.7·4.7·4.7, P2 11.3·11.5·11.5 / 11.6·9.4·9.4, W1/W2] for Snap and Rapid: proceed always, still-escalation Snap-only, advice exactly-when-soft with the numbers printed; non-soft/unmeasured/mixed vectors quiet; the recorded ladder data re-asserted — usable reads exist below the old floor band; rectClipFinding on card-shaped/short/top-anchored rects + describePrepRect prefix stability). **Suites: 1085 checks (1022 + 63 new), zero failures, all 37 files. Client + server typecheck clean.** The harness pipeline has no gate to mirror (the floor lived at capture time, before the pipeline it mirrors; Pass 33 changes no reader/matcher code the pipeline executes — detect.ts changes are diagnostics-only functions the harness doesn't import, focus.ts isn't imported by the harness, App.tsx isn't either), so no `pipeline.ts` mirror change was needed.
- **FINAL BATTERY (changed code, `results/p33-final.json`): per-record semantic diff (verdict / name / set / number) against the Pass 32 control (`results/p32-final.json`) over all 156 records — TWO changed records, both the known deadline-flip class, ZERO wrong settles in either run:** `exif-zedruu-6-oriented` correct-printing → correct-name-only (same card; control settled via bottom/image evidence with NO title read at 2807ms, final pooled a title read and settled name-only at 3779ms — the evidence race resolving the other

way in a slower window; the exif-zedruu family flipped identically in Pass 30) and `userphoto-1000044050_3_pwav` correct-name → no-identification (control's decisive "Pyroblast" title read landed at 2781ms; in this window the record ran 4167ms and the read never landed inside its caps — Pass 30's entry documents this exact record flipping in BOTH directions with this exact run-time signature, and two isolated rechecks in this window also read no-identification at ~4300ms under visible CPU throttling; no pipeline-executed file changed in this pass, so there is no code path by which Pass 33 could cause it). Verdict distribution otherwise identical (correct-printing 106 → 105 is the zedruu record moving within correct classes).

- **PHONE-ONLY REMAINDER (Andrew's):** Snap **Stingerquill Annex**, **Retrieve the Esper**, and **Eye of Jace** (still owed), same dim room, Light on Auto or On. SUCCESS LOOKS LIKE: an identification — artwork, OCR, or both — or an honest FULL-pipeline no-read whose panel shows title/bottom/match/image stage lines and, when the frames were soft, the focus advice attached. NEVER AGAIN a capture+detect-only panel. Watch for: (1) the Focus line on soft Snaps — the still-escalation note with the still's sharpness next to the video numbers (or the printed admission that this browser exposes no ImageCapture stills); (2) the new full-frame image in "What the scanner read", gold outline showing the read area against the whole card; (3) a clip finding on the Rect line if a fallback rect sits short; (4) the watchdog clock fix finally getting its on-device exercise if a soft attempt runs long. If a Snap still fails, its panel now separates every cause by itself: Focus line (softness + AF state + still measurement), Light line, Stages line (which channels ran), Rect line (which rect, and whether it may clip).

Pass 34 (2026-11-10 late night, Andrew's verdict on Pass 33: "Its working better again. Can you calibration more better finding the whole card. Its good but i need to gather move the phone to get it to find the card top part and the bottom needs better detection." + three screenshots — the Pass 33 live view (Esper tracked, outline hugging it), the Rapid fragment showing the degraded-guard demotion text, and the S1 Snap panel)

- **CONFIRMED WORKING, NOT DISTURBED:** the Pass 33 contract is phone-confirmed by this batch — soft bursts run the full pipeline (S1's stages: title 3.5s · bottom 99ms · match 2.0s), the still escalation fired and printed ("hi-res still escalation: still sharpness 7.3 vs video best 7.5" — **stills do NOT beat video in his rig; the softness is room/sensor-bound, recorded and closed**), the full-frame-with-read-area panel view works, focus advice attaches as advice. Pass 32's watchdog untouched (scanflow.ts not modified).
- **ITEM 1 — THE DECISIVE VERDICT THAT SETTLED "NO PLAUSIBLE NAME" (S1: Snap, Retrieve the Esper FIN #88, STILL path, rect tracked (0.03,0.18)-(0.80,0.94); panel prints "Image match: Retrieve the Esper d=44 (art 20 + card 24) · gate: decisive (d=44 ≤ 44, margin ∞ ≥ 8) · best of 3 frames"; settled outcome "No plausible name could be read from that card", evidence "title (burst)", raw 'A LRBLERVe the Esper & iC · 3 · EE | bottom: DE'). LOCALISED — TWO COOPERATING DEFECTS. (1) THE VERDICT WAS NEVER JUDGED:** the fusion consult in `identifyBurst` was single-shot, gated on the image channel landing inside its sanctioned wait (12s; the service's measured p90 round trip is 17.2s, so late landings are routine). S1's wait expired → the consult block was skipped with img null (after publishing

the "timed out" line) → Rescue 2 ran on its fresh 3.5s grant (hence title 3.5s) → DURING Rescue 2 the channel landed and the completion handler published the decisive line with outcome=null → the pool-exhausted judgement ran with imagePresentCandidate === null → kind:"none" → the "No plausible name" header. THE TELL: S1's image line carries NO outcome suffix — `describeImageChannel` appends one for every judged outcome, so the surviving line provably came from the completion handler, not the consult; the evidence line likewise stayed at its initial "title (burst)" — no fusion branch ever ran. RULED OUT: the degraded-guard demotion (that is the R1 Rapid fragment from the same session — its line DOES carry the suffix "...dim capture: decisive needs margin ≥ 12 and stable frames") and a Did-you-mean demotion with a contradictory header (uncertain outcomes render the suggestion sheet; S1's header is the `none` panel). **(2) THE GUARD'S STABILITY LEG WOULD HAVE DEMOTED IT ANYWAY:** S1's candidate (d=44, medianDist 50, dim) is exactly the demoted shape. But ranked lists are name-level — margin ∞ means the channel proposed exactly ONE name; there is no rival for cross-frame instability to arbitrate between. Every matrix-convicted dim impostor sat at a FINITE margin (8–10) with the true card as runner-up; all three on-device margin-∞ demotions were the TRUE card (Tome Blast SOS 135 d=36 median 56; Last Gasp FRA 56 d=42; this Esper d=44). **FIXES:** (a) **late consult** (App.tsx) — the consult body is factored into a `consultFusion(img)` closure inside identifyBurst; just before the pool-exhausted judgement, if the consult never ran and the channel has since landed with candidates, it runs then (shielded +8s like the first pass), so a late decisive verdict is judged on the full pool — OCR absence is not a contradiction. (b) **guard refinement** (fusion.ts) — the degraded guard's stability leg now arbitrates CONTESTED verdicts only (`stable = median == null || median ≤ 44 || margin is ∞`); the margin leg (dim decisive needs margin ≥ 12) is unchanged, so contested unstable dim verdicts and the margin-8–10 impostor class still present in both modes. The exemption is MODE-INDEPENDENT (the measured margin-∞ population is 2 Rapid + 1 Snap, uniformly true cards): the Borg-Queen demote-then-confirm flow survives for contested verdicts; an uncontested true verdict settles one attempt sooner — Andrew's Pass 23 rule working as stated. The S1 vector replays through `fuseChannels` to settle-image in both modes in tests/pass34.test.ts, alongside the impostor/contested pins.

- **ITEM 2 — WHOLE-CARD FINDING (his ask: the finder must lock the WHOLE card — top part + bottom — without him moving the phone). MEASURED** on new dim box fixtures (`tests/harness/prep_p34.py` + `probe-p34-edges.ts` , results `p34-edges.json` : light tray interior, card seated LOW at a true-5:7 seat (0.11,0.20)-(0.79,0.879), top border under a shadow gradient, bottom edge on the tray's bright seam, ×0.5–1.0 exposures — his Light-Off rig; cards: Retrieve the Esper FIN 88, Stingerquill Annex FRA 190, Swamp ONE 274): **the TOP edge is the lost edge** — raw detector top error **+0.140…+0.254 of frame height** in the shadow scenes (esper-shadowtop +0.140, swamp-shadowtop +0.175, esper-dimlow +0.169, swamp-dimlow +0.167, annex-dimlow +0.254); the **bottom is never lost** (+0.002…+0.006 raw, seam scenes included — the bright seam sits BELOW the card and Pass 28's trim keeps it out) and **sides never** (−0.005/+0.003). The historical 70/30 aspect growth then made it worse in both directions: top still **+0.028…+0.064 short** (title centroid at 0.02–0.03 of the read crop vs 0.09 control; **−0.02 — outside the crop — for annex-dimlow**) while pushing the exact bottom **+0.056…+0.092 into the tray** (bottom line dragged up to 0.81–0.87 of the crop). The

tracker faithfully stabilised the short rect — a lock is only as good as the detections folded into it. **FIX — SHOULDER-ANCHORED COMPLETION (detect.ts):** the detector now measures the texture density just outside each edge of its span, normalised by the span's interior median (`Detection.shoulders`, `edgeShouldersFromProfiles`): a correctly-placed edge has void beyond it (~0), a shadow-lost edge has the card's faded continuation beyond it. Measured separation on the fixtures: **lost tops 0.23–0.62, correct tops 0.00–0.02, opposite bottoms 0.00–0.01**; thresholds SHOULDER_STRONG 0.14 / WEAK 0.08 sit inside the gaps. `completeCardAspect` (with `correctCardAspect` as its wrapper — every existing caller byte-compatible) places the WHOLE aspect deficit on the lost edge when one side is strong and the opposite weak; both-strong (busy playmat) or ambiguous shoulders keep the historical 70/30 / symmetric split. Applied at both prepare sites and all four App normalisation sites (preview outline, calibrateBox, probes), so detections, the tracker lock, the live outline, and calibration all carry the completed rect — outline/read parity by construction. **AFTER (same probe): every shadow scene reads at control quality — read-rect top −0.016…−0.008, bottom +0.013…+0.017, title@ 0.09, bottom@ 0.92** — and bright controls are unchanged. **PRINTED:** `PreparedCard.completion` → the Rect line states "rect completed (top extended ~N%)" whenever anchored completion fired (describePrepRect; pinned in pass34.test.ts). **TRACKER GATE (brief item d):** `rectPlausibleVsSeat` passes one-edge-short rects up to ~35–40% single-edge loss (area ratio + seat coverage) and rejects ~45%+ — the gate was never the dim-rect problem; pinned explicitly (30%/35% top-short pass, 30% bottom-short passes, 45% fails, the Borg Queen quadrant fails). **PASS 30 INTERACTION VERIFIED** (probe-p30-dark re-run): the quadrant detection now completes upward ((0.30,0.52)→(0.30,0.40) corrected) but its area vs seat stays 0.42 < 0.6 — still gate-rejected, tracker never adopts it, the fallback chain still reads the seat at cover 1.00 on every tick. Completion improves sub-rects; the plausibility gate remains the arbiter.

- **FIXTURES + SUITES.** New: `tests/pass34.test.ts` (**31 checks, all pass** — the S1 vector settles in Snap and Rapid; judged vs unjudged panel-line suffix (the S1 tell); contested-impostor guards intact; the Borg-shape uncontested demotion now settles; shoulder measurement on synthetic profiles incl. the frame-flush edge; anchored completion top/bottom/left + both-strong and ambiguous fallbacks + wrapper parity; the completion print; the tracker-gate pins), `tests/harness/prep_p34.py`, `tests/harness/probe-p34-edges.ts`, `tests/harness/results/p34-edges.json`. **Suites: 1116 checks (1085 + 31 new), zero failures, all 38 files. Client + server typecheck clean.**
- **FINAL BATTERY (changed code, `results/p34-final.json`) vs same-session baseline on untouched code (`results/p34-baseline.json`): 21 of 156 records changed verdict text — and a same-window A/B control proved the window, not the code, dominates this diff.** Distribution moved favourably (correct-printing 83 → 93, no-identification 44 → 34; the single WRONG record, `dist-treasure-keeper-40`, is IDENTICAL in both runs — zero new wrong settles). Of the 5 apparent regressions (correct → no-identification), 2 (both Prism Array) re-settled correctly in isolated rechecks; the other 3 (num-courier-bat-box40, hand-bolrac-dim-angle, exif-zedruu-1-raw) failed CONSISTENTLY in isolation at 4.6–5.2s — and then failed IDENTICALLY (×2 each, same 4.7–5.2s signature) with the Pass 34 shoulder change toggled

OFF at the prepare sites (`results/p34-control-*.json`): the documented deadline-starvation flip class in a slow machine window, reproduced without the change. The 16 improvements carry the mirror signature (baseline runs at the 4.4–5.4s caps settling at 1.7–3.5s in the final window) and are likewise window-flavoured; neither direction is claimed as a Pass 34 effect. The battery's harness pipeline does not import fusion.ts/App.tsx, so Item 1 is pinned by its unit fixtures; Item 2's rect-level effect is proven by the edge probe above.

- **PHONE-ONLY REMAINDER (Andrew's):** the finder test, in the dim box with the phone HELD STILL (no moving it to help the edges): **Retrieve the Esper** and **Stingerquill Annex** Snaps (Light Auto or On, as before). SUCCESS LOOKS LIKE: the live outline locks the WHOLE card — top border and bottom line inside it — from the still phone; the Rect line may state "rect completed (top extended ~N%)" when the top was shadow-lost and completed; and S1's exact failure cannot recur — a decisive artwork verdict that lands late is now judged before the attempt settles (watch for an Esper Snap whose image line ends "→ settled by image" even when the service was slow). **Eye of Jace** retest still owed whenever he gets to it.

Long-term data behavior- **Refresh policy**: Card-names index cached client-side until user taps Refresh (or first load). Collection/decks are server DB source of truth, refetched via react-query on open and invalidated on writes. Card prices are fetched at add time and stored on the collection row (purchase/scan-time price snapshot); collection value sums stored foil-aware prices.

- **Growth**: collection_items rows (one per scryfall_id+foil+condition), decks + deck_cards rows. Duplicate scan of same card+foil increments quantity.
- **Ordering**: Collection newest first by default, searchable/filterable. Decks list by updated date.
- **Time semantics**: downloaded-at date for names index shown date-only; collection added_at stored as timestamp, rendered viewer-local.

Rejected approaches

- **Tried**: Downloading Scryfall bulk JSON (all cards full data, hundreds of MB) **Why rejected**: Far too large for phone/browser storage; the names catalog (~<1MB) gives instant local matching and full details need only one fetch per identified card.
- **Tried**: Client-side Tesseract.js in the browser **Why rejected (v1–Pass 2)**: The web-artifact platform requires all model/vision work to run through server actions (`ctx.inference`); client-side ML libraries are not permitted in this runtime. The title-band-only crop, pooled treatments and no-veto matching from the original spec are preserved exactly — only the reader itself moved server-side. Matching of the read text against the local name index still happens 100% on-device and instantly. **REVERSED in Pass 3 (2026-11-08)** by explicit user direction after timed evidence (16–20s/card) proved the server vision read is the bottleneck:

on-device Tesseract.js is now the primary path and the server read is the fallback only. Recorded here so the reversal is not mistaken for an oversight.

- **Tried**: Largest-centred-rectangle crop / whole-card OCR / confidence gates **Why rejected**: Proven failure modes from AetherForge — diverged from guide box, debris vetoed correct reads. Guide-mapped crop + title-band-only + no veto implemented instead.